

SIEMENS

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CT Examination

Operator Manual – SOMATOM Scope
syngo CT VC50

CT Examination

Operator Manual – SOMATOM Scope

syngo CT VC50

Legend



Indicates a hint

Is used to provide information on how to avoid operating errors or information emphasizing important details



Indicates the solution of a problem

Is used to provide troubleshooting information or answers to frequently asked questions



Indicates a list item



Indicates a prerequisite

Is used for a condition that has to be fulfilled before starting a particular operation



Indicates a one-step operation



Indicates steps within operating sequences

Italic

Is used for references and for table or figure titles



Is used to identify a link to related information as well as previous or next steps

Bold

Is used to identify window titles, menu items, function names, buttons, and keys, for example, the Save button

Blue

Is used to emphasize **particularly** important sections of the text

Courier

Is used for on-screen output of the system including code-related elements or commands

Courier

Is used to identify inputs you need to provide

Menu > Menu Item

Is used for the navigation to a certain submenu entry

<variable>

Is used to identify variables or parameters, for example, within a string



CAUTION

CAUTION

Used with the safety alert symbol, indicates a hazardous situation which, if not avoided, could result in minor or moderate injury or material damage.

CAUTION consists of the following elements:

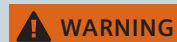
- Information about the nature of a hazardous situation
- Consequences of not avoiding a hazardous situation
- Methods of avoiding a hazardous situation

WARNING

Indicates a hazardous situation which, if not avoided, could result in death or serious injury.

WARNING consists of the following elements:

- Information about the nature of a hazardous situation
 - Consequences of not avoiding a hazardous situation
 - Methods of avoiding a hazardous situation
-



Legend

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1 Legal Notice

Siemens reserves the right to modify the design and specifications contained herein without prior notice. Please contact your local Siemens Sales representative for the most current information.

CAUTION

- ◆ Federal law of the United States of America restricts this device to sale by or on the order of a physician. (21 CFR 801.109(b)(1))

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Siemens reserves the right to modify this software. Please contact Siemens Service for the most current information.

The products and features described in this documentation may not be commercially available in all countries. Due to regulatory reasons, their future availability cannot be guaranteed. Please contact your local Siemens Sales representative for further details.

The parameters mentioned herein were specified to the best of our knowledge. We assume no responsibility whatsoever for the correctness of this information. Variations may prove necessary for individual patients. The treating physician bears the sole responsibility for all the parameters selected.

Third-party software should not be added to your system except by Siemens-authorized personnel. Customer takes full responsibility for any damage or loss of data resulting from the use of unauthorized third-party software. Siemens Healthcare GmbH makes no warranty with respect to third-party software.

1.1 Availability of Products and Features

Due to international regulatory restrictions, some products and features described in this document are not commercially available in all countries. Features may be blocked or enabled by optional software licenses. Future availability of these features cannot be guaranteed. Some features may not be explicitly marked as optional. Please contact your local Siemens Healthineers representative for further details.

1.2 System and Software Representations

The photographs, graphics, figures, and medical images used in this documentation are examples only. Their actual display and design may be slightly different on your system.

1.3 Regulatory Information

1.3.1 Regulation Compliance

Compliance to local or governmental regulations which may govern the installation, operation, and/or maintenance of the system is the sole responsibility of the medical facility.

1.3.2 Legal Manufacturer



Siemens Medical Solutions USA, Inc.

Molecular Imaging

2501 N. Barrington Road

Hoffman Estates, IL 60192

USA

1.3.3 EU Authorized Representative

EC	REP
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Siemens Healthcare GmbH
Henkestr. 127
91052 Erlangen
Germany

2 Introduction

The SOMATOM CT system is a component of your Symbia system and is used to generate tomographic images of the organs of a patient for diagnostic purposes in accordance with the computed tomography procedures. The tomographic images generated with this system can be viewed on a monitor, processed, filmed, and exported.

SOMATOM Scope can be operated in the x-ray department of a hospital or doctor's office. The CT system must be operated in rooms that are designated for medical use.

2.1 Authorized users

SOMATOM Scope must be used by persons with the necessary specialist knowledge, for example, doctors, trained radiologists, or trained technologists, after appropriate application training.

21 CFR 801.109(b)(1) CAUTION: "Federal law restricts this device to sale by or on the order of a physician"

2.2 Additional information on pediatric imaging

Be careful when scanning patients outside of the typical adult size range. For more details and further literature on pediatric imaging, please check the FDA's Pediatric x-ray Imaging web page:

<http://www.fda.gov/radiation-emittingproducts/radiationemittingproductsandprocedures/medicalimaging/ucm298899.htm>

2.3 Names and parameters

All names and data of patients and institutions that are used in the user documentation are entirely fictional.

Any resemblance to names of existing people or organizations past or present is entirely coincidental.

All parameters and images shown in the user documentation are examples. Only the parameters displayed by your system are definite.

Configuration dependent designations, such as names of drives, network nodes, and databases that are used in the user documentation are usually not the same as the designations to be found on a particular installation of the system in a clinical environment.

2.4 Medical practice

The information presented in the documentation is for illustration only and is not intended to be relied upon by the reader for instruction as to the practice of medicine.

Any health care practitioner reading this information is reminded that they must use their own learning, training, and expertise in dealing with their individual patients. This material does not substitute for that duty and is not intended by Siemens Healthineers, to be used for any purpose in that regard.

The drugs and doses mentioned are consistent with the approval labeling for uses and/or indications of the drug. The treating physician bears the sole responsibility for the diagnosis and treatment of patients, including drugs and doses prescribed in connection with such use.

2.5 Trademarks

Product names, logos, brands, and other trademarks featured, or referred to, in the Instructions for Use are the property of their respective trademark holders.

Exemplary list of trademarks used in the Instructions for Use:

- SOMATOM®
- syngo®
- SmartSelect®
- Adobe Systems® (for example, SVG-Viewer, Reader, Flash)
- Camtasia®

- List&Label™
- Microsoft® (for example, Windows 10, Paint, Excel, WordPad, Windows Media Player)
- Oracle and Java are trademarks or registered trademarks of Oracle America, Inc.



3 Safety advice

The basis for ensuring the safety of people and equipment are the statutory stipulations and guidelines as well as the information given in the user documentation.

The SOMATOM system was designed and built so that it cannot cause injury if used properly.

3.1 Basic safety rules



CAUTION

Operation of the system by non trained users!

Incorrect diagnosis or treatment due to misinterpretation of image information.

- ◆ The SOMATOM system must only be used by persons with the necessary specialist knowledge, for example, physicians, trained radiologists, or trained technologists, after an appropriate application training.



CAUTION

User is not instructed in how to operate the applications!

Wrong basis for diagnosis.

- ◆ The operator must be qualified to use the applications.

Read the user documentation carefully and observe the instructions. This applies especially to guidelines that deal with function tests, mechanical safety, and radiation protection.

CAUTION

Not observing the Instructions for Use of the system, system options and accessories!

Injury of the patient.

- ◆ Always use the Instructions for Use in conjunction with the Instructions for Use of the particular units used.
- ◆ Follow the safety instructions.

CAUTION

Not observing the Instructions for Use of the software and its applications!

Wrong basis for diagnosis.

- ◆ Always use this Instructions for Use in conjunction with all Instructions for Use provided.
- ◆ Follow the safety instructions.

Relevant statutory stipulations must be observed during installation, operation, and maintenance. Supplementary safety information is provided in the System Owner Manual:

- Technical specification
- Standards and statutory regulations
- Dosimetry and imaging performance report
- Location of labels

3.1.1 General safety practice

As an operator, you must observe certain basic rules.

■ Patient registration

Incorrect assignment of the patient's name to patient images can cause incorrect diagnoses.

Make sure that you enter the personal data of the patient correctly. This applies especially to patient position, sex and date of birth.

■ Patient positioning

The patient must always be positioned on the patient table. *Never* position the patient directly in the gantry!

Motion artifacts and/or injuries of the patient may occur if he/she moves uncontrollably during acquisition. Immobilize the patient, if necessary.

■ Restraint straps

The restraint straps are not permanently attached to the table. Thus, they cannot prevent the patient from falling off the table.

Make sure that the patient, especially a restless patient, does not fall off the table.

■ Special care

Take special care when examining infants, small children, obese patients or patients who do not keep still or have accident injuries. Take the same care when using positioning accessories.

■ System movements

When the gantry or the patient table is moved, the patient or other people could be injured.

- Make sure that the patient and other people are not in danger before you execute system movements or scans.
- Ensure that no objects can obstruct movement.

■ Information

The better the patient is informed, the easier he or she will find it to follow the rules. Give the patient enough information about the examination you are about to perform.

■ During scanning

Despite the careful design of the CT scanner, it is not possible to completely eliminate the risk of injury (danger due to collision, crushing, radiation). Never leave the CT scanner unsupervised during an examination.

■ Communication

The patient can communicate with you during the examination through the intercom system. Leave the intercom system switched on during the examination (**Listen to patient** key).

■ Stopping system movements and radiation

All system movements and radiation can be interrupted in case of an emergency.

- Make sure that you know where the **STOP** key is located. You will find **STOP** keys on the gantry and on the control box.
- Always observe the patient directly. Press the nearest **STOP** key in case of danger.

After you have pressed a **STOP** or **EMERGENCY OFF** key or if a power failure has occurred, no motorized system movements are possible. However, you can pull the patient out of the opening of the gantry manually.

■ Artifacts

Various influences can cause a distorted image. If you are not sure, repeat the acquisition, if necessary.

■ Documentation

Faults in the computer system can lead to loss of data. Document or export the patient images and studies *as fast as possible*.

■ Warning messages

Your system monitors important resources like memory or disk space. Warning messages may appear.

To avoid system behavior like freezing or delayed response time to user due to lack of memory resources, immediately restart your system if possible or at least close some applications and background activities.

3.1.2 Proper and improper use

The SOMATOM system is suitable for acquiring tomographic images of patients for diagnostic purposes using the computed tomography procedure. A patient should be examined by doctors in accordance with the methods and precautions described in the user documentation.

Improper use means if the systems or parts of the systems are used for other purposes.



Siemens Healthineers is not responsible for the loss of patient data and will refuse warranty in case of unauthorized (improper) use of the equipment (for example, surfing on the internet).

■ Switching on and off

Improper shutdown can damage the equipment and cause loss of data (on the computer). Always switch the unit on and off in the recommended way.

■ Function checks / Maintenance intervals

Only a regularly checked and serviced unit is safe. Perform the prescribed function tests. Make sure that maintenance is performed at the prescribed intervals. See System Owner Manual.

■ Placing objects on surfaces

Improper placement of objects on parts of the system may cause contamination and/or damage that may impair scanning results and/or cause image artifacts.

Do not place objects on any parts of the system that are not intended to be supporting surfaces.

■ Liquids, chemicals, heavy objects, and food

Food and drinks *must not* be consumed in the vicinity of the console or placed on the console.

In particular near the console (keyboard, control box), you must be careful with liquids and food. They can get into the equipment and cause malfunctions.

■ Accessories

Defective or not released accessories can cause artifacts, injuries to the patient and operating personnel or damage to the equipment.

- Only use accessories approved by Siemens Healthineers.
- Replace defective accessories with new original accessories immediately.



You can expand a SOMATOM standard system individually and adjust it to your clinical requirements. Please refer to the listing in your purchase contract for information about the expansion levels of your system.

3.1.3 In case of a malfunction

CAUTION

Malfunction of the system or the equipment!

X-ray not, or only partially, usable.

- ◆ Contact your local service organization if a malfunction of the system or the equipment occurs.

3.1.4 System disposal

CAUTION

Improper disposal of the system or parts of the system!

Pollution of the environment.

- ◆ System components hazardous to persons or the environment must be disposed of with care and in compliance with legally binding ordinances.
- ◆ Examples of environmentally relevant components are accumulators and batteries, transformers, capacitors, monitor picture tubes, and phantoms.
- ◆ For details contact your local customer service representative or your Siemens Healthineers Country Organization.

 CAUTION

Incorrect disposal!

Pollution of the environment.

- ◆ Please contact your Representative at Siemens Healthineers when ordering information about disassembly of the unit.



For more information, refer to System Owner Manual.

3.2 Personal safety

Despite the careful design of the CT scanner, it is not possible to completely eliminate the risk of injury (danger due to collision, crushing, radiation).

■ Before a scan

The patient must be positioned in such a way that he or she cannot collide with or be injured by table top movements or the tilting movement of the gantry.

■ During a scan

As long as the table or scanning unit is moving during scanning, the patient must always be visible and the intercom system must be switched on to hear the patient.

3.2.1 Electrical shock

Interior parts of the system carry voltage as long as the system is connected to the power supply.

 CAUTION

The system carries line voltage!

Electric shock or burn from high short-circuit current.

- ◆ Never open components of the system.
- ◆ Contact, and leave all repairs to, the Customer Service at Siemens Healthineers.
- ◆ Make sure that no objects, for example, necklaces, paperclips, or liquids can get into the interior of the system (electric shock, short circuit).

 WARNING

Cleaning of parts of the system while the system is connected to the power supply!

Electric shock due to possible contact with line voltage.

- ◆ Always switch the system off at the main power switch before cleaning or disinfecting.

The monitor bears a risk of electrical shock.

 WARNING

Unauthorized manipulation on or improper use of the system!

Electric shock

- ◆ Never open the monitor.
- ◆ Contact, and leave all repairs to, the Customer Service at Siemens Healthineers.
- ◆ Never place cups, glasses or other vessels containing liquid on or near the monitor, in case of accidental spillage.
- ◆ Make sure that no objects, for example, necklaces, paperclips, or liquids can get into the interior of the device (electric shock, short circuit).

 WARNING

Cleaning of the monitor housing during operation!

Electric shock

- ◆ Only clean the housing when the monitor is switched off.

Uninterruptible Power Supply batteries must be handled by qualified service personnel.

 WARNING

Wrong handling of batteries!

Electric shock or burn from high short-circuit current.

- ◆ Observe proper precautions.
- ◆ Servicing should be performed by qualified service personnel knowledgeable of batteries and required precautions.
- ◆ Keep unauthorized personnel away from batteries.

3.2.2 Safety of laser light marker

The patient must be instructed not to look into the laser light marker. If necessary, the user must ensure that the patient's eyes are protected (for example, if the patient is not responsive or if the eyelid closing reflex is impaired).

The patient must not wear glasses or contact lenses.

 CAUTION

Looking into laser beam with optical instruments!

Loss of sight possible.

- ◆ Do not look directly into the laser beam.



Laser product: do not view directly with optical instruments into the laser beam.

3.2.3 Safety of electronics used on patients

Electronic devices used on patients must be taken care of.

WARNING

Using a combination of different electronic devices on one patient!

The total leakage current may exceed safety limits.

- ◆ Do not use more electronic devices on the patient than absolutely necessary.

3.2.4 Hygiene standards

To minimize the risk of infection, all parts must be kept clean at all times.

CAUTION

Insufficient cleaning or disinfection of the equipment!

Injury to the patient or the personnel (bio hazard).

- ◆ Always clean or disinfect the equipment after use.
- ◆ Always observe the instructions for cleaning and disinfecting.
- ◆ Make sure that the table and the accessories are clean and covered with paper, if possible.

Blood and contrast medium are health risks. Take safety precautions when removing blood or residual contrast medium.

3.2.5 Patients with implants

Possible interference from the CT with implants must be clarified prior to examination.

**CAUTION**

Scanning patients with implanted devices such as pacemakers or neuro stimulators!

Interferences may cause malfunctions of the implanted device.

- ◆ Observe the patient closely during examination.

3.2.6 Patient monitoring

As long as the table or scanning unit is moving during scanning, the patient must always be visible and the intercom system must be switched on to hear the patient. Press the **Hear Patient** key to open the listening connection.



Special care is required if contrast medium is injected intravenously during an examination with table feed (for example, spiral scans).

**CAUTION**

Unintentional patient movement!

Injury to the patient.

- ◆ Always fix and observe the patient during the measurement.

** CAUTION**

Unobserved movement of the patient table or gantry!

Collision of the patient with the gantry.

- ◆ Monitor the patient continuously as long as the table top and gantry are moving.
- ◆ Take special care with the tilt of the gantry other than 0 degree or a table height other than the isocenter for the patient.

** CAUTION**

Unobserved movement of the patient table or gantry when using accessories and other supports!

Collision of the patient with the gantry.

- ◆ Monitor the patient continuously during table movement and gantry movement.
- ◆ Take special care of the patient when tilting the gantry.
- ◆ Follow the marking on the accessories.

3.2.7 Tripping hazard

There is a risk of stumbling when accessories are connected via cables.

** CAUTION**

Tripping of user and other persons!

Injury to the patient, the personnel, or other persons.

- ◆ Make sure that cables are installed in such a way that nobody can stumble over them.

3.2.8 Electrostatic discharge

Under normal conditions, the unit is resistant against electrostatic discharges. Refer to the System Owner Manual.

In extreme cases, faults cannot be ruled out. However, no dangerous operating states for the patient or the user will result from this.

3.3 Mechanical safety

The CT system must be operated in rooms that are designated for medical use. The room temperature and the humidity of the air must not exceed the values given in System Owner Manual.

Modifications or upgrades of the system must be made in compliance with all legal stipulations by the Siemens Healthineers service or other authorized personnel.

If damage or defects should occur to or on the system (patient table, gantry), on add-ons or accessories, safety of operation is no longer guaranteed. Watch out for such damage and have these parts repaired or replaced *immediately*.



CAUTION

Unexpected movement or malfunction of the system or equipment!

Injury to the patient or personnel. Damage to the system or equipment.

- ◆ Contact, and leave all repairs to, the Customer Service at Siemens Healthineers.

3.3.1 Fire and explosion protection

In the event of a fire, you must know how to use fire extinguishers and where the escape routes are. We recommend using CO₂ fire extinguishers.

CO₂ fire extinguishers must be located in easily accessible and visible locations.

 WARNING

Fire inside or in the vicinity of the system!

Injury to the patient or the personnel, or damage to the equipment. Gas poisoning due to burning plastic.

- ◆ Switch off the acquisition system in the event of fire.
- ◆ Make sure that you and the patient know where the escape routes are.
- ◆ Make sure that you know where the fire extinguishers are located and familiarize yourself with the use of them.

The system and accessories are not suitable for operation or storage in areas with explosive gases. The same applies to the accessories.

3.3.2 System operating conditions

 CAUTION

System is operated outside of the specified temperature range!

Wrong diagnosis possible (artifacts may occur).

- ◆ If the working conditions specified cannot be met, the room must be equipped with air-conditioning or the system must be shut down.

The system must be operated in rooms that are designated for medical use. The room temperature and the humidity of the air must not exceed the values given in System Owner Manual.

■ Ventilation openings

Blocked ventilation slots can cause damage to the equipment and loss of data (computer). Make sure that the ventilation openings on the equipment are *not* covered, for example, by cloths.

- Power connection

The power supply must be provided to all products operated as part of an X-ray system through a fixed wiring connection and a multipole interrupting device provided by the customer. The equipment must be installed according to specification DIN VDE 0100, Part 710 or must meet the respective national regulations.

- Site on/off switch

In compliance with the MDD (Medical Device Directive) an on/off switch must be installed on site by the customer.

The on/off state must be visible.

- Radiation warning lamps

Radiation warning lamps must be installed on all doors of the examination room. They must be visible from all areas where radiation can be released.

A warning signal sounds when radiation is being generated.

3.3.3 Temperature monitoring of X-ray tube

The temperature of the X-ray tube assembly is permanently monitored and calculated in advance (this function is running in the background). If the temperature rises above the limit value, a warning is given.

If necessary, scanning is interrupted until the required cooling period has elapsed.



This means that you might have to wait a certain time before you can continue with the examination as planned.

3.3.4 System overheating

The possible causes of overheating are:

- Ambient temperature too high
- Ventilation openings covered
- Defective cooling system
- Dirty air filters

If certain parts of the equipment overheat, a warning is displayed on the monitor. In this case, finish the current measurement as quickly as possible.



For exchange of the air filters please refer to System Owner Manual.

3.3.5 Electromagnetic compatibility

The system complies with the EMC stipulations. Certain items of equipment, for example, radio telephones ("mobiles") exceed the limit values of the EMC stipulations. In extreme cases, use of such items can cause interference.

Equipment such as monitors could be disturbed by magnetic fields (for example, MR unit). To some extent they can also interfere with ECG and EEG units.

Do not use a radio telephone or similar equipment in the vicinity of the system.



For more information, refer to the System Owner Manual.

3.3.6 Equipment modifications

Modifications or upgrades of the system must be made in compliance with all legal stipulations by the Siemens Healthineers service or other authorized personnel.



CAUTION

Impermissible or faulty manipulations or changes of the hardware or software can cause the system to malfunction!

Injury of the patient or damage to the equipment, or both.

- ◆ Do not open or remove the cover of the equipment nor install third-party software.

**CAUTION**

Impermissible or faulty manipulations, or changes of the software or hardware, or connection of the system to a network!

Unauthorized access; damage to the equipment.

- ◆ Make sure that all necessary precautions with respect to the existing level of security are considered when adding a functionality or altering the shipped configuration.
- ◆ You are not permitted to open or remove the housing of the equipment or to install third-party software.

As the manufacturer, assembler, installer or importer of the system, Siemens Healthineers does not accept liability for the safety functions, reliability or performance of the system, if one of the following circumstances applies:

- Installation, upgrade, readjustment, modification, repair or upgrading by persons not authorized to do so by Siemens Healthineers.
- Components not properly replaced by the original parts from Siemens Healthineers.
- Electrical installation in the CT room that does not comply with the requirements of DIN VDE 0100-710 or other binding codes of practice.
- Operation of the system in a way which deviates from the instructions given in the user documentation.

We therefore recommend that you obtain a certificate of conformity containing the following information:

- Type, extent and date of the work performed on the system
- Names of all those involved in the work (and their companies)
- Their signatures

Please note that this does *not* imply that the repairs are authorized. Siemens Healthineers does *not* accept liability for repairs that are carried out without our written permission.

If you are planning to install equipment of other manufacturers, you must obtain information about potential dangers in connecting or using systems or equipment of other manufacturers. This information can be drawn from the system specification.

If this information is not sufficient, you must consult the manufacturer of such systems/equipment or a specialist about the following topics:

- Reliability and performance of the systems/equipment
- Potential safety risks for people and equipment

If you connect additional equipment, you have to ensure that the configuration complies with standards and statutory regulations. For more information, refer to *System Owner Manual - Standards and statutory regulations*.



- For more information, refer to System Owner Manual.
- Please contact your Siemens Healthineers representative for more information.
- You can expand a SOMATOM standard system individually and adjust it to your clinical requirements. Please refer to the listing in your purchase contract for information about the expansion levels of your system.

3.3.7 Monitor failure

If no image is displayed on the monitor when the system is switched on, the monitor is likely to be defective. Call Siemens Healthineers customer service.

 CAUTION

Monitor failure!

Uncontrolled system.

- ◆ Do not make any more entries via the keyboard.
- ◆ Interrupt the examination.
- ◆ If necessary, press a **STOP** key or, in case of an emergency, an **EMERGENCY OFF** button.

3.3.8 Infusion tube tension

The infusion lines must not be under tensile stress. Make sure that the infusion lines are long enough and cannot become caught or crushed by table movements.

 CAUTION

Use of short infusion tubes!

Tensile stress on the infusion tubes when moving the table top. Tubes can get caught.

- ◆ Only use infusion tubes that are long enough.

3.4 Radiation safety

For your own and the patient's protection, you must adhere to the radiation protection regulations.

CAUTION

Radiation in the examination room after the **Start** key has been pressed!

Undesired radiation exposure.

- ◆ Leave the examination room before initiating scanning or for interventional CT examination, wear protective clothing.

Special equipment is part of the radiation protection regulations.

- Radiation control area (→ Page 42 *Radiation control area*)
- Radiation warning lamps (→ Page 42 *Radiation warning lamps*)
- Radiation shielding (→ Page 43 *Radiation shielding*)



For the dose distribution information, please refer to System Owner Manual.

3.4.1 Radiation control area

The console is located outside the radiation control area. The patient can be observed through a lead glass window. The gantry and the patient table can be operated by remote control.

When scanning, operate the system from the console whenever possible.

3.4.2 Radiation warning lamps

The radiation warning lamps on the gantry operator panels, on the control box as well as site radiation warning lamps to be provided by the customer must light up if scanning has been triggered.



If a radiation warning lamp does not light up once a scan has been released, please contact Siemens Healthineers service.

The color of the illuminated indicator indicates the operating state of the system:

- *Blinking Green*: ready for scanning



- *Yellow*: radiation warning



A warning signal sounds when radiation is being generated.

3.4.3 Radiation shielding

If scans have to be released from inside the control area or if accompanying personnel have to remain there, safety measures have to be taken (lead aprons, protective walls, and so on).

Anyone who has to be near the patient during scanning must observe the following precautions:

- Wear protective clothing (lead apron, and so on).
- Wear a PEN dosimeter and/or film badge.
- Stay in the zone shielded by the system, i.e., to the side of the gantry or behind a mobile protective wall.

3.4.4 Radiation protection for patients

The physician is responsible for protecting the patient from unnecessary radiation.

- Always use a gonadal shield, if possible.
- Scan pregnant and breast feeding women cautiously, perform it under the direction of doctors of necessary.
- Avoid repeated scan for babies and children, use child protocols and apply necessary protection.
- Use dose reduction applications, such as CARE features.

You can avoid repeating a measurement and so reduce the radiation load on the patient by taking certain precautions.

- Inform the patient sufficiently. The patient must move as little as possible during scanning and follow the breathing instructions (to avoid motion artifacts).
- Ask the patient to remove all jewelry, hair clips, and so on (to avoid metal artifacts).
- Make sure that the system is clean. Residual contrast medium can cause artifacts.
- Observe the calibration and maintenance instructions in the user documentation.

3.4.5 Deterministic radiation effects

Certain modes of operation allow selections of scan parameters that may lead to a peripheral CTDI₁₀₀ of more than 1 Gy. This dose may exceed the threshold for deterministic radiation effects on the patient's skin or eye lenses. For general dose information on the CT system, refer to the System Owner Manual.

3.4.6 Radiation in CT intervention

CARE Vision CT uses scan parameters and operating conditions which are unique and may require additional care and radiation protection measures. In any case the scan time should be kept as short as possible to avoid unnecessary exposure. Users should wear protective clothing to reduce exposure.

■ Radiation exposure to patients

CARE Vision CT applies continuous exposure at moderate mA levels. Due to the potentially long scan times and the limited scan volume, the dose for certain slices may increase to levels significantly higher than those known from standard CT applications. The patient's exposure level is usually estimated by CTDIvol which is designed to give the average dose in the scanned volume.

Before starting the scan, the dose rate (CTDIvol in mGy per second) is displayed on the monitor. During the CARE Vision CT scan, the accumulated dose (CTDIvol) reflecting the patient exposure is displayed on the monitor. The display scale ranges from zero to 2000 mGy. If the table is shifted during the exam, the accumulated dose will be distributed to different slices and will be lower than indicated by the display. When a new scan is loaded, the dose display again starts from zero.

■ Radiation exposure to personnel

During the procedure the physician is in the scan room and close to the exposed scan plane. Special care should be taken to avoid excessive and unnecessary radiation exposure.

■ Protection against primary x-ray exposure

Direct exposure to the x-ray beam should be avoided. Any body parts in the scan plane may, in the worst case, receive approximately the accumulated dose as shown on the CTDI display.

■ Protection against stray radiation

The x-ray beam is limited to the imaged scan volume, but a significant portion of the x-rays is scattered and distributed in the scan room (stray radiation). The measured dose distribution in the scan room is shown on diagrams. See System Owner Manual.

3.5 Software related safety

The SOMATOM Scope scanner is equipped with the Siemens Healthineers multi-modality software syngo® based on the operating system Windows 10®.

Please note that your system is delivered with currently known precautions regarding data security. Data is stored in a predefined directory by default. Data access is possible via **File Browser** only. For security reasons, the operating system is not accessible anymore.

CAUTION

Unauthorized access may affect system performance and data security!

Lowered system performance up to a non-operational system. Loss of data security including loss of all patient data.

- ◆ Enable your system administrator to ensure network security and the security of operational infrastructure, for example, desktop visualization.
- ◆ Consult manuals for secure setup.
- ◆ Perform system updates as required.
- ◆ Run your medical device in protected network environments only and do not connect it to public networks directly.
- ◆ Set up firewalls.
- ◆ Set up user account password protection.
- ◆ Prevent configuration files from being changed by users.
- ◆ Update virus protection software as required.

3.5.1 Software operating system

The SOMATOM Scope scanner is equipped with the Siemens Healthineers multi-modality software syngo® based on the operating system Windows 10®.

 CAUTION

Unauthorized access to the operating system!

Software manipulation

- ◆ Users are not authorized to work on operating system level. Any software manipulation is forbidden.

 CAUTION

Unauthorized access to operating system!

Software manipulation

- ◆ Users are working on operating system level at their own risk. Siemens Healthineers does not take any responsibilities for adverse effects of the system generated by such actions. Any software manipulation is forbidden.

 CAUTION

Native Windows configuration dialog box does not disappear while radiation is started. The dialog box remains visible, as it is a dialog box displayed by "Windows" and not a *syngo* dialog box!

This may cause confusion.

- ◆ Avoid opening this dialog box while radiation can be applied. The computer and the *syngo* system should never be configured during an acquisition.

 CAUTION

The operating system may not support all characters for the received DICOM document!

The display of the name, for example, the patient name may be incomplete or misleading.

- ◆ *syngo* will not change the used characters, for example, minority characters, even if it cannot display them.
- ◆ Use other attributes for identification, if possible.

3.5.2 Virus prevention and protection

While Siemens Healthineers takes measures to prevent computer viruses from being transmitted via Internet/Intranet, we cannot guarantee that hacker attacks or attachments do not contain computer virus code. Therefore, to enhance virus prevention and protection, the following precautions should always be taken.

 CAUTION

System infected by computer virus!

Manipulation of the software.

- ◆ *Never* use a data medium that could contain a computer virus, for example game CD.
- ◆ Check disks for viruses *before* using them.

■ Virus prevention

As a user of a networked PC, you should be aware of a possible infection with computer viruses or other malicious software. The focus lies on prevention of such an infection:

- All precautions have been taken to ensure data security and to prevent access to your *syngo* system with malicious intent.
- Data traffic through all unnecessary ports or "doors" is blocked. Data exchange is restricted to defined paths only.

■ Virus protection

- Use technical network defenses.
- Prepare policies, procedures, and user training.
- Restrict physical access whenever possible.
- Reduce logical interconnections to the minimum.
- Establish secure remote access for servicing.

■ Suitable network topology

A good and secure network architecture is able to reject most attacks from virus, worms, and so on. The network architects and IT administrators of a hospital shall use a suitable topology with the best protection of the medical devices connected.

- Medical devices shall always be connected within sub-networks, separated from the rest of the office equipment by a firewall PC.
- Mail programs that may receive virus-infected attachments or browser software with connection to the Internet are a potential risk for all systems connected to the same subnet. We recommend to prohibit this type of data transfer into the subnets of medical devices.

3.5.3 Software licenses

Each time you start up the system the software checks whether it is correctly installed. Check if the operating system starts up without displaying a license error message.

Some of your scan protocols or software application modules are protected by licenses. A central license manager handles the software licensing. If the license of an application expires, a warning message appears when the application starts. It remains on top until you click **OK**. Read the message carefully. It provides information about the license expiration period. The warning message contains a **Don't show me** check box. You may select the check box to prevent the warning message coming up again. Contact your system administrator or Siemens Healthineers service.

When a scan protocol you stored is loaded into Chronicle with a missing license, the protocol parameters may be adapted, or you are informed. You are also informed about the consequences for scanning and reconstruction.

■ What happens if a license expires?

If, for example, the license for the use of archive drives expires, a message box will remind you.

- The corresponding archive and export menu commands and icons will not be available.
- The list entries in the **Export to**, **Format** and **Eject from** dialog boxes are not available.
- Media inserted in the corresponding device will not be mounted and not be visible for browsing. As a consequence, data import is not possible.

■ 3D licenses

The 3D package contains the licenses for several 3D functions. If one or more of these functions are not licensed or if the licenses have expired, icons and menu commands of these functions will not be available.

If the license of the 3D package is not available at all, the **3D** task card shows the text: **No license exists for 3D.**

■ Security license

The management consoles of the security system are not accessible as long as the security license is not installed. You can only activate the security system in the **Service** software after the security license has been installed. With the license, the management consoles are accessible, regardless of whether the security system is switched on or off.



This documentation describes all software options, even though they may not necessarily be installed on your system. Contact your local sales representative if you wish to acquire an option.

3.5.4 Proper use of *syngo*

syngo is suitable for displaying and processing medical images of patients for diagnostic purposes using medical imaging systems. The manufacturer does not accept liability with regard to safety functions, reliability, or performance of the system in case of improper use.

**CAUTION**

Using *syngo* directly on a public or private network may lead to insufficient system performance!

Possible degradation of system performance or unexpected system behavior.

- ◆ Only use *syngo* in a secure and load-adapted network.

This medical software package is for authorized use only, and it is for use only by personnel with the necessary specialist knowledge, for example, physicians or technologists.

3.5.5 *syngo* Security Package

Based on diagnostic images and medical reports, decisions have to be made that affect the health of patients. Therefore, it is necessary to protect such sensitive documents from unauthorized access and to record all actions on the data.

■ Security Package

When the *syngo* Security Package is switched on, access to the *syngo* software package is protected by user identification and password. The system administrator sets a unique password for each user. Depending on the security policy in your hospital, the users may be allowed to change their password.

Access to functions as well as to data is only permitted to authenticated and authorized users. Each user is identifiable at all times. The workstation only allows authorized users. Once you log onto the workstation, your user profile determines which applications and patient data you can access.

If you are logged onto the workstation and try to access a restricted task card or patient data, you will receive an error message indicating that you do not have access to that patient or application. Though you may be able to view all of the task cards available across your institution, you may not have the permissions to work with all of them.

■ Audit trail

When the *syngo* Security Package is switched on, the activities of all users are recorded in an audit trail – including the identity of each user. The audit trail can only be seen by authorized users. Your system administrator is responsible for archiving the audit trail on a regular basis.

By activating the *syngo* security system, the customer service technician turns your system into a system with high data protection level. To secure your system, you will have to instruct a service technician as to which security options you want to be enabled.

CAUTION

After activating the security system, access is limited to only the defined users!

Limited access to the system.

- ◆ Make sure you have read and completed all preparatory steps.
- ◆ Back up your complete system as done after installation before activating the security system.

 CAUTION

Behavior of secured systems

The hospital's security policy also affects the behavior of the *syngo* system in certain cases (for example, password strength requirements, enabled empty passwords, or locking of an account after a specific number of failed logins).

- ◆ Establish a user model for your hospital and verify it before the security system is activated.
- ◆ Establish a proper procedure for emergency access.
- ◆ You may enable the service page configuration, by which emergency login without password is possible provided the entered user name belongs to "Emergency access" group
- ◆ Always back up your system before enabling the security system and before any major changes.
- ◆ Inform all users about any changes and settings. They should contact you or any other administrator immediately in case of problems.

It is recommended not to remove the predefined privileges of the administrative accounts because otherwise the administrator may not be able to solve system problems like disk is full (Bypass and archiving audit trail needed) or no license available (Bypass and change password for remote service needed). At least user management privilege should be granted. (Remark: system privileges assigned to build-in groups will not be removed without explicit user confirmation.)

 CAUTION

User access may be prevented due to forgotten or unknown accounts or passwords, account lockout policy, or wrong setup, for example, in case of an emergency!

Inaccessible system

- ◆ Define a general user account for emergency access and assign it to a group and a role both called **Emergency Access**.
- ◆ Define a local user account for emergencies. The password for this account should never expire.
- ◆ Do not allow users to change the password for the emergency account.
- ◆ The users shall contact administrators immediately in case of problems.

 CAUTION

Protection of data is changed!

The access to the data is no longer possible.

- ◆ If not enough permissions for data access are available, the person responsible for user management in hospital should add users to the groups until access is possible.

 CAUTION

The system blocks when the audit trail is filled due to too many records, or too much drive space covered!

In this case, it is not possible to work with the system. Emergency access is also not possible.

- ◆ To prevent system blockage, you must archive the log files and remove them from the audit trail on a regular basis.
- ◆ When the audit trail blocks, user with the **SecuritySystem > BypassBlockedSystem** and the **AuditTrail > Archive** privileges can login to the system and delete or archive the log files.

3.5.6 Safety on Expert-i

Expert-i is a technology that allows you to sit at a computer (the client computer) and connect to the remote *syngo* workstations. Expert-i works on the local area network (LAN) in your hospital. For example, you can connect to a *syngo* workstation from another image reading room and have access to the exam data, as though you were in front the *syngo* workstation.

 CAUTION

The display quality at the remote assistance workplace cannot be guaranteed!

Wrong basis for diagnosis.

- ◆ If the remote assistance workplace shall be used for diagnostic purposes, ensure that all necessary regulatory and legal requirements for the monitor are fulfilled.

! CAUTION

Misuse of data handling!

Wrong diagnosis

- ◆ Make sure that only a trained user gets remote access to the system.

! CAUTION

Remote and uncontrolled patient movement!

Possible injury to the patient by moving parts.

- ◆ Do not leave the *syngo* Acquisition Workplace unattended if full access is enabled. Local user still has full responsibility for appropriate usage of system.



Only use remote access in high performance networks with powerful computers to avoid insufficient display quality.

3.5.7 Secure networking

Since a high number of cyber attacks are conducted over the network, the CT system should be operated in a secure network environment. This also prevents a possible spread of malware, in case it has infiltrated the network.

The CT system offers safeguard measures against known network attacks, however, it is recommended to observe the following hints:

- Use **subnetting**. Operate the CT system and its associated DICOM nodes in a physical subnet or a VLAN. Keep this network as small as possible and open only necessary ports to the outside, for example, DICOM port 104. Usually, it makes sense to spread the subnet over some associated devices, such as a radiology department.
- Do not make network ports accessible to the outside. Network ports in corridors, waiting areas, or other publicly accessible places are easily abused by hackers.
- Do not use wireless LAN.



It is recommended to consult a network security specialist for installations with high security requirements.

3.5.8 Time synchronization with the network

When the system is in a Windows domain, an additional NTP server needs to be provided for time synchronization. This setting is typically performed upon request by the customer. The service technician needs the name or IP address of the hospital's NTP server.



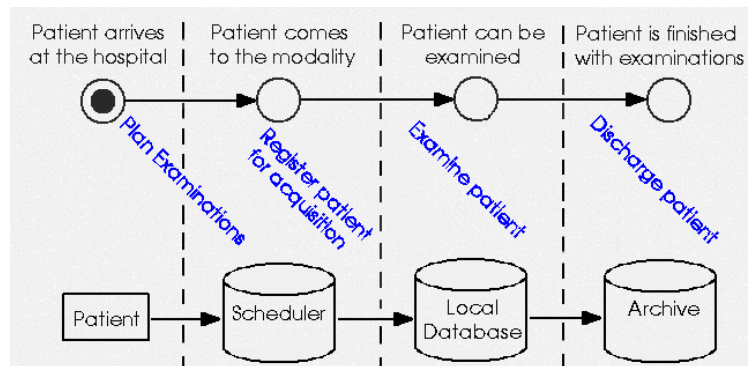
A public (Internet) NTP server is not recommended since, for security reasons, the system should not have a direct Internet connection.

In addition, the service technician needs to know the current time zone of the installation and whether the automatic daylight time (summer time) switch should occur (if the specific time zone supports it).

4 Examination basics

Your SOMATOM Scope scanner is a whole body CT scanner especially intended for spiral scans. These are obtained by a continuous and quick rotation of the X-ray detector system along with simultaneous motion of the patient table.

It may be helpful to outline basic clinical planning and to sketch the concept of the underlying databases derived from these assumptions.



Basically, it can be assumed that for each set of examinations, the patients go through several stages:

- First, they arrive at the hospital or practice where their examination is planned. For example the patients may come to a reception which is connected with the modalities via HIS/RIS, or they may go directly to a particular modality.
- Secondly, they go to one of the modalities where their examinations are actually performed and the relevant information for diagnosis is produced. Prior to the examination, images acquired during previous examinations can be transferred from the digital archive to your workstation.
- Thirdly, the images are acquired and stored in the database.
- At the end of the examination, the patients leave the modality and the information is archived.

If several examinations are planned, the patients may subsequently go to the next modality. The following 3 databases correspond with these patient stages:

- Scheduler
- Local database
- Archive

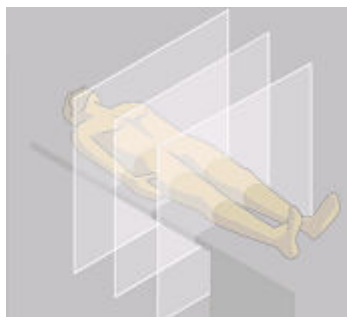
4.1 CT scan mode

A tomo scan is in one of the scan modes as indicated by the watermark on the **Routine** sub-task card.

- Spiral
- Sequence
- Multiscan
- CARE Vision
- Control Scan

4.1.1 Sequence

In the sequence mode, scan data is acquired by the sequential slicing of a scan range. The range can be scanned with or without gaps depending on the **Feed** value set on the **Scan** sub-task card.



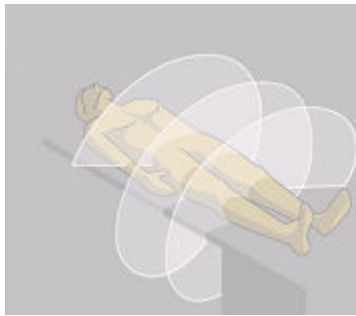
Two applications of the sequence mode are biopsy and Add scan.

A biopsy scan is used for CT intervention and made without table feed.

An Add scan is the acquisitions of one or more slice images in the sequence mode. The Add scan is inserted immediately after and as an extension to a scanned range. The parameters of the Add scan are derived from the preceding scan range. You can also change the Add scan's parameters, if necessary.

4.1.2 Spiral

A spiral scan covers a volume with continuous X-ray exposure and table feed. Compared to the sequence mode, a spiral scan covers a longer range within less time. Because the scan acquires a continuous data stream, there are several ways in which you can reconstruct the acquired data, for example, by using a different slice width.



4.1.3 Multiscan

A multiscan is done with continuous X-ray imaging and no table feed.

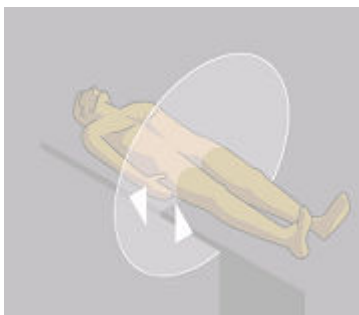
A multiscan works generally for perfusion studies (body perfusion & neuro perfusion). The perfusion protocols are available in the **Specials** group in Patient Model Dialog.



4.1.4 CARE Vision

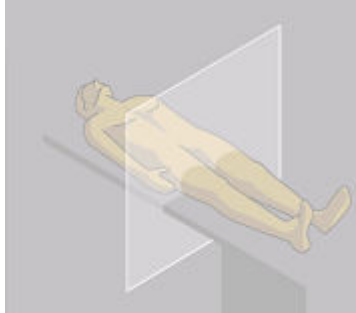
CARE Vision is a fluoroscopic mode for interventions like drainages under CT control. The physician executes a scan by *holding down* the scanning footswitch for a desired period and then release it. Radiation is on during this period. Images at the current table position and in different points of time are acquired. Therefore, you can observe the movement of the surgical instrument.

You use a CARE Vision scan protocol from the **Specials** group in Patient Model Dialog.



4.1.5 Control Scan

A control scan is a single acquisition of one slice image done before a diagnostic scan, using parameters derived from the diagnostic scan that is to follow. You cannot change most parameters of a control scan. Using the image acquired by a control scan, you can check whether the parameters (for example, mAs) are appropriate for the diagnostic scan.



4.2 CT basic parameters

A CT examination is controlled by scan parameters and recon parameters.

- Scan parameters include tube voltage in kV, tube current in mAs, acquisition (collimation), pitch in the spiral mode, feed in the sequence mode, and so on.
- Recon parameters include slice width, increment, window, kernel, Field of View (FoV) and so on.

4.2.1 Slice width and acquisition

The slice width is the thickness of the reconstructed tomographic images. It determines the image definition (spatial resolution). As a rule, the slice width should correspond to the smallest anatomic structure to be displayed.

A thicker slice leads to poorer image quality in spatial resolution. A thinner slice produces good image definition for better separation of neighboring structures, for example, in the inner ear.

The slice width is chosen on the **Routine** or **Scan** sub-task card in combination with scan **Acquisition**. Acquisition indicates the X-ray detector's physical construction used for a data acquisition, and consists of the multiplication of two factors:

- The number of the detector's physical slices used for a scan
- The width of one physical slice

The multiplication of these two factors equals to the z-coverage of a scan.

4.2.2 Pitch

The pitch parameter is used for a spiral scan.

$\text{Pitch} = \text{table feed per rotation} / \text{z-coverage}$

For a pitch less than 1, there is an overlap for data acquisition.

4.2.3 Feed

The **Feed** parameter (table feed) is used in the sequence mode. The table feed is equal to the z-coverage by default. You can, however, scan with gaps between slices for some rare diagnostic issues. Scanning with gaps bears the risk of not covering all anatomical sections. To scan with gaps in the sequence mode, set up **Feed** on the **Scan** sub-task card to be greater than the z-coverage.



When you set **Feed** to be zero (0), multiple scans will be performed at the same table position.

4.3 Dose information

In radiology, the term dose or patient dose refers to the energy deposited in the body of the patient by ionizing radiation.

4.3.1 Dose indication

The CT Dose Index volume (CTDIvol) and the Dose Length Product (DLP) are calculated and appear on the **Routine** and **Scan** sub-task cards for a selected scan entry in Chronicle.

CTDIvol: 3.85 mGy (32 cm) DLP: 107.24 mGycm

4.3.2 CARE Dose4D

CARE Dose4D optimizes the scan dose by adapting the X-ray tube current to the anatomy of a patient.

You specify the **Quality ref. mAs** value on the **Scan** sub-task card. This value is used to adjust the image quality to the diagnostic requirements or to the individual preference of a radiologist. CARE Dose4D controls the actual varying mAs applied during scanning.

**CAUTION**

mA proposal of CARE Dose not adequate due to wrong attenuation data from topogram!

X-ray not or only partially usable.

- ◆ Check plausibility of the automatically selected scan parameters before scanning.

CARE Dose4D combines three methods for dose adaptation:

- Dose adaptation to the patient size
- Dose adaptation along the z-axis
- Angular modulation

Dose adaptation to patient size

Larger patient sizes lead to more attenuation and a reduced number of X-ray quanta reaching the detector, resulting in increased noise in images. This adverse effect can be limited by adjusting the dose according to the body size.

A Quality Reference mAs is used to define a baseline image quality based on the selected scan protocols. The baseline image quality refers to a patient weighing 70 kg to 80 kg (155 lbs to 180 lbs).

The actual patient attenuation is determined from the topogram.

- Adult slim
- Adult obese
- Child

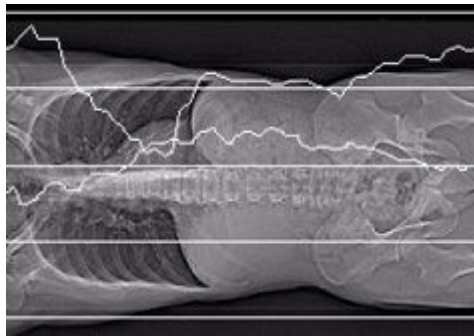
Depending on the actual patient size, CARE Dose4D automatically modifies the tube current during acquisition.



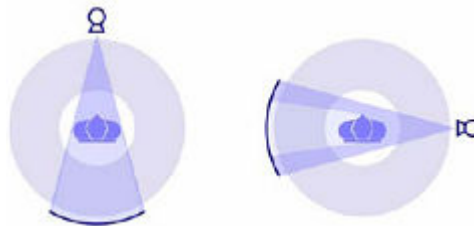
The **Quality ref. mAs** should be changed only for image quality purposes and *not* according to the size of an individual patient.

Dose adaptation along z-axis The X-ray attenuation profile along the z-axis is decided based on a single topogram. The profile consists of both the measured attenuation in the direction of the X-ray projection and the calculated attenuation in the direction perpendicular to the X-ray projection.

Based on the attenuation profile, the tube current values are calculated for every table position, resulting in an optimal dose in the acquisition of every slice image.



Angular modulation Without modulation, the attenuation of the X-ray beam for elliptical body regions such as the shoulder varies strongly with the tube rotation angle.



The angular modulation adapts the tube current over each single tube rotation according to the angular attenuation profile:

- For the low attenuation areas, the tube current is reduced to avoid excessive dose.
- For the high attenuation areas, the tube current is increased to reduce image noise.

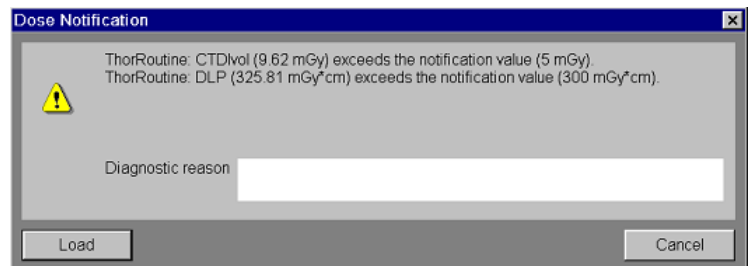
The angular attenuation profile is measured during each single tube rotation. The appropriate modulation is applied for the next rotation with a delay of only half a rotation so that consistent image quality can be achieved.

4.3.3 Dose Notification

In Scan Protocol Assistant, you can set up the thresholds of CTDIvol and DLP for a scan range.

During an examination, when you click **Load** for the scan range, its CTDIvol and DLP values are checked. If they exceed the thresholds, a **Dose Notification** dialog box appears. It shows the name of the scan range, its CTDIvol or DLP value that exceeds the threshold, and the threshold value.

On the **Routine** and **Scan** sub-task cards, the CTDIvol or DLP value that exceeds the threshold is highlighted in red.



If you want to proceed with the current dose values, type your diagnostic reason and click **Load**. Otherwise, click **Cancel** to stop loading.



Your operation will be recorded and saved to H:\siteData\DoseLogs. Ensure that you back up the data in case of a system upgrade.

4.3.4 Dose Alert

During the exam, anytime you click **Load** for a scan, its CTDIvol and DLP are accumulated to the total CTDIvol and total DLP of the exam.

If a threshold is exceeded, a **Dose Alert** dialog box appears. It shows that the accumulated dose of the examination exceeds the configured alert values either for CT DIvol or DLP with the next scan range or the next scan ranges.

You need to enter user name, diagnostic reason, and password (if set in Local Service) before you can proceed with the current dose values.

To set a password for **Dose Alert**, click **Configuration > Application** in Local Service.

In **Examination Configuration**, you can set up the alert values of CT DIvol and DLP.

- Different exam dose thresholds can be set up for adult and child. The lower alert value will be used if Chronicle includes both adult and child entries.
- If no alert values are configured, **Dose Alert** dialog box does not appear.



- In examinations where there is a clinical requirement to exceed the alert value and where a repeated acknowledgment of the **Dose Alert** dialog box would severely hamper the workflow, you may activate the **Do not show again for the remaining examination** checkbox.
- Your operation will be recorded and saved to H:\siteData\DoseLogs. Ensure that you back up the data in case of a system upgrade.

4.4 Image artifacts

Even with a carefully manufactured, calibrated and maintained CT system, patterns can sometimes appear in images which are not part of the computed tomography image. Such patterns are called *artifacts*.

For best image quality results, we strongly recommend to position the patient such that the organ to be examined is in the center of the scan field. Exceptions are examinations of the aorta or the trachea, which should not be positioned in the very center in order to avoid ring artifacts.

Artifacts can have various causes which can in general be divided into two groups: *measurement related* artifacts and *system-related* artifacts.

4.4.1 Measurement related artifacts

The most significant measurement related artifacts are the following:

- Artifacts related to patient table accessories
- Partial volume effects
- Beam hardening
- Metal artifacts
- Motion artifacts

Artifacts of patient table accessories

Certain patient table accessories may cause image artifacts:

- If you use the table top extension image artifacts might occur.
- Be aware that the edges of the straps may cause artifacts.
- Overhanging material flaps

CAUTION

Flaps outside of the scan range!

Image artifacts.

- ◆ Always use patient restraint straps to fix the flaps on the patient.

CAUTION

Flaps at the head end outside of the scan range!

Image artifacts.

- ◆ Put these flaps under the patient.
- ◆ Try not to scan the footend region.

Partial volume effects Thick slices are often preferred instead of thin slices. This helps to increase signal to noise ratio and to limit the examination time by reducing the number of slices. However, this may result in partial volume effects. These are caused by high density tissues (such as a bone) or objects (such as pacemakers) projecting only partly into the slice plane. One such well-known artifact is the so-called Hounsfield bar between the petrous bones. Other regions of the skull are also problematic with regard to partial volume effects which are visible as dark streaks. Additional correction and / or special parameter settings (head modes) help to minimize these effects.

- To avoid partial volume effects, use thinner slices.

Beam hardening When X-ray beams pass through tissue, the average energy of the radiation spectrum is shifted towards higher energy. This is known as *beam hardening*. It depends on the material density and thickness of the tissue through which the X-ray beam passes.

For body parts similar in density to water, for example, soft tissue, the effect on the CT image can be compensated. In areas of high density such as hips, shoulders or lower skull, hypodense streaks and areas with reduced CT values can be seen. Along the sides of the upper part of the skull, the so-called “cupping effect” can be seen as a brightening of the CT values. The cupping effect also occurs in rare cases such as infant skull tomograms, due to residual extra focal radiation.

Additional corrections and/or special kernel parameter settings (head modes) help to minimize these effects. Yet, they cannot fully eliminate them.

Metal artifacts Metal objects such as tooth fillings, surgical clips, jewelry, hair clips, belts or hip prostheses, and so on, can cause extreme forms of beam hardening artifacts. Depending on density and size, they can lead to total absorption of the radiation. This results in corresponding strong black or white streaks or star-shaped artifacts.

Motion artifacts Movement of organs in the slice or displacement of the entire slice during scanning likewise causes bright and dark artifacts. These are usually seen as streaks or in areas of low density. To reduce motion artifacts use the automatic patient instructions (API) function for breathing and swallowing commands.

! CAUTION

Motion-related artifacts may lead to suboptimal detection performance!

Wrong basis for diagnosis.

- ◆ Use single-breath hold technique for image acquisition.

Greater difficulties are encountered with peristalsis. Heartbeat artifacts are also problematic.

4.4.2 System related artifacts

System related artifacts may be due to the following:

- Uncalibrated system
- Detector deviations

Uncalibrated system

If the CT system is not properly setup (for example, no check-up performed) CT scale displacements and inhomogeneities can result. This may result just after switching on the unit, during warm-up to operational temperature (calibration), or due to an extended length of service of the X-ray tube. The same applies to defects in the measurement system.

The daily quality checkup usually detects and reports such discrepancies. Additional calibrations may also be necessary if large changes in temperature occur in the room.

! CAUTION

When you restart the system, the detector has not yet reached operating temperature!

Wrong diagnosis due to image artifacts.

- ◆ Calibrate the system as part of the checkup. Repeat calibration if ring artifacts occur.

Other system defects or calibration deviations cause streaks or (partial) rings to appear in the CT image.

Detector deviations In CT systems with a rotating combined tube-detector system even minimal deviations of individual detector channels from the original calibration level may lead to rings or partial ring artifact structures in the CT image. The closer such channels lie to the detector center, the greater are these effects.

In an extreme case, rings may appear concentrated as a blurred spot in the center of the scan field (the scan field center appears at the image center if the **Center X** and **Y** image parameters are both zero). Such rings are easily recognizable, but a blurred spot in the center of the scan field might lead to inaccurate diagnoses.

4.4.3 Scan repetition due to image artifacts

Measurement related or system related artifacts cannot always be excluded. Therefore the scan should be repeated if a tomographic structure appears questionable. This should be done after slightly shifting the patient's position within the slice, for example, by changing vertical position of the table by more than 5 mm. The result can be used to exclude system-dependent long-term or temporary equipment faults as the artifact cause.

CAUTION

Artifacts affecting the diagnosis are evident or suspected in a patient image, or the patient may have moved during scanning!

Improper diagnosis possible.

- ◆ Scanning must, under all circumstances, be repeated with a slight shift in patient position.

Persisting object related artifacts can be considered measurement-related with a high probability. They should be recognizable as such by an experienced radiologist.

4.5 Scan protocols

A scan protocol pre-defines examination steps. Scan protocols are the building blocks to set up the examination procedure. Siemens Healthineers provides you with a large set of scan protocols that cover most of your routine CT examinations. You can also create your own scan protocols.

- You can create, edit, or delete scan protocols using **Scan Protocol Assistant**.
- During patient registration, you can select a scan protocol from the **Study** list in the **Patient Registration** dialog box.
- At any time before an exam is closed, you can modify the examination procedure by selecting scan protocols in **Patient Model Dialog**.



Siemens scan protocols provide an optimized set of default parameters balancing image quality, scan time, and patient dose. They are defined to be applied directly to your examination after minor adaptations of, for example, the language of the Automatic Patient Instructions. Special clinical conditions and applications may require the adaptation of the default recommendations. As the final diagnosis is the sole responsibility of the user, the user always has to make sure that the chosen scan protocol meets the requirements of the examination.

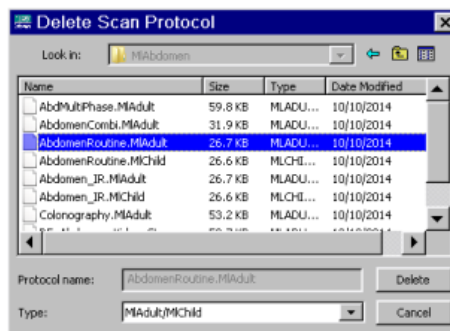
4.5.1 Creating custom scan protocols

You can create a custom scan protocol based on either a Siemens default protocol or an existing custom protocol. When you modify a Siemens default protocol, the modified protocol becomes a custom protocol. You can create a custom scan protocol in the following ways:

- Load a base protocol into Chronicle. Then, you can manipulate Chronicle, for example by adding/deleting exam steps or changing exam parameters. Finally, select **Edit > Save Scan Protocol** to save Chronicle in a new scan protocol.
- Use **Scan Protocol Assistant** to edit a scan protocol.

4.5.2 Deleting scan protocols

- 1 Select **Edit > Delete Scan Protocol** from the main menu.
- 2 In the **Delete Scan Protocol** window, double-click the folder of a protocol group. For example, abdomen protocols can be found in the **MIAbdomen** folder.
- 3 Select the protocols you want to delete.



- 4 Click **Delete**.



- You can select multiple protocols to be deleted.
- You need to enter a password to delete a scan protocol, if the password is set in **Configuration > Application** in Local Service.

4.5.3 Average CT

The CT standard spiral mode acquires only a subset of all patient based movements (caused by breathing, heart beat) during an examination. Compared to the radiotherapy data, CT data has no patient or organ movements.

Average CT mode allows you to acquire a CT volume that shows organ motion as a blurred image. The system provides a spiral mode with special parameters. With the very low pitch factor, the system can cover the full breathing cycle with one blurry image. This mode is only possible with special scan protocols marked with the suffix "Avg".

The average CT mode helps the radio-oncologists in radiotherapy treatment planning to visualize the tumor excursion for a better understanding of the target volume. With average CT, you can identify structures, such as lung lesions, during breathing.

On the **Scan** subtask card, the **Average CT** option is available. When using an average CT scan protocol, the option is automatically selected.

Only certain slice thicknesses are available for average CT. Collimation and rotation time cannot be changed. The **Series Description** is extended by the suffix "AvgCT".

CAUTION

Usage of the Average CT images for normal CT diagnosis!

Wrong diagnosis.

- ◆ The image generated by the Average CT mode shall not be used in normal CT diagnosis because of the possible artifacts and low resolution.



- In order to reach the dose as low as possible, the minimum tube voltage of 80 kV and the minimum tube current is used for average CT. By default, CARE Dose4D is switched off in an average CT scan protocol.
- If average CT is activated, the images are marked by "CQ!" to indicate clearly that images generated in this mode should not be used for the purpose of CT diagnosis.

4.5.4 Scan Protocol Assistant

Scan Protocol Assistant is a step-based wizard that guides you to set up scan protocols. The wizard contains up to 5 steps for various tasks.

You can work with two types of protocols:

- Siemens protocols
- Custom protocols



Use of Scan Protocol Assistant requires a license.

4.5.5 Importing scan protocols from teamplay

You can import scan protocols from teamplay.

✓ The scan protocols have been copied from the source medium to the `\SiteData\teamplay\ScanProtocols\Import` folder.

- 1 In Step 1 of the Scan Protocol Assistant, click **Import protocol updates from teamplay**.
- 2 In Step 2, select the scan protocols to import.
- 3 Click **Import**.



- 4 Confirm and finalize the task.



- Before using a scan protocol, it must be imported to the target scanner. Scan protocols or protocol trees can be imported to scanners of the same model type with the same or newer software version.
- When new scan protocols are available for importing, the users at the scanner will be notified to use the Scan Protocol Assistant to import them. If the user does not recognize the notification icon on the status bar () or does not take any action, a dialog box with a reminder message will be displayed after the next system startup.

5 Planning an examination

Proper patient positioning & instructions will smooth your exam workflow and avoid exam delays.

■ Patient registration

Before you examine a patient, you need to register the patient. Patient registration means that you provide the CT system with the patient information that is necessary for the examination, such as the patient's name, patient ID, date of birth, and sex.

■ Selecting a scan protocol

A scan protocol pre-defines exam steps. Scan protocols are the building blocks to set up the examination procedure. After you click the **Exam** button in the **Patient Registration** dialog box. Patient Model Dialog appears automatically for you to select a scan protocol.

■ Patient positioning

You place the patient on the patient table and position him or her for the examination. For safe and comfortable positioning of the patient, positioning aids are available.

5.1 Patient registration

Before you examine a patient, you need to register the patient. Patient registration means that you provide the CT system with the patient information that is necessary for the examination, such as the patient's name, patient ID, date of birth, and sex.

Patient name, patient ID, date of birth and sex are used for unique identification of a patient in the databases or of exported data on devices.



When registering patients in Chinese or Japanese language, the latin alphabet without special characters (ASCII) must be used for technical reasons.

5.1.1 The Patient Registration window

Click the **Patient Registration** icon or press the patient registration key on the symbol keypad (Num 0) to open the **Patient Registration** dialog box.

Patient Registration dialog box (full-size view)

You can switch to the compact view of the registration dialog box by clicking the **Fold** button (▲). Click the **Unfold** button (▼) to switch back to the full-size view of the dialog box. When being opened a second time, the dialog box appears in the same view as being lastly closed.

Patient Registration dialog box (compact view)



- The items in **bold** must be filled in. Otherwise, the **Preregister** and the **Exam** buttons are not available.
- An easier way to fill in the patient's birth date is to enter the age and then click into the **Date of birth** box. The patient's year of birth will be calculated and automatically filled in, together with the current month and day. You just need to change the current month and day to match the patient's birth date.
- To obtain a patient ID easily, click in the **Patient ID** box then click in any other box. An ID will be displayed automatically.

5.1.2 Registering an emergency patient

If a patient is in a critical condition and therefore has to be treated and examined immediately, you can use the emergency registration.

- 1 Click the **Patient Register** icon.



- 2 Click **Emergency** in the **Patient Registration** dialog box.

The emergency protocol is automatically added into Chronicle. The dialog box closes automatically. Proceed to scan the patient



- The emergency protocol is defined in **Scan Protocol Assistant**.
- Emergency patients are registered with provisional data. The data needs to be corrected and completed after the emergency exam.
- You cannot perform the emergency registration while a scan is being loaded or is in progress.

5.1.3 Registering a new patient

- 1 Click the **Patient Register** icon.



- 2 Enter the patient information in the **Patient Registration** dialog box.
- 3 Click **Exam**.

The **Patient Model Dialog** box appears, in which you select the scan protocol for a patient.

5.2 Patient positioning

You position a patient on the patient table for a scan. The patient should be positioned comfortably in an anatomically correct position. Positioning aids can be used to make it easier for the patient to rest securely and calmly. For best image quality, the body part to be examined should be as near to the gantry isocenter as possible.

- ✓ The patient table is lowered so that it is easy for the patient to sit down or to be placed on.
- 1 Ask the patient to lie down in an appropriate position and use positioning aids if necessary.
 - Supine or prone
 - Head or feet first
 - Left or right lateral
 - 2 Position the body part to be examined in the middle of the table top.
 - 3 Turn on the laser light marker to visualize the gantry isocenter and the radiation exposure plane.



- 4 Align the radiation exposure plane at the body position where you want to start a scan, by moving the patient table into the gantry opening.



- 5 Adjust the table height so that the body part to be examined is vertically as near to the gantry isocenter as possible.



- 6 Turn off the laser light marker.



From now on, the patient must lie still.



- Jewelry, glasses, prostheses, and so on, must be removed from the body part to be examined.
- Keep an eye on the patient in case of any injury while moving the table.
- Prevent crushing of infusion lines, respiratory tubes, catheters, and ECG cables while moving the table.

5.2.1 Patient positioning safety

If the system is operated correctly and the patient is correctly positioned, there is very little risk of injury to the patient or personnel.

The patient must be positioned in such a way that he or she cannot collide with or be injured by table top movements or the tilting movement of the gantry.

- Immobilize the arms of the patient with the straps supplied to rule out collisions and injury to the hands under the table top.
- Special caution is required with obese, unconscious, apathetic, unresponsive, or pediatric patients. With obese patients, make sure that no collision occurs.

CAUTION

Incorrect patient positioning!

Injury to the patient by moving parts.

- ◆ Make sure that neither the patient's clothing nor hair can get caught in mechanical parts.
- ◆ Make sure that infusion lines and respiration tubes, catheters and ECG cables cannot get caught in the space between the table top and the side parts. These components must not be put under tensile stress in any other way.
- ◆ Make sure that patient bedding cannot get caught by moving parts of the patient table.
- ◆ Use positioning aids as described.

Before you start the examination convince yourself that the patient is not endangered either by the movement of the table or by the tilt of the gantry.

 CAUTION

Unintentional patient movement!

Contusion of patient extremities at patient table and gantry.

- ◆ Always fix and observe the patient during system movements.

 CAUTION

Blinking of displays!

Seizures may occur in epileptic patients.

- ◆ Pay special attention to those patients.

5.2.2 Patient instructions

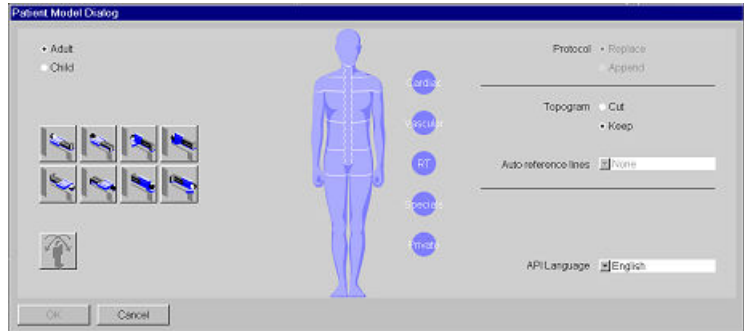
If the patient must hold his or her breath during CT acquisitions, please ensure he or she is instructed how to prior to examination. This especially applies to examinations of the thorax and abdomen. However, the slight movement that respiration inevitably causes, can also lead to artifacts in other regions.

- Explain to the patient before the examination how he or she should breathe.
- Explain to the patient that he or she must hold his or her breath for a relatively long time.
- Use the intercom system to give the patient breathing instructions.

5.3 Patient Model Dialog

In Patient Model Dialog, you select scan protocols for an exam. You can also set up the patient position, auto reference lines, and the API language for an exam.

5 Planning an examination



5.3.1 How is Patient Model Dialog accessed

Patient Model Dialog appears in the following situations:

- At the beginning of an exam

After you click the **Exam** button in the **Patient Registration** dialog box, Patient Model Dialog appears automatically for you to select a scan protocol.

- Before an exam is closed

You can open Patient Model Dialog anytime by clicking the **Select or change exam** icon in the lower-left corner of the **Examination** task card.



5.3.2 Selecting a scan protocol

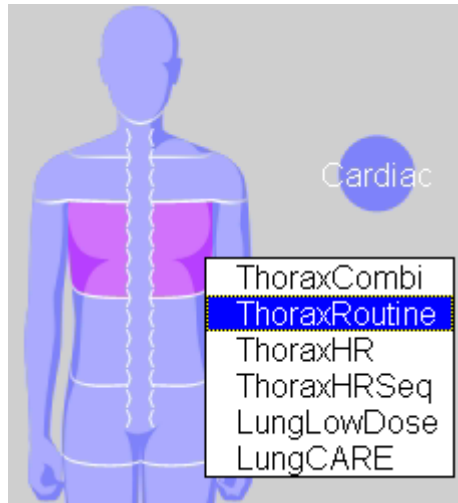
- 1 In Patient Model Dialog, ensure that the correct patient size is clicked: **Adult** or **Child**.

2 If exam steps already exist in Chronicle, do *one* of the following:

- To append a scan protocol after the last exam step, click **Append**. Click **Cut** if you want to remove the topo scan (if any) of the protocol for appending; otherwise, click **Keep**.
- To replace the non-executed exam steps with a scan protocol, click **Replace**. Click **Cut** if you want to remove the topo scan (if any) of the protocol for replacing; otherwise, click **Keep**.

3 Do *one* of the following:

- To exam a body region, point to the body region and select a scan protocol.



- To perform a special exam, point to a non-anatomical group (for example, **Vascular**) on the right side and select a scan protocol.

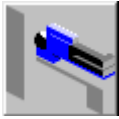


It is *not* recommended to combine adult scan protocols with child scan protocols.

5.3.3 Setting up a patient position

Patient positions are indicated by the position icons in Patient Model Dialog. The patient position for an exam is indicated by the clicked position icon.

5 Planning an examination

Icon	Patient position
	Head first - supine (H-SP)
	Head first - prone (H-PR)
	Head first - right lateral (H-RL)
	Head first - left lateral (H-LL)
	Feet first - supine (F-SP)
	Feet first - prone (F-PR)
	Feet first - right lateral (F-RL)
	Feet first - left lateral (F-LL)

A wrongly selected position will result in invalid orientation information on CT images.



CAUTION

Wrong entry of patient position!

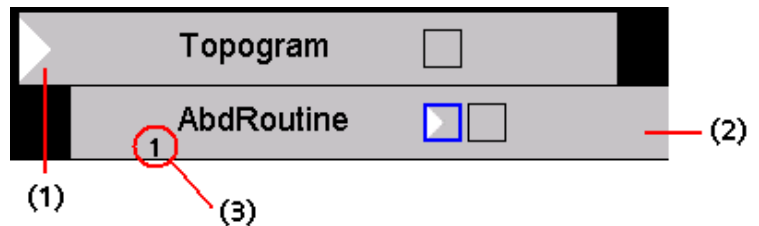
Wrong basis for diagnosis.

◆ Make sure that the patient position is correct.

- ◆ In Patient Model Dialog, click the position icon corresponding to the current position of the patient on the patient table.

5.4 Chronicle

Chronicle is a list of exam steps. It controls the entire examination procedure.



Chronicle

(1) White arrow: the exam step to be executed

(2) Selected exam step

(3) Scan step No.

- If an exam step is selected, it is shifted to the right side. The parameters of the step appear on the neighboring sub-task cards, for example, the **Routine** sub-task.
- The exam step to be executed is marked by a white arrow.
- Exam steps are executed one after another, in the order they are listed.

5 Planning an examination



For performance reasons, it is *not* recommended to have more than 20 non-performed scan steps in Chronicle.

6 Routine examination

A CT examination consists of the following steps:

- Preparing an examination

- Patient registration
- Patient positioning

- Loading a scan protocol

A scan protocol pre-defines exam steps. After you click the **Exam** button in the **Patient Registration** dialog box, Patient Model Dialog appears automatically for you to select a scan protocol. The scan protocol is then loaded into the **Examination** task card.

- Scanning a patient

- Acquiring a topogram
- Acquiring tomograms

- Reconstructing images

A CT scan collects the image raw data. From the raw data, it is possible to reconstruct several sets of diagnostic images with different recon parameters. Image reconstruction is done in recon jobs. Every scan comes with at least one default recon job. if necessary, you can create additional recon jobs to reconstruct more images.

- Evaluating images

- Documenting exam results

6.1 General safety of CT examinations

CAUTION

Superimposition by subfunctions, for example, SOMATOM life!

Scan parameters are superimposed. Dose not as desired.

- ◆ Close subfunctions before scan start.

CAUTION

Changed scan parameters are unsuitable!

Dose not as desired.

- ◆ Verify the changed scan parameters before scanning.

CAUTION

Wrong table feed direction!

X-ray not, or only partially, usable.

- ◆ Always keep an eye on the patient.
- ◆ Stop scanning in case of wrong table feed direction.

CAUTION

Wrong correction tables are used for reconstruction!

X-ray not, or only partially, usable.

- ◆ Run daily quality tests.

The **Examination** task card offers you several functions regarding the dose safety of a scan:

- If the dose efficiency of a scan is below 70%, you are informed by a message box when you load the scan.
- When more than one scan is planned without table feed in the sequence mode, you need to confirm this in a message box.

- If you stop a scan shortly after its x-ray is activated, the acquired raw data of the scan may not be sufficient for images reconstruction. The scan does not produce any images but it yields dose whose information will be recorded in relevant reports.

The Chronicle entry for the stopped scan will be automatically repeated.

6.2 Acquiring a topogram

The first step in a CT examination is typically a topo scan. The resulting image - a topogram - appears in the topo segment (upper-left segment) on the **Examination** task card.

A topogram is represented in following ways in the topo segment:

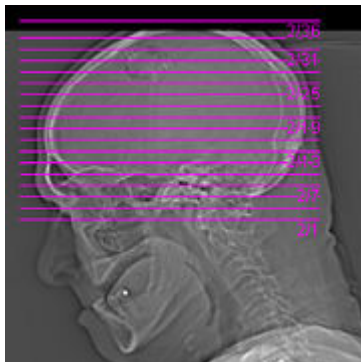
- AP view

The X-ray tube position is **Top** or **Bottom**. A topo scan produces a topogram in the anterior-posterior view.



■ Lateral view

Using the **Lateral** tube position, a topo scan produces a topogram in the lateral view. The patient table is always at the right side of the topogram.



A topogram is used for the following purposes:

- To plan the ranges for scanning and reconstructing tomograms
- To watch the progress of a tomo scan
- To document reference lines



If desired, you can acquire tomograms without a topogram.

6.2.1 Starting a topo scan

- ✓ The patient table has been moved to the starting position of the topo scan.
- 1 Close the door of the scan room.
 - 2 In Chronicle, select a topo entry and check the exam parameters on the sub-task cards.

**CAUTION**

Changed scan parameters are unsuitable!

Dose not as desired.

- ◆ Verify the changed scan parameters before scanning.

- 3 Click **Load**.



- 4 Press **Start** on CT control box when prompted.



Radiation starts. The topogram appears in the topo segment.



The topogram is displayed in real-time to indicate the scan progress. As soon as the desired portion of the topogram appears, you can suspend the topo scan, thereby reducing the patient's exposure to radiation.

6.2.2 Suspending a topo scan

- ✓ A topo scan is in progress.
- ◆ Do *any* of the following:
 - Click **Suspend**.



- Press **Suspend** on CT control box.

The topo scan is stopped. Part of the topogram remains in the topo segment.



If you suspend a topo scan during the scan start delay, the scan is unloaded.

6.2.3 Linking two topo scans

When you link two topo scans together, you can acquire a pair of topograms in different views (anterior-posterior view/lateral view) or different directions (head to feet/feet to head) with the same scan range. This helps you to plan the scan range or pre-plan the recon range for the following tomo scan more accurately, especially for body regions that are smaller than the body's contour, such as the heart or spine disc. All Siemens default cardio and SpineSeq scan protocols are provided with two linked topo scans.

- ✓ There is a topo scan entry in Chronicle.
- 1 Insert a new topo scan entry immediately after the existing one.
 - 2 Select the newly-created topo scan entry.
 - 3 Select the **Attach to previous Topogram** check box on the **Routine** sub-task card.

A dual topo indicator () is displayed, which indicates that the two topo scans are linked. The current topo scan becomes the secondary topo scan and the previous topo scan becomes the primary topo scan.

The following parameter settings of the secondary topo scan are automatically adapted to the primary topo scan:

- **Topogram length:** same as the primary topo scan. You can change the topogram length, if required. Once it has been changed manually, it will no longer automatically replicate the topogram length from the primary topo scan.
- **Tube position:** different from the primary topo scan. For example, if the tube position of the primary topo scan is **Top** or **Bottom**, the tube position of the secondary topo scan is automatically updated to **Lateral**. However, you can change the tube position, if required.
- **Direction:** opposite to the primary topo scan. You can change the direction, if required.



- To unlink two topo scans, clear the **Attach to previous Topogram** check box on the **Routine** sub-task card of the secondary topo scan.
- If you remove the primary topo scan entry from Chronicle, the linked secondary topo scan entry will be removed as well.

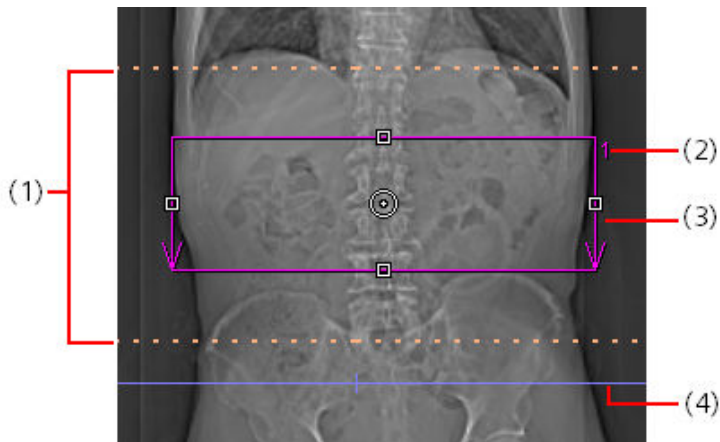
6.3 Planning an exam range

The exam range should be planned on a topogram to cover the desired anatomic region for diagnosis. First of all, you can zoom or pan the topogram for a better view. Several ways help you to define a range efficiently.

- ◆ Do one of the following:
 - Use the **Recon region** list on the **Recon** sub-task card for automatic region detection, if available.
 - Move or resize the range object in the topo segment to define a range approximately.
 - Edit the range parameters on sub-task card to define a range precisely.
 - Ctrl-drag a range object to enable or disable auto FoV, if applicable.
 - Shift-drag a range object to prevent operations on a range group.

6.3.1 About an exam range

The range of a tomo scan is graphically represented in the topo segment.



Graphic objects in the topo segment

- (1) Radiation exposure range
- (2) Scan step No.
- (3) Exam range
- (4) Table position

Colors are used to identify the status of a range:

- Magenta: the active (selected) range
- White: all other (unselected) ranges
- Yellow: an invalid range



- The exposure range is larger than the exam range. The exposure range is shown by default. You can hide it in Examination Configuration.
- When planning a head scan with two ranges without an overlap it may occur that after the planning of the tilt, an overlap of the scans is visible. To solve this problem, you can adjust the second range manually in the topo segment.

6.3.2 Exam range object

An exam range is graphically represented with rectangles, cutlines, or parallelograms in the topo segment.

The upper and lower borders (decided by scan **Begin** & scan **End**) of a rectangle indicate the scan range. Left and right borders (decided by **Center X** & **FoV**) indicate the image recon scope of the selected recon job. Arrows indicate the scan direction.

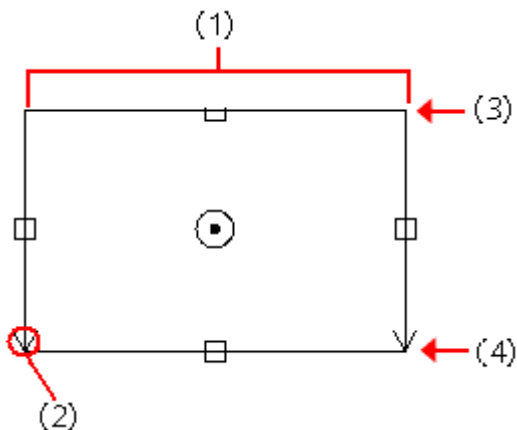


Illustration of an exam range object

- (1) FoV
- (2) Scan direction
- (3) Scan begin position
- (4) Scan end position

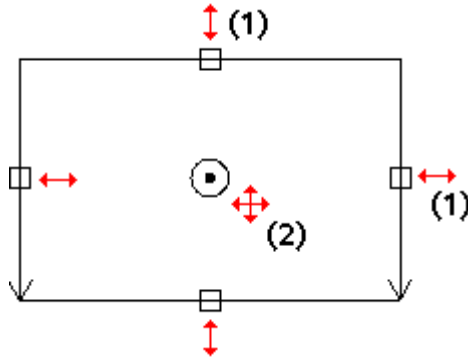
If the patient table does not move, the exam range is graphically represented as a cutline.



If the gantry is tilted, the exam range is represented as a parallelogram on a lateral topogram. Inclination of the parallelogram corresponds to **Tilt**.

To select an exam range object, you can click it in the topo segment, or select the corresponding tomo entry in Chronicle. Handles appear around a selected exam range object, allowing you to manipulate it.

To manipulate an exam range object, use drag-and-drop operations at different places, as shown below:



Manipulation of an exam range object

- (1) Resize
- (2) Move

The following table lists the manipulation of an exam range object:

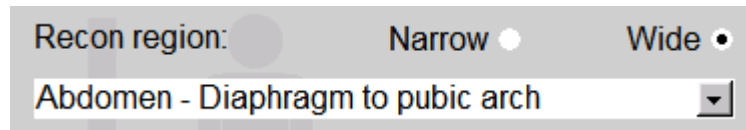
To...	Point at...	Cursor shape
Move range object	The pivot / area near the four borders	
Change range object width	Left/right handles	
Change range object length	Upper/lower handles	
Title range object (on a lateral topogram only)	Corners / upper & lower borders	



When a range object is resized or moved, the corresponding range parameters are updated on sub-task cards, and vice versa.

6.3.3 FAST Planning

The FAST Planning function automatically locates a desired body region on a topogram. On the **Recon** sub-task card, you can select the body region from the **Recon region** list. To turn off FAST Planning, select **None** from the list.



Depending on the patient physique, **Narrow** or **Wide** can be selected on the **Recon** sub-task card. The **Narrow** option does not include soft tissue. The **Wide** option includes soft tissue.



- Once you change a range manually during an exam, FAST Planning will be turned off.
- Auto FoV cannot be used for a range for which FAST Planning is available.

6.4 API & image comments

The API (Automatic Patient Instruction) function plays recorded instructions automatically to guide a patient through an exam. Each API recording consists of two pieces of text: text A and text B.

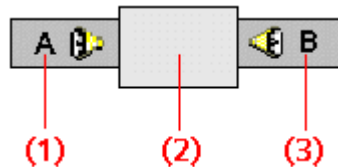
Text A, for example, "Breathe in and hold breath", is played before a scan. Text B, for example, "Breathe again", is played after the scan or when the scan is suspended.

You can comment an image with pieces of text. The comments appear at the lower border within the segment showing the image.

6.4.1 About API

The API (Automatic Patient Instruction) function plays recorded instructions automatically to guide a patient through an exam. Each API recording consists of two pieces of text: text A and text B.

Text A, for example, “Breathe in and hold breath”, is played before a scan. Text B, for example, “Breathe again”, is played after the scan or when the scan is suspended.



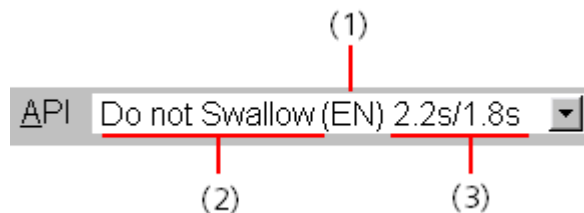
Automatic Patient Instruction

- (1) Text A
- (2) Scan
- (3) Text B

The system is delivered with pre-defined API recordings in multiple languages. You can also create custom API recordings in the **API/Comment Setup** window.

6.4.2 Creating or editing an API recording

The following figure illustrates an API recording.



API recording

- (1) API language
- (2) API name
- (3) Duration of text A and text B

1 Choose **Setup > API/Comment**.

2 In the **API/Comment Setup** window, select an API language from the **Language** list.

3 In the **API** list, do *one* of the following:

- To create an API recording, select **New Entry**, overwrite with a new name, and press **Enter**.
- To edit an API recording, select its name.

4 Click **Record**.

5 On CT control box, hold down the **Talk to Patient** key and speak clearly into the microphone to record text A. Release the key to complete the recording.

6 Repeat step 5 to record text B.

The API recording is created or edited.



If an API is frequently used and it is changed for one entry, users can apply the new API to all subsequent entries. The API text is not forwarded to either the first scan entry for dual spiral dual energy for single tube or the second scan entry for dual spiral dual energy for single tube.

6.4.3 Using API

To use API during a scan, you need perform the following tasks:

- Selecting an API recording
- Turning on/off API for an exam

Selecting an API recording

- 1 In Chronicle, select the scan step to which you want to apply an API recording.
- 2 On the **Scan** sub-task card, select the API language in the **Language** list if necessary.
- 3 In the **API** list, select the name of the API recording.

An API icon (a speaker) appears on the scan entry.





- The API recording will be played only if you also switch on the API function for an exam (→ Page 103 *Turning on/off API for an exam*).
- To prevent the API recording from being played, select **None** from the **API** list.

Turning on/off API for an exam

The **API** button controls whether your selected API recordings are played in an exam.

API button	Status	Description
	Clicked	Any scan step for which you specified an API recording will play the recording during the exam.
	Un-clicked	API recordings will <i>not</i> be played, even if they have been specified for scan steps.

◆ Do one of the following:

- To switch on API, make sure that the **API** button is clicked.
- To switch off API, make sure that the **API** button is un-clicked.



If none of the scan steps are specified with an API recording, the **API** button is unavailable.

6.5 Acquiring tomograms

Tomograms are the cross-sectional views of a body region. You acquired tomograms for diagnosis, typically in either spiral or sequence scan mode.

6.5.1 Starting a tomo scan

- ✓ The exam range of a tomo scan has been planned properly.

- 1 In Chronicle, select a tomo entry and check corresponding exam parameters on sub-task cards.

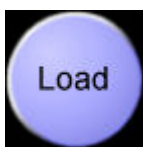
CAUTION

Changed scan parameters are unsuitable!

Dose not as desired.

- ◆ Verify the changed scan parameters before scanning.

- 2 Click **Load**.



After a scan has been loaded, any user interaction on sub-task cards should be avoided.

- 3 If prompted, hold down **Move** on CT control box to move the patient table to the scan starting position.



- 4 Press **Start** on CT control box.



Radiation starts. Tomograms appear in the tomo segment.



The blue line in the topo segment indicates the position of the patient table.

6.5.2 Suspending a tomo scan

✓ A tomo scan is in progress.

◆ Do *any* of the following:

- Click **Suspend**.



- Press **Suspend** on CT control box.

The tomo scan is stopped.



If you suspend a tomo scan during the scan start delay, the scan is unloaded.

6.5.3 Repeating a tomo scan

1 In Chronicle, right-click a tomo entry.

2 Choose **Repeat** from the shortcut menu.

The tomo entry is repeated before the next non-executed scan step.

6.6 Common recon parameters

Some parameters for CT images reconstruction:

- Slice width
- Recon increment

- Window
- Kernel

6.6.1 Slice width

The slice width is the thickness of the reconstructed tomographic images. It determines the image definition (spatial resolution). As a rule, the slice width should correspond to the smallest anatomic structure to be displayed.

A thicker slice reduces the number of images to be reconstructed and may speed up an examination, but results in poorer image definition. A thinner slice produces good image definition for better separation of neighboring structures, for example, in the inner ear.

On the **Recon** sub-task card, you can select the slice width of a recon job in the **Slice** list. This recon slice is no thinner than the scan slice defined in the scan collimation.

6.6.2 Recon increment

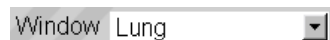
The recon increment defines the distance between recon images.

If the increment is smaller than the slice width, the images are reconstructed with overlaps. This improves the image quality for 3D post-processing. If the increment is larger than the slice width, the images are reconstructed with gaps in between.

6.6.3 Window

Window values (window width and window center) define the brightness and contrast of recon images, based on organ-specific CT values. Appropriate window values produce images with an optimized appearance for a particular organ.

On the **Recon** sub-task card, you can choose the organ-specific window values in the **Window** list.



6.6.4 Kernel

On the **Recon** sub-task card, kernels are selected in the **Kernel** list, for example, B90s. A kernel is a recon parameter affecting image sharpness and noise.

Each kernel's name starts with a capitalized letter representing the body region or a special application. **B** stands for body; **H** stands for head; **U** stands for Ultra High Resolution; **C** stands for child head; **S** stands for a special application (for example, *syngo* Osteo CT).

The image sharpness is defined by the number in the kernel's name. The higher the number, the sharper the image.

Image appearance	Typical kernel number
Smooth	10-20
Medium	30-50
Sharp	60-70
High resolution	80-90

6.6.5 Iterative metal artifact reduction

Metal objects, such as metal implants or metal markers, can cause beam hardening artifacts. Depending on density and size, these metal objects can lead to a total absorption of radiation resulting in strong black or white streaks or star-shaped artifacts. Such artifacts can be reduced by the iterative Metal Artifact Reduction (iMAR) algorithm.

iMAR can be used in conjunction with several other recon options:

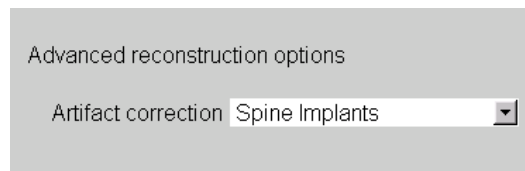
- iMAR works for both axial and 3D recon.
- iMAR works with extended FoV.
- iMAR works with an Iterative Reconstruction method.



The images reconstructed with iMAR may not be exactly the same as the images reconstructed using a conventional method. Therefore, it is recommended to use both reconstruction methods.

- ✓ A recon job to which you want to apply iMAR is selected.
- 1 On the **Recon** sub-task card, make sure an iMAR-compatible kernel is selected (for example, B41s).
- 2 Click the **iMAR** icon ()

The iMAR options appear.



iMAR supports the following implants:

- Neuro coils (iMARne)
- Dental fillings (iMARde)
- Spine implants (iMARsp)
- Shoulder implants (iMARsh)
- Pacemaker (iMARpa)
- Thoracic coils (iMARth)
- Hip implants (iMARhi)
- Extremity implants (iMARex)

The parenthesized abbreviations are made up of "iMAR" and the first two characters of the implant name, for example "iMARne" for neuro coils. The abbreviation will be inserted into the **Series description** box and the **Comments** box for recon jobs using iMAR.

- 3 In the **Artifact correction** list, select the implant type.



- Use of iMAR requires a license.
- Use of iMAR can be defined in a scan protocol.
- Do not use iMAR images for *syngo* Osteo evaluation.
- iMAR is not applicable to some special scan types, such as Control scan, CARE Vision, Bolus Tracking, and TestBolus.

6.6.6 Iterative reconstruction

Depending on your scanner type and license, one of the following iterative reconstruction methods is available.

- IRIS
- SAFIRE

IRIS Iterative Reconstruction in Image Space (IRIS) an image-based iterative reconstruction method using a special algorithm. Compared with the common reconstruction method based on the same image raw data, the reconstruction with the IRIS algorithm produces images with less noise and enhanced image sharpness.

To activate IRIS, click the **IRIS** check box on the **Recon** sub-task card.



When a recon job is set up to use IRIS, the recon job number is marked in blue.



When you select the **IRIS** check box, the **Kernel** list on the **Recon** sub-task card changes to the **Algorithm** list. An algorithm's name starts with the character *I* or *J*, for example, I30s or J40. Algorithm I corresponds to kernel B. Algorithm J corresponds to kernel H.



- Use of IRIS requires a license.
- IRIS *cannot* work with extended CT scale.
- Do *not* use IRIS images for evaluation with the *syngo* Osteo or CaScoring applications.

About SAFIRE

Sinogram Affirmed Iterative Reconstruction (SAFIRE) is a raw data-based iterative reconstruction method using a special algorithm. Compared with the common reconstruction method based on the same image raw data, the reconstruction with the SAFIRE algorithm produces images with less noise and enhanced image sharpness.

To activate SAFIRE, click the **SAFIRE** check box on the **Recon** sub-task card. Then, specify a value in the **Strength** box. The higher the strength value, the more artificial the images become.



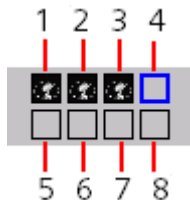
When you select the **SAFIRE** check box, the **Kernel** list on the **Recon** sub-task card changes to the **Algorithm** list. An algorithm's name starts with the character *I*, *J* or *Q*, for example, I30s or J40. Algorithm I corresponds to kernel B. Algorithm J corresponds to kernel H. Algorithm Q corresponds to kernel D.



- Use of SAFIRE requires a license.
- SAFIRE *cannot* work with extended CT scale.
- Do *not* use SAFIRE images for evaluation with the *syngo* Osteo or CaScoring applications.
- Depending on your system type, if you apply SAFIRE in sequence scan, the display of reconstructed images may have some delays during scan.

6.7 Working with recon jobs

A recon job defines a group of parameters used to generate a set of diagnostic images from the image raw data acquired by a scan. Recon jobs of a scan are represented by the recon icons on a scan entry, sorted from left to right, and from the first line to the second line.



When you click a recon icon, its corresponding recon job number is selected on the **Recon** sub-task card, where the recon parameters are available.



On the **Auto Tasking** sub-task card, if you select the **Auto recon** check box for a recon job before a scan, the recon job is automatically executed after the scan.

6.7.1 Status of a recon job

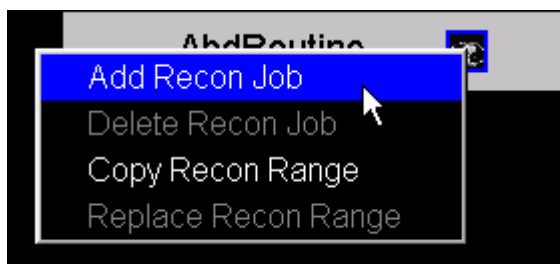
A different recon job icon represents the status of a recon job.

Status	Icon	Description
Open (axial)		The axial recon job has been created, but not executed.
Open (3D)		The 3D recon job has been created, but not executed.
RTR planned		All images of a recon job will be reconstructed parallel to a scan (realtime reconstruction). As soon as the scan is done, the recon job is finished.
RTD planned		This icon is only available for the default (the first) recon job of a scan. To ensure realtime display of images in the tomo segment, one image per gantry rotation will be reconstructed during a scan. However, gaps are left between the recon images.

Status	Icon	Description
RTD processed		This icon is only available for the default (the first) recon job of a scan. The recon job has been processed automatically for realtime display during a scan. After the scan, further execution of the recon job is necessary for a complete set of recon images.
Running		The recon job is running (being executed). Images are being reconstructed.
Queued		The recon job has been queued for execution.
Suspended		The recon job has been suspended.
Finished		The recon job is completed.
Finished (256 matrix)		Only for the recon job using the 256 matrix option of a cardiac spiral scan. The recon job is completed.

6.7.2 Creating a recon job

- 1 In Chronicle, select a scan entry.
- 2 On the entry, right-click any recon icon and choose **Add Recon Job** from the shortcut menu.



A new recon icon appears and is selected.



- You can also click one of the recon job numbers (1 to 8) on the **Recon** sub-task card. If the clicked number is greater than the quantity of existing recon jobs, one more recon job is added.
- You can add up to 8 recon jobs in one batch. If you want to reconstruct more than 8 sets of images, delete some of the finished recon jobs. The corresponding images will *not* be deleted from the local database in Patient Browser. Then, recon jobs can be added again.

6.7.3 Starting a recon job

You can start an open recon job, a RTD processed recon job, or a suspended recon job, after a scan.

✓ No recon job is running.

1 In Chronicle, click the icon of the recon job you want to start.

2 Click **Recon**.



The recon job is started. Recon images appear one after another in the tomo segment. A blue progress bar appears right to the tomo segment.

6.8 Performing axial recon

An axial reconstruction produces slice images in the axial view.

- First, create a recon job and select **Axial** as the recon type on the **Recon** sub-task card.
- Then, plan an axial recon range. (→ Page 116 *Planning axial recon range*)

Other recon parameters, such as slice width and recon increment, are given default values for the recon job. If necessary, you can edit these parameters.

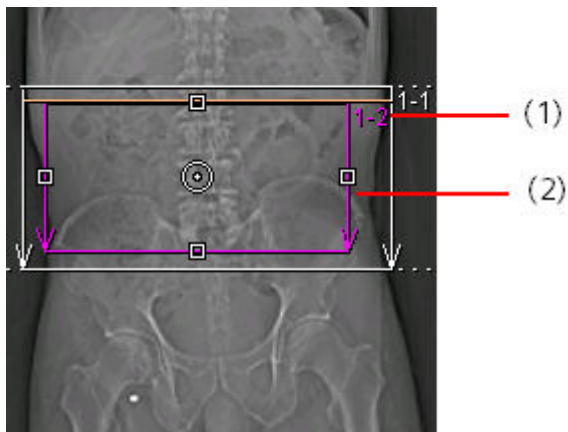
Before the image is reconstructed, you can click the **Overview** button on the **Recon** sub-task card to reconstruct a test image. This is used to check whether recon parameters are suitable for a proper image quality.

- Finally, start the recon job. Images are reconstructed and appear one after another in the tomo segment.

6.8.1 About axial recon range

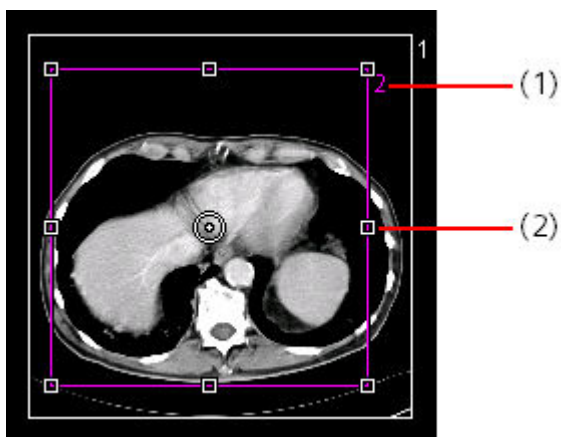
After a scan is done and you click the icon for one of the scan's unperformed axial recon jobs, two graphic objects representing the job's recon range appear in magenta in the topo and tomo segments, respectively.

You can manipulate these graphic objects in order to plan a recon range.



Topo segment after a scan

- (1) Scan step No. and recon job No. (e.g. 1-2)
- (2) Graphic object of a recon range in the topo segment



Tomo segment after a scan

- (1) Recon job No. (e.g. 2)
- (2) Graphic object of a recon range in the tomo segment

6.8.2 Planning axial recon range

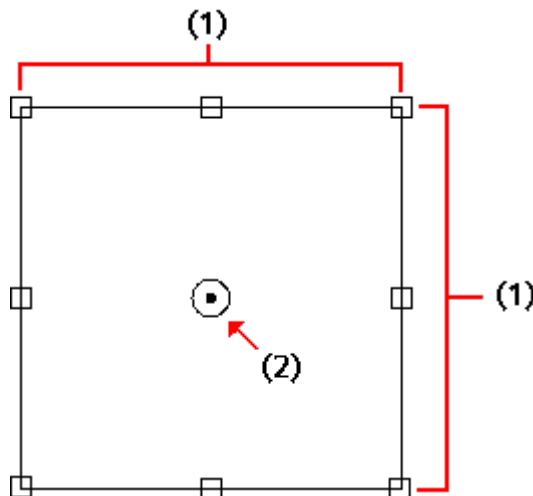
There are several possibilities for you to plan a recon range:

- To plan a range quickly but approximately, you can manipulate range objects in the topo and tomo segments.
- To plan a range precisely, you can edit range parameters on the **Recon** sub-task card.
- You can use the **Recon region** list on the **Recon** sub-task card for automatic region detection, if available.
- You can also copy an axial recon range from another axial recon job.

6.8.3 Recon object in tomo segment

In the tomo segment, a recon range is graphically represented as a square.

The side length of the square indicates the recon FoV (defined by the **FoV** parameter). The coordinates of the square's center point correspond to the **Center X** and **Center Y** parameters.



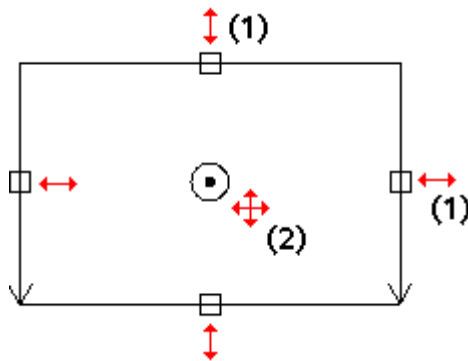
Recon range object in tomo segment

- (1) FoV
- (2) Center X & Center Y

6.8.4 Editing axial range objects

After a scan is done and you click the icon for one of the scan's unperformed axial recon jobs, two graphic objects representing the recon job's range appear in magenta in the topo and tomo segments, respectively. You can manipulate these graphic objects in order to plan an axial recon range.

The following figure shows the drag-and-drop operations you can perform to a recon range object in the topo segment:



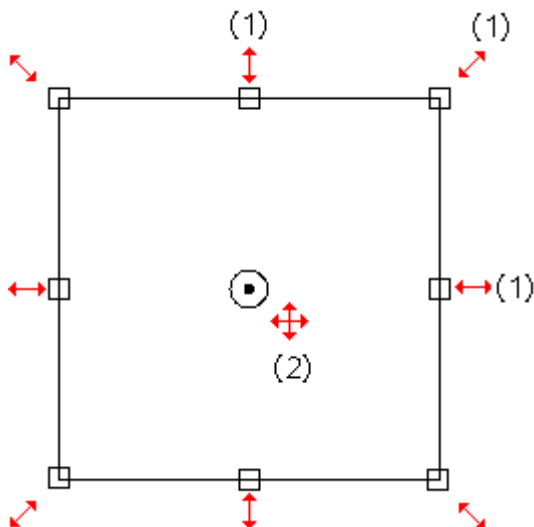
Changing recon range in topo segment

- (1) Resize
- (2) Move

The following table lists the drag-and-drop operations to a recon range object in the topo segment:

To...	Point at...	Cursor shape
Move the range	The pivot / area near the object borders	
Change FoV	Left and right handles	
Change the recon starting and ending positions	Upper and lower handles	

The following figure shows the drag-and-drop operations you can perform to a recon range object in the tomo segment:



Changing recon range in tomo segment

- (1) Resize
- (2) Move

The following table lists the drag-and-drop operations to a recon range object in the tomo segment:

To...	Point at...	Cursor shape
Move the range	The pivot / area near the object borders	
Change the FoV	Any handle	



- When a range object is resized or moved, the corresponding range parameters are updated on the sub-task cards, and vice versa.
- Zoom or pan the topogram or tomograms for a better view.
- Use the **Recon region** list on the **Recon** sub-task card for automatic region detection, if available.
- **Ctrl**-drag a range object to enable/disable auto FoV, if applicable.
- **Shift**-drag a range object to prevent operations on a range group.

6.8.5 Using extended FoV

The extended FoV (Field of View) is used when the standard FoV cannot cover the body region for which you want to reconstruct images.

CAUTION

Reduced image quality caused by extended field of view!

Wrong diagnosis due to insufficient image information.

- ◆ For diagnosis, do not use the image area outside the regular field of view. It has reduced quality and may contain artifacts.

The images reconstructed with the extended FoV are marked with a red circle, indicating the standard FoV range. To delete this red circle, select it and press the **Delete** key.



◆ On the **Recon** sub-task card, select the **Extended FoV** check box.



- The image area outside of the red circle has a reduced image quality and is *not* suitable for diagnostic purposes.
- Use of extended FoV requires a license.
- If you apply extended FoV (field of view) for interventional (for example, Biopsy) scan, please be aware that the image quality might be impacted. Normal FoV is recommended for interventional scan.

6.8.6 Using HD FoV

Actual analysis shows a HU deviation of nearly ± 500 HU in the Extended Field of View of 700 mm. With the Enhanced Extended Field of View (HD FoV) algorithm, this situation should be improved. Detecting the patient's contour of the body in combination with an iterative reconstruction, an HU stability of approximately ± 20 HU in the Extended Field of View up to 700 mm can be achieved.



The HU deviation is measured with Gammex 467 phantom; clinical results may vary.

The images reconstructed with HD FoV are marked with a dotted red circle, indicating the good quality range. To delete this red circle, select it and press the **Delete** key. Outside the dotted red circle to the FoV of 700 mm, the HU deviation is ± 20 HU. The image quality outside the FoV of 700 mm is not assured.



- ◆ On the **Recon** sub-task card, select the **HD FoV** check box.

When the **HD FoV** check box is selected, the FoV default value and the upper/lower limits of FoV will change.

When **HD FoV** is enabled and is not possible any longer because of tilting, a dialog opens informing the user that **Extended FoV** is automatically used instead.



- The image area outside the red circle has a reduced image quality and is not suitable for diagnostic purposes.
- While the quality of the image outside of the field of measurement (FoM) is improved, the reconstruction time, especially the first image time, increases significantly.
- Use of HD FoV requires a license. If both **Extended FoV** and **HD FoV** are enabled, **HD FoV** should be more effective and is used.
- HD FoV is used mainly for RT planning and improves the contouring and HU accuracy outside the FoV of 500 mm.

6.8.7 Using extended CT scale

Using an extended CT scale, a wider range of CT values (from -10240 to +30710 Hounsfield units) can be distinguished on images. When this option is selected, objects with a higher attenuation, such as metal implants, are shown within the gray scale instead of in white.

- ◆ On the **Recon** sub-task card, select the **Extended CT scale** check box.

6.9 Performing 3D recon

To improve workflow efficiency, you can perform 3D image reconstruction directly on the **Recon** sub-task card. This is more convenient than evaluation using the separate **3D** task card. You can reconstruct both MPR and MIP images.

Before you plan a 3D reconstruction, make sure that a spiral scan has been finished and at least one axial recon job has been finished. The axial recon images are used as the base images for planning a 3D reconstruction.

- Firstly, create a recon job and select **3D** as the recon job type on the **Recon** sub-task card. An open 3D recon job is indicated by a 3D recon job icon.



On the 3D recon toolbar, select the desired planning base in the **Planning base** list. The axial recon jobs that have been finished and that are listed before the 3D recon job are available for you to select.

- Secondly, you need to plan the 3D recon orientation. In a standard 3D orientation, you can reconstruct coronal, axial, or sagittal images. If you want to reconstruct images at sloping slices, you need to define an oblique axis.

Next, plan the 3D recon volume by setting up the recon starting & ending positions, and by setting up the recon FoV.

Other recon parameters, such as slice width and recon increment, are given default values for a recon job. If necessary, you can edit these parameters. Before the image is reconstructed, you can click the **Preview Image** icon on the 3D recon toolbar, and check the preview image to decide whether current recon parameters are appropriate.

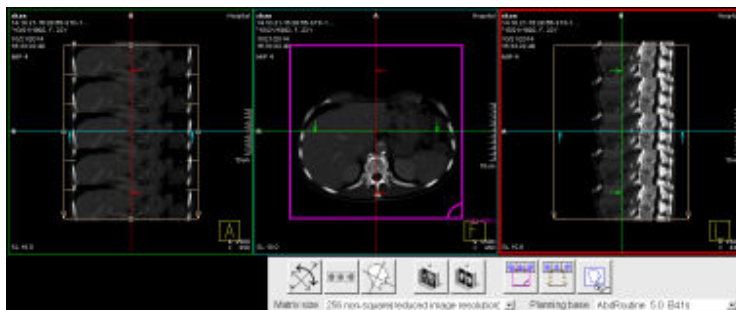
- Finally, start the 3D recon job.



- Use of 3D recon on the **Recon** sub-task card requires a license.
- 3D recon can be done using the raw data acquired only by spiral scans without gantry tilt.

6.9.1 3D recon segments

When the **3D** recon job type is selected on the **Recon** sub-task card, the 3D recon segments appear instead of the standard topo and tomo segments.



Each 3D segment shows one plane of a scanned image volume. The following views are shown from left to right:

- Coronal view
- Axial view
- Sagittal view

6.9.2 Using a standard 3D axis

- ◆ In the **Recon axis** list of the **Recon** sub-task card, do *one* of the following:
 - Select **axial** if you want to reconstruct axial images. The second segment (in the axial view) is marked as the FoV segment.
 - Select **coronal** if you want to reconstruct coronal images. The first segment (in the coronal view) is marked as the FoV segment.
 - Select **sagittal** if you want to reconstruct sagittal images. The third segment (in the sagittal view) is marked as the FoV segment.

6.9.3 Setting up an oblique axis

If you want to reconstruct images at sloping slices, you need to define an oblique axis.

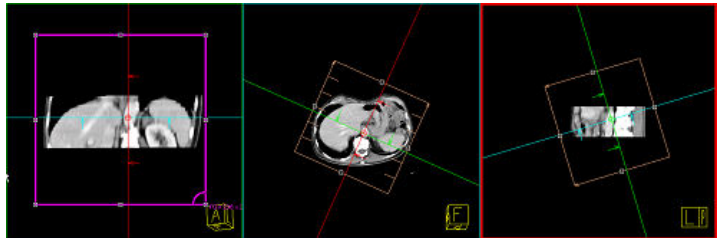
- 1 On the 3D recon toolbar, make sure that the **Free Mode** icon is activated.



- 2 Make sure that the **Hide Reference Lines** icon is not activated so that reference lines are shown.



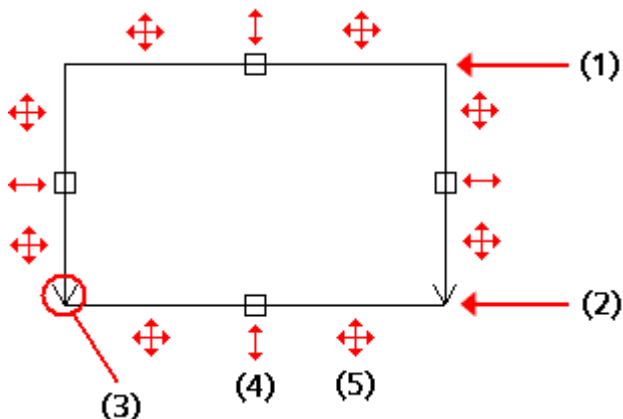
- 3 In the reference segments, position the mouse pointer close to one of the reference lines. The cursor shape changes into a rotation icon (↻).
- 4 Rotate the reference lines by dragging them clockwise or counterclockwise, until the FoV segment shows the image view that you want.



- As soon as any reference line is rotated, the value in the **Recon axis** list changes to **oblique**.
- If the **Free Mode** icon is not clicked, you cannot rotate the reference lines.

6.9.4 Setting 3D recon start and end positions

The recon starting & ending positions are graphically represented in the reference segments by the borders perpendicular to the recon direction, as shown in the following figure:



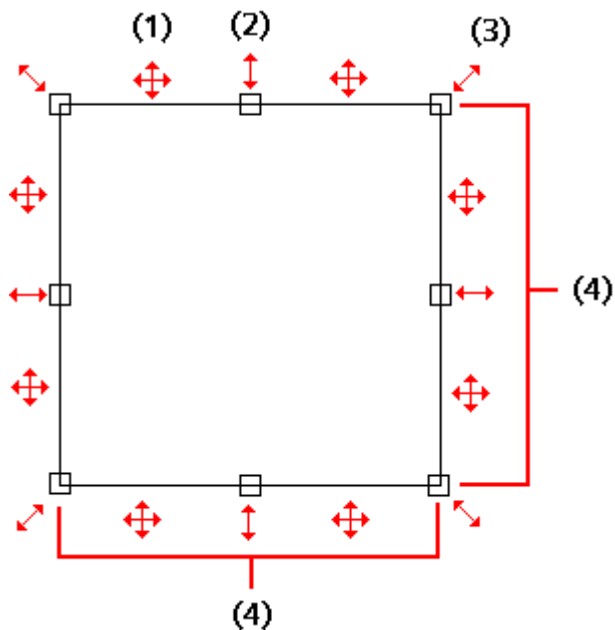
3D recon range object in a reference segment

- (1) Recon starting position
- (2) Recon ending position
- (3) Recon direction
- (4) Resize
- (5) Move

◆ In any of the reference segments, drag the borders for the recon starting & ending positions.

6.9.5 Setting up 3D recon FoV

The FoV of a 3D recon job is graphically represented by the FoV object in the FoV segment, as illustrated by the following figure:



3D recon range object in the FoV segment

- (1) Move
- (2) Resize
- (3) Resize
- (4) FoV

◆ Do one of the following:

- To set up FoV quickly but approximately, manipulate (drag to move or resize) the FoV object in the FoV segment.
- To set up FoV precisely, enter values in the **FoV** area on the **Recon** sub-task card.

FoV 192 mm x 192 mm



Images are typically reconstructed in squares, for example, 512 by 512 pixels. If necessary, you can reconstruct non-square images, by selecting an option from the **Matrix size** list on the 3D recon toolbar: **512 non-square** or **256 non-square**. Please note that the 256 non-square option leads to a reduced image resolution.

6.9.6 Using FAST 3D

FAST 3D helps you to plan a 3D recon range more efficiently by automatically completing the following tasks:

- Detecting the shape of a scanned body part and its z-axis centerline
- Using the centerline as the 3D recon axis
- Setting the 3D recon volume (recon starting position, recon ending position, & FoV) approximately
- Panning or zooming images in 3D segments

As a result, you see images that can be automatically rotated to fit the patient's anatomy. In 3D segments, recon range boxes and the FoV box approximately cover the body region, resulting in no black recon images. The coronal, axial, and sagittal images are placed in the segment center and properly sized.



FAST 3D can be turned on in a scan protocol. If a protocol has FAST 3D off, you can still activate it on the **Recon** sub-task card after the protocol is loaded into **Examination**. For any new 3D recon job you add from the **Recon** sub-task card, FAST 3D is turned on by default.



If FAST 3D is on for a 3D recon job before a scan, the 3D alignment takes place automatically as soon as you switch to the 3D segments after the scan. You can trigger this automatic 3D alignment any time by activating the **FAST 3D** option on the **Recon** sub-task card.



- FAST 3D results are well suited to be automatically reconstructed. It is also possible to tilt the reconstruction axis manually to further improve the alignment.
- If **Auto recon** has been selected in **Auto Tasking**, FAST 3D is activated by default.
- Use of FAST 3D requires a license.
- If there are a large number of images (>1000) in the 3D recon planning base series, the FAST 3D algorithm may fail during the 3D recon job due to *syngo* IPT performance limitations.

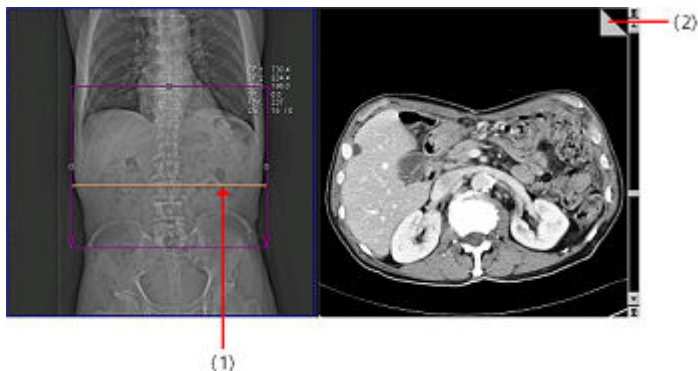
6.10 Images evaluation

In the **Examination** task card, you can perform basic images evaluation after scanning a patient:

- Scrolling through tomograms and view reference lines
- Using the **Graphics** and **Image Manipulation** toolbars
- Sending images to the *syngo* **Viewing** task card.
- Sending images to CT postprocessing applications.

6.10.1 Scrolling through tomograms

An orange line on the topogram indicates the position along the z-axis for the axial image currently appearing in the tomo segment.



Scrolling through tomograms

(1) Z-axis position for the currently shown tomogram

(2) Dog-ear

◆ In the tomo segment, do *one* of the following:

- Click the outer part of the dog-ear to scroll forwards through images.
- Click the inner part of the dog-ear to scroll backwards through images.



You can also scroll through images by rotating the wheel button of your mouse.

6.10.2 The Graphics toolbar

The **Graphics** toolbar provides graphic tools for the evaluation of images.

! CAUTION

Distance and angle measurements in the topogram!

Incorrect measurement values due to the projection technique used.

- ◆ Only perform distance measurements in the topogram in the longitudinal direction (head-foot direction).
- ◆ Do not perform angle measurements.

To open the toolbar, choose **Image > Graphics** from the main menu.

■ Measure distances



■ Draw a Region of Interest (ROI) in a circle



■ Draw an irregular shape to evaluate a ROI



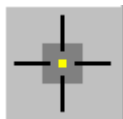
■ Write annotation text on images



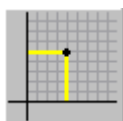
- Measure angles



- Measure the HU value of a small area in pixels



- Set measurement points on crosshair



6.11 Closing a scan

- ◆ Click the **Close current patient** icon.



If E-Logbook is activated, it appears in the sub-task card area. Enter the info about an examination and click **OK**.

The current examination is closed.

6.12 Troubleshooting exam

The system informs you when there are scan conflicts. Functionalities such as Scan Assistant help you solve the conflicts.

6.12.1 Exam parameters conflict

The system detects any invalid exam parameters that are highlighted in yellow. To solve an invalid parameter, adjust its value until the highlight disappears.

6.12.2 Using FAST Adjust

The FAST Adjust function assists you in solving the following issues by adjusting scan parameters automatically:

- X-ray tube cooling delay
- Impossible range
- Peak exposure limit
- eMode conflict

If the FAST Adjust function is available, the **Adjust** button appears left to the **Load** button.

The Adjust button	Description
	The FAST Adjust function is available, but not used. Rest the mouse pointer on the button to view the defined adjusting strategies for scan parameters.
	The FAST Adjust function is used. Rest the mouse pointer on the button to view how scan parameters have been adjusted.

The **Adjust** button does not appear in the blow-up view.

- ◆ Click the **Adjust** button.



The corresponding scan parameters are adapted.



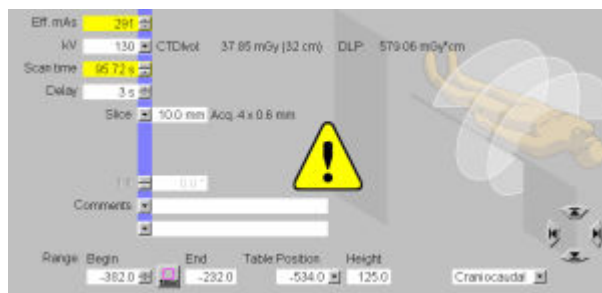
If a problem is not solved by using the **Adjust** button, try to adapt scan parameters manually or use Scan Assistant, if available.

6.12.3 Using Scan Assistant

Scan Assistant assists you in solving the following issues:

- X-ray tube cooling delay
- Impossible range

Scan Assistant is indicated by a warning triangle appearing on the **Routine** sub-task card.



- 1 Click the warning triangle.
- 2 In the **Scan Assistant** window, drag the sliders from the red zone to the green zone for the following parameters.
 - Scan delay
 - Scan time
 - Quality ref. mAs / Eff. mAs



- Moving one slider may affect the boundaries of the green and red zones of the other sliders.
- Scan Assistant is only available for a spiral scan.
- Click the **Proposals** button to get further help.

7 Special examination

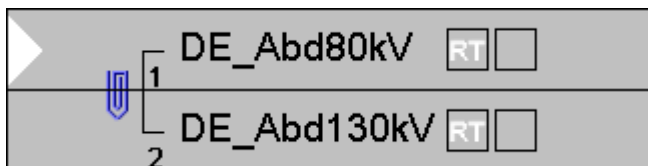
This section describes information about the following special examinations :

- Dual Spiral Dual Energy
- Veterinary usage

7.1 Dual Spiral Dual Energy

The attenuation coefficients of different materials differ under CT scans of varying energies. This can be exploited to obtain additional information about the chemical composition of body tissues, for example, of kidney stones.

Dual Spiral Dual Energy supports a dual-energy exam on a single X-ray tube CT system. A Dual Spiral Dual Energy exam consists of two consecutive spiral scans of different voltages.



The same range is scanned twice with different tube voltages: the first spiral scan uses 80kV; the second spiral scan uses 130kV. The scan data sets (80kV and 130kV) can be evaluated by using the *syngo* Dual Spiral Dual Energy application to visualize information about anatomical and pathological structures.

CAUTION

Images of Single Source Dual Energy: Inappropriate image results due to insufficient image registration!

No diagnosis possible.

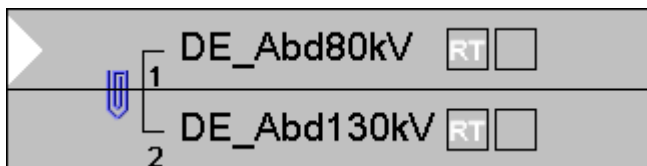
- ◆ Make sure that the patient does not move between the scans.



- Dual Spiral Dual Energy protocols, whose name starts with "DE", are available from the various body regions (for example, from the Abdomen region) in Patient Model Dialog, and are protected by a license.
- The *syngo* Dual Spiral Dual Energy application is available from the *syngo.via* software.

7.2 Dual Spiral Dual Energy exam range

A Dual Spiral Dual Energy exam range is a special kind of auto-range, as shown by the sign of a paper clip.



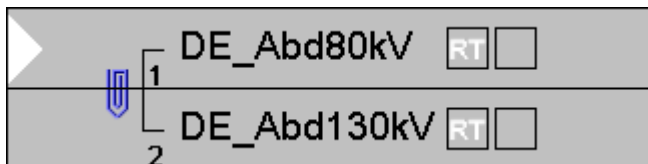
The two spiral scans in a Dual Spiral Dual Energy exam are bound together and cannot be split. Exam parameters of the second spiral are inherited from exam parameters of the first spiral, and cannot be changed. Therefore, edit the exam parameters of the first spiral (for example, scan range **Begin & End**), if necessary.



The first spiral scan uses 80kV and the second spiral scan uses 130kV.

7.3 Paper clip for Dual Spiral Dual Energy

A special kind of auto-range, as shown by the sign of a paper clip, indicates Dual Spiral Dual Energy exam (→ Page 135 *Dual Spiral Dual Energy*).



7.4 Automatic postprocessing

On the **Auto Tasking** parameter card, you can define the way in which a reconstructed series is automatically postprocessed. Depending on the type of scan, not all options may be available. **Auto Tasking** parameters are set individually for each reconstruction job.

7.4.1 Monoenergetic postprocessing

With **Monoenergetic**, you can calculate images that are equivalent to images acquired with a monoenergetic X-ray beam of selectable energy. By adjusting the energy value of the image output, you can optimize the image contrast between different materials.

7.4.2 Creating monoenergetic images

CAUTION

Wrong selection of energy level of monoenergetic images by the user!

Wrong diagnosis

- ◆ Use the correct energy level: Low energy level for visualization of iodine. High energy level for soft tissue visualization.

- 1 Click the **Auto Tasking** parameter card.
- 2 From the **Auto postprocessing** selection list, select a Dual Energy processing algorithm. For example, **DE Mono 40keV** or **DE Mono 120keV**.

- If you select **DE Mono < 70keV** as the postprocessing algorithm, two series, **DE Mono 40keV** and **DE Mono 50keV**, will be calculated when a reconstruction is finished.
- If you select **DE Mono > 100keV** as the postprocessing algorithm, three series, **DE Mono 120keV**, **DE Mono 140keV**, and **DE Mono 190keV**, will be calculated when a reconstruction is finished.

Automatic archiving of these series is possible when auto transfer is configured.

You can check the status in the status bar at the bottom of the task card. Message can be as follows:

- FAST DE Results completed. Verify image registration.
- FAST DE Results in progress.

If you have issues during monoenergetic postprocessing, follow the instructions below for each message.

»	FAST DE Results failed: Insufficient memory. ◆ Create a reconstruction job with fewer images. ◆ Close an application and start a new reconstruction.
»	FAST DE Results failed: Different number of images at low/high kV. ◆ Start a reconstruction with a shorter reconstruction range.
i	■ The monoenergetic algorithms are only available for Dual Spiral Dual Energy scan recon jobs. ■ The monoenergetic postprocessing items are not available for 3D reconstruction jobs. ■ When the number of images reconstructed exceeds the threshold 500, the No. of images box on the Recon task card is displayed with a yellow background.

7.4.3 Configuring monoenergetic images

In the **FAST DE Results Configuration** dialog box, you can configure the dual energy algorithm parameters.

- 1 Select **Options > Configuration** from the main menu.

The configuration panel is displayed.

- 2 Double-click the **FAST DE Results** icon.

The **FAST DE Results Configuration** dialog box appears.

- 3 For **Global Settings**, select the **Display Boundary of Analyzed Area** check box if you want to show a red ring around the evaluable volume region.

- 4 On the **Algorithm Parameters** tab, modify the algorithm parameters of the selected application class according to your requirements.

Minimum [HU] and **Maximum [HU]**: HU range included in the calculation

Range: range of the smoothing filter in units of the pixel size

- 5 Click **Apply** to save the configuration and keep the dialog box open.

– or –

Click **OK** to save the configuration and close the dialog box.



The algorithm parameters are dependent on the computation used by the various application classes. Modification of these parameters is recommended only if the images are very noisy or if the HU values fall outside the expected norm.

8 Contrast examination

CT examinations often involve the intravenous (IV) injection of contrast medium. Such contrast examinations improve the characterization of tissue and lesions, as well as increase the opacity of vessels.

There are 3 methods to carry out a contrast examination:

- Bolus Tracking
- TestBolus
- Fixed delay: the radiologist estimates the scan delay for the contrast examination.

8.1 Contrast timing and volume

For contrast examinations, the exact timing for a scan is essential. The scan must be performed when the organ or vessel of interest has the optimal contrast enhancement. Therefore, it is essential to initiate the scan with an appropriate delay.

In multi-phase examinations, the countdown of the delay starts after a preceding phase has finished. In the three-phase liver examination, for example, the scan delay can be set to 600s.

CAUTION

Misunderstanding of reference point for start delay in bolus examination!

Faulty synchronization between bolus and scanning.

- ◆ Make sure that the chosen delay time fits to the characteristics of the bolus injection.

 CAUTION

Wrong start delay!

X-ray not, or only partially, usable.

- ◆ For acquisition with contrast medium, select the flooding time of the contrast medium as the delay.

The amount of contrast medium needed depends on the patient's weight, circulatory condition, and scan duration. In general, no more than 3 ml per kg of body weight for adults and 2 ml per kg of body weight for children should be applied. The scan delay also plays a role. For a CTA study (the arterial phase), the principle is to keep the contrast medium flowing for the scan duration. Thus, the contrast volume can be calculated using the following formula:

Contrast volume = (scan delay + scan duration) × flow rate

8.2 Bolus Tracking

Bolus Tracking monitors the flow of contrast medium in a vessel, and can trigger a diagnostic spiral scan when the vessel enhancement reaches an optimum level.



- Use of Bolus Tracking requires a license.
- The Bolus Tracking feature is also referred to as CARE Bolus.

8.2.1 Workflow of Bolus Tracking

Before you perform Bolus Tracking, make sure that a topogram has been acquired as usual, and that the parameters for a diagnostic spiral scan are set up properly.

First, insert the Bolus Tracking steps. You can insert the Bolus Tracking steps by right-clicking a spiral entry in Chronicle and choosing **Bolus Tracking** from the shortcut menu. The Bolus Tracking steps are added immediately before the spiral entry.

- Pre-monitoring



- IV bolus



- Monitoring



Then, prepare for Bolus Tracking.

In the pre-monitoring step, you acquire an axial image at a reference slice. The image acquisition uses the sequence mode without any contrast medium.

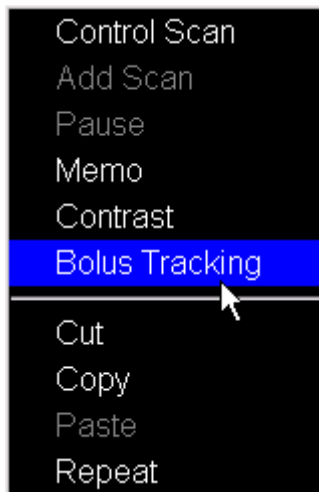
The IV Bolus step provides a pause for the injection of contrast medium. In this step, you place a trigger ROI inside the cross-sectional view of a vessel on the pre-monitoring image. Contrast enhancement in the vessel is indicated by an increase in HU at the ROI. You also activate the auto-trigger option and specify a trigger threshold in HU.

Finally, start Bolus Tracking.

You must start the monitoring step of image acquisitions and start the injection of contrast medium simultaneously. Scans at the reference slice and the HU evaluation of the trigger ROI are carried out repeatedly. When the trigger ROI's HU value reaches the trigger threshold, the monitoring step stops automatically and its next spiral scan is triggered.

8.2.2 Inserting Bolus Tracking

- 1 In Chronicle, right-click a spiral entry.
- 2 Choose **Bolus Tracking** from the shortcut menu.



Bolus Tracking steps are added immediately before the spiral entry:

- Pre-monitoring
- IV Bolus
- Monitoring

An injector symbol appears on the monitoring entry, indicating that the injection of contrast medium is needed.

A bracket symbol ([]) connects the monitoring entry to the next spiral scan, indicating that these two scan steps are performed with one manual start.



- The pre-monitoring and the monitoring parameters are derived from the following spiral scan. Therefore, you need to set up the parameters of the spiral scan properly before inserting the Bolus Tracking steps.
- Bolus Tracking steps can also be inserted by choosing **Insert > Bolus Tracking** from the main menu.

8.2.3 Premonitoring

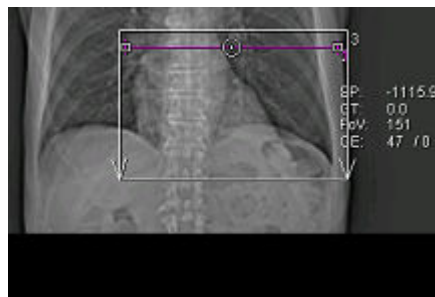
In the premonitoring step, you acquire an axial image at a reference slice. The image acquisition uses the sequence mode without any contrast medium.

Positioning a pre-monitoring scan

The pre-monitoring scan is a single acquisition of a slice image in the sequence mode for which no contrast medium is used. You need to position the scan properly to acquire the cross-sectional view of a vessel, as required for Bolus Tracking.

✓ A topogram has been acquired.

- 1 In Chronicle, select a pre-monitoring entry.
- 2 To place the pre-monitoring cutline at the proper position, do *one* of the following operations on the topogram:



- To move the cutline, drag the line center.
- To change the width of the cutline, drag the side handles.



- When you insert Bolus Tracking steps before a spiral scan, the pre-monitoring & the monitoring cutlines are positioned, by default, slightly ahead of the starting position of the spiral scan. The patient table stays at the cutlines for the Bolus Tracking procedure. After the procedure, the table does not have to move to a new position to initiate the spiral scan. When you drag a cutline towards this default position, the cutline will snap to the position.
- You can define the snapping distance in **Bolus Tracking Configuration**.

Starting a pre-monitoring scan ✓ You have positioned the pre-monitoring scan properly.

1 In Chronicle, select a pre-monitoring entry.

2 Click **Load**.



3 Move the patient table, following the guidance of the yellow message bubble.

4 Press **START** on CT control box.



The pre-monitoring image appears in the tomo segment. The **IV bolus** entry becomes selected in Chronicle.

8.2.4 IV Bolus

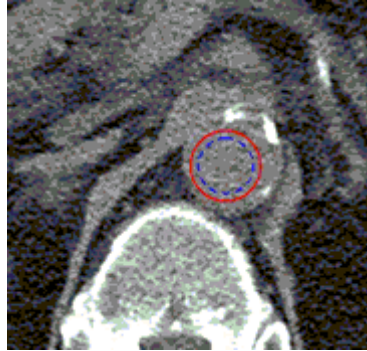
The IV Bolus step provides a pause for the injection of contrast medium. In this step, you place a trigger ROI inside the cross-sectional view of a vessel on the pre-monitoring image. Contrast enhancement in the vessel is indicated by an increase in HU at the ROI. You also activate the auto-trigger option and specify a trigger threshold in HU.

Placing trigger ROI ✓ The pre-monitoring image has been acquired.

1 In the tomo segment, zoom in or pan the pre-monitoring image for a clear view of the vessel to be monitored.

2 Click at the central point of the vessel.

The trigger ROI appears.



3 Make sure that the ROI is positioned well inside the vessel, by moving or resizing the ROI, if necessary.

4 Click the **Accept** button.

The ROI is defined.

Specifying trigger threshold

1 In Chronicle, select an **IV Bolus** entry.

2 Make sure that the **Auto trigger** check box is selected and specify the threshold in HU in the **Level** box.

Auto trigger ☒ Level
 Algorithm

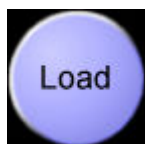


- You need to enter a differential HU value, that is, the relative density increase caused by the CM.
- If you do not select **Auto trigger**, you have to manually start the spiral scan after Bolus Tracking (→ Page 149 *Starting spiral manually after Bolus Tracking*).
- You can change the trigger threshold graphically by dragging the blue line on the **Trigger** sub-task card of the **Monitoring** entry. The threshold value appears to the right of the blue line.

8.2.5 Monitoring

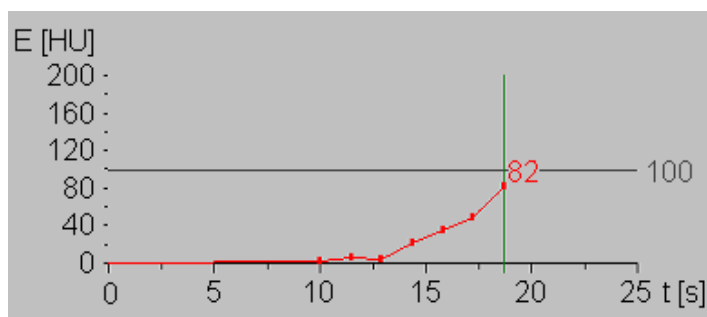
You must start the monitoring step of image acquisitions and start the injection of contrast medium simultaneously. Scans at the reference slice and the HU evaluation of the trigger ROI are carried out repeatedly. When the trigger ROI's HU value reaches the trigger threshold, the monitoring step stops automatically and its next spiral scan is triggered.

- Starting monitoring** ✓ The trigger ROI has been defined (→ Page 146 *Placing trigger ROI*).
 ✓ Auto trigger is activated and the trigger threshold has been defined (→ Page 147 *Specifying trigger threshold*).
 1 In Chronicle, select a **Monitoring** entry.
 2 Click **Load**.



- 3 Press **START** on CT control box to initiate the monitoring step of image acquisitions. Meanwhile, start the injection of contrast medium via your injector.

The system switches to the **Trigger** sub-task card, and shows the time-density curve for the contrast enhancement under each image acquisition.



The countdown of executed image acquisitions appears on the **Monitoring** entry. The tomo segment shows the real-time monitoring images one by one.

As soon as the curve reaches the threshold as indicated by the blue line on the **Trigger** sub-task card, the monitoring step is stopped and the next spiral scan starts automatically.



During the monitoring process, any user interaction should be avoided.

Starting spiral manually after Bolus Tracking

This action is activated in Examination Configuration. You may start a spiral scan manually after Bolus Tracking in the following situations:

- **Auto trigger** is *not* selected in the IV Bolus step. You want to have a manual control of when to initiate the spiral scan.
 - **Auto trigger** is selected. However, during the monitoring process, you decide to start the spiral scan in advance to the scan trigger because you have detected the optimal vessel enhancement by observing the images in the tomo segment.
 - **Auto trigger** is selected. However, the spiral scan is not triggered due to an improperly placed trigger ROI, a trigger threshold that is too high, or movement of the patient. In these cases, you either repeat the monitoring step or start the spiral scan manually.
- ◆ To manually start the spiral scan, do *one* of the following:
- During the monitoring process, click the **Start Spiral** button or press **START** on CT control box. The spiral scan starts immediately.
 - If the monitoring step does not trigger the next spiral scan, follow the guidance of the yellow message bubble to press **START** on CT control box.

8.3 CARE Contrast

CARE Contrast provides the option to use the injector coupled mode in a contrast examination. You can connect an injector to your SOMATOM CT scanner so that the communication between both devices is synchronized. You can start or stop a scan either via the CT scanner or via the injector. If you press **START** on CT control box, both devices are started simultaneously.

8.3.1 CARE Contrast safety

Use of CARE Contrast requires a license. Refer to the operator manual of your injector manufacturer for more information about scanning in the coupled mode. Siemens Healthineers will not be responsible for any problems resulting from use of unapproved injectors.

CAUTION

Scanning with coupled contrast medium injection!

Contrast medium not, or only partially, usable.

- ◆ Follow the instructions in the Instructions for Use of your bolus injector.

CAUTION

Scanning with coupled contrast medium injection!

Undesired radiation

- ◆ Take care of the modes of scanning and CM injection.

CAUTION

Unintended activation of coupled mode!

Undesired radiation

- ◆ Verify that uncoupled mode is activated before starting the injection.

CAUTION

Coupled start is not activated!

Injection will not be started automatically, if coupled start is not activated.

- ◆ Check in **Scan** subtask card whether a coupled start is activated or not.



If the injector is not connected properly, the **Check Injector** message is displayed.

8.3.2 Working with CARE Contrast

- 1 In Chronicle, select a scan step that will be executed with the injection of contrast medium and click the **Scan** sub-task card.
- 2 In the **Scan Start** list, select *one* of the following:
 - **Injector coupled (start button)**: the START key of the CT scanner starts the injector.
 - **Injector coupled (foot switch)**: the foot switch of the CT scanner starts the injector.
- 3 Load the scan step.

The system prompts you to arm the injector if it is not loaded properly. When the yellow message bubble shows **Press START**, both the injector and the CT scanner are ready for the examination.
- 4 Do *any* of the following:
 - Press **START** or the foot switch at the CT scanner.
 - Press **START** on the injector panel outside the scan room or directly at the injector inside the scan room.

The CT scan and the injection of contrast medium (CM) are started simultaneously.



- When you press **STOP** on CT control box or on the injector panel, CT radiation and the CM injection are stopped immediately.
- When a Control scan is inserted or a coupled range is repeated, the **Scan Start** will be reset to the uncoupled state.
- After you have set up the injector coupled mode for a scan step, you can save the current entries in Chronicle as a custom scan protocol for future use, by choosing **Edit > Save Scan protocol** from main menu.

8.4 Bolus injector

The system provides a connector for released injectors. Only use injectors that are released for your system. Please contact your Siemens Healthineers representative for more information.

The injector is connected to the CT system via a permanently installed cable. The installation of the cable must be done by a Siemens Healthineers service technician.

For installing a permanent connection to the interface unit of rack mounted injectors, an interface connection line is available at the console.



- For further instructions concerning the injector, please refer to the operator manuals of the injector manufacturers.
- In some countries, the bolus injector is only supplied with the additional monitor.

9 Cardiac examination

The motion of the heart makes it difficult to capture static images. A cardiac exam monitors a patient's electrocardiograph (ECG) and uses the ECG signals to control image acquisition. The ECG signals ensure that images are reconstructed during the same cardiac phase. This suppresses motion artifacts.



Cardiac examinations are available in two modes:

- ECG-gated spiral
- ECG-triggered sequence



The availability of the cardiac function depends on your system configuration.

9.1 Cardiac exam safety

WARNING

ECG is used as survival or diagnostic equipment!

ECG leads to wrong diagnosis.

- ◆ The ECG monitor may only be used in conjunction with the HeartView/Cardiac CT option. It is not intended for monitoring the patient.

WARNING

Rate meters may continue to count the pacemaker rate during occurrences of cardiac arrest or arrhythmia!

ECG leads to wrong diagnosis.

- ◆ Keep pacemaker patients under close surveillance.

CAUTION

Use of defibrillator!

Patient injury (burn) and destroying the ECG unit.

- ◆ The CT plate is grounded. Follow the instructions in the Instructions for Use of the defibrillator before using it.

CAUTION

Use of unsuitable ECG cables or electrodes!

Wrong diagnosis possible.

- ◆ Only original Siemens recommended ECG cables shall be used.

**CAUTION**

Contact between conductive parts of the ECG electrodes or other conductive parts!

Heart arrhythmia.

- ◆ To ensure patient safety, the conductive parts of the ECG electrodes (including associated connectors) and other patient-applied parts, should not contact other conductive parts, including earth ground, at any time.

9.2 Preparing a cardiac examination

To prepare a cardiac examination, you need the following basic steps:

- Patient positioning for cardiac
- Placing ECG electrodes
- Observing ECG signals
- Defining heart range

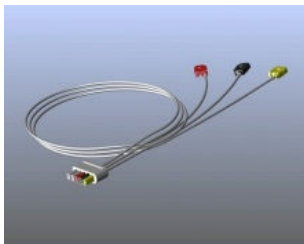
9.2.1 Patient positioning for cardiac

It is important that you position the patient well before a cardiac examination.

- Center the patient in the scan field.
- Immobilize the patient using positioning accessories, if necessary.
- Ensure that the patient's torso is straight.
- Use knee support to improve the patient's comfort.
- Warn the patient of the warm sensation if a contrast agent is going to be administered.

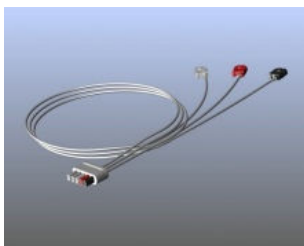
9.2.2 ECG electrodes

The ECG electrodes are connected to the receptacle at the patient table. When applying the ECG electrodes to the patient, the color coding according to IEC or AHA is recommended.

**Europe version**

Color coding according to IEC

- *Yellow electrode* – on the left mid-clavicular line, directly below the clavicle
- *Red electrode* – on the right mid-clavicular line, directly below the clavicle
- *Black electrode* – on the right mid-clavicular line, 6th or 7th intercostal space

**USA version**

Color coding according to AHA

- *Black electrode* – on the left mid-clavicular line, directly below the clavicle
- *Red electrode* – on the left mid-clavicular line, 6th or 7th intercostal space
- *White electrode* – on the right mid-clavicular line, directly below the clavicle



The ECG cable should be cleaned with isopropyl alcohol or another mild disinfectant only. Other agents, especially substances which contain acetone should not be used. This accessory may not be steam sterilized.

9.2.3 Placing ECG electrodes

After a cardio protocol is loaded and before the ECG cable is connected, you see an illustrative image on the gantry front panel to guide you how to place ECG electrodes. Please be aware that different color schemes are used in Europe and the United States.

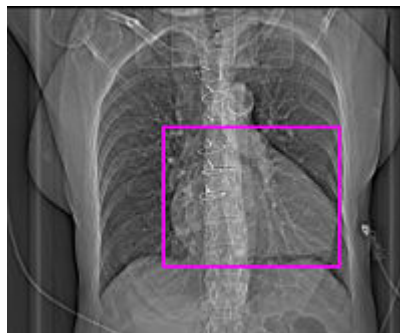
- 1 Attach the ECG electrodes to the patient's body.
- 2 Plug in the ECG cables to the ECG sockets on the CT patient table.
- 3 Arrange the ECG cables properly along the patient's side and away from the scan field.



- Use ultrasound gel to improve ECG signal conduction.
- If the ECG electrodes have not been connected properly such that ECG signals are not available, a message box appears. Connect the ECG electrodes and click **Try again** on the message box.

9.2.4 Defining heart range

- ◆ On a topogram, place a scan range over the heart from the carina to the apex.



9.2.5 Using synthetic ECG

With synthetic trigger/sync, in the case of ECG signal loss during a scan, the system calculates the sync pulses by averaging over the last few measured heart beats. A simulated ECG is then used to continue the scan.

CAUTION

Missing ECG-Sync due to the disconnection of the patient's ECG-electrodes!

X-ray not, or only partially, usable.

- ◆ Check the ECG electrodes before scanning.
- ◆ A contrast enhanced scan should not be aborted but operated with synthetic ECG-data.

The synthetic trigger/sync option is activated by default for a cardiac scan. It is recommended that you keep this option always activated for a scan using contrast medium. If this option is deactivated, a scan will be aborted in the case of ECG signal loss during the scan.

- 1 Choose **Options > Configuration**.
- 2 Click the **HeartView** icon.
- 3 On the **General** tab, select the **ECG Synthetic Trigger/Sync** check box.

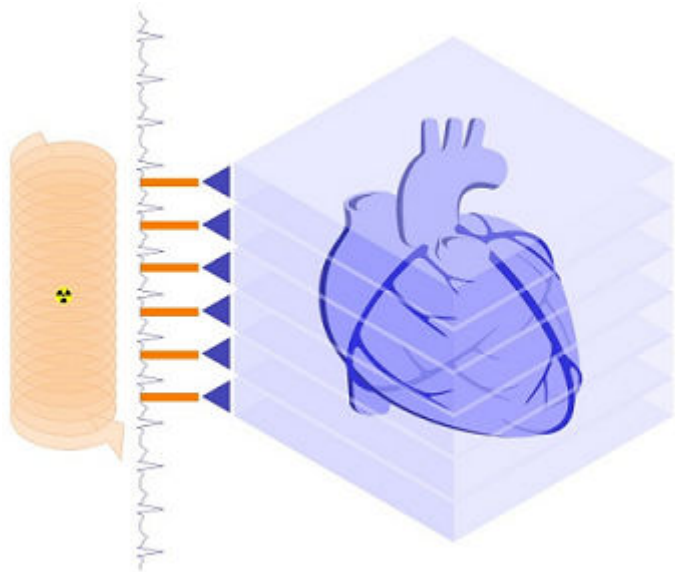
☒ ECG Synthetic Trigger / Sync



If the synthetic trigger/sync option is activated and ECG pulsing is on, the pulsing will be switched off if the ECG signal is lost during a scan. Then, the scan dose will be always at the 100% level.

9.3 CorCTA spiral

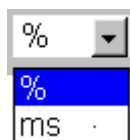
Retrospective ECG gating uses the spiral mode for imaging. Continuous exposure to radiation is applied for the entire cardiac cycle over several consecutive heart beats. The ECG is recorded simultaneously with the spiral data acquisition to allow for retrospective selection of the cardiac phase for images reconstruction.



The ECG-gated spiral mode is recommended for patients with arrhythmia or a higher heart rate.

9.3.1 Setting up ECG pulsing

- 1 On the **Trigger** sub-task card, select the **ECG Pulsing** check box.
- 2 In the **ECG Pulsing** area, select % (relative time) or ms (absolute time) as the unit in the third list.



- 3 In the first and second boxes, specify the starting point and the ending point of the pulsing window.

☒ ECG Pulsing 60  to 70  % 



CAUTION

Wrong setting in display!

Ionizing radiation

- ◆ Set the ECG pulsing window to clinically relevant value.

9.3.2 CorCTA spiral scan prerequisites

Make sure that the following pre-requisites are fulfilled before you start a CorCTA spiral scan:

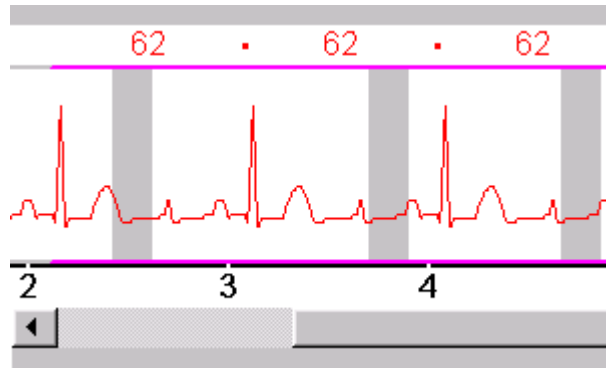
- The patient is positioned properly.
- ECG electrodes are placed properly and stable ECG signals are available.
- A proper cardiac scan protocol has been chosen.
- Bolus timing has been planned.
- A topogram is available, on which the heart range is defined for the CorCTA scan.
- ECG pulsing has been set up.

9.3.3 CorCTA spiral scan information

First, please observe the pre-requisites of a CorCTA spiral scan (→ Page 160 *CorCTA spiral scan prerequisites*).

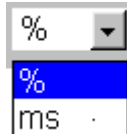
A coronary CTA scan is carried out in a similar way as a typical contrast-enhanced examination. For a CTA scan, bolus timing is important. The scan should be initiated when the vessel of interest has an optimal contrast enhancement.

After a cardiac scan is done in the spiral mode, the ECG that is used for the cardiac scan is updated on the **Trigger** sub-task card. Continuous pink lines appear to indicate the uninterrupted radiation exposure over a few heart beats.



9.3.4 Selecting a cardiac recon phase

- 1 In the **Phase Start** area of the **Trigger** sub-task card, select % (relative time) or ms (absolute time) as the unit in the second list.



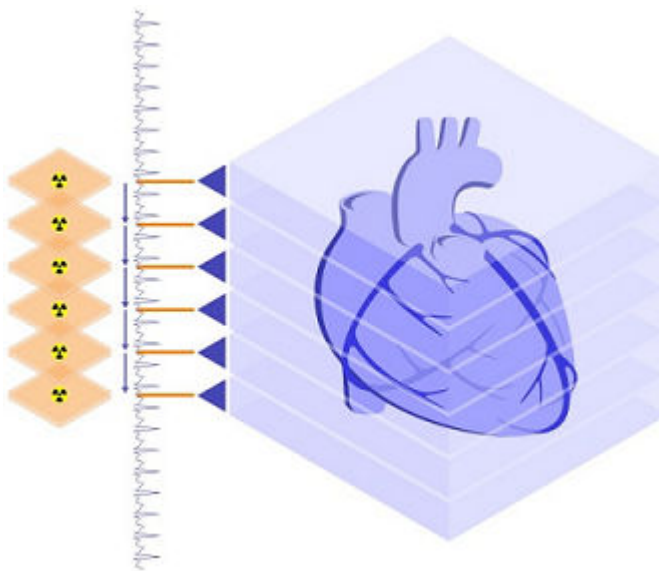
- 2 Specify the phase starting time in the first box.



A negative value (unit: %) defines starting points before R-peaks.

9.4 CorCTA sequence

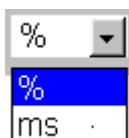
Prospective ECG triggering uses the sequence mode for imaging. The appropriate cardiac phase is used to trigger a scan. To time the trigger, the mean RR interval of the three preceding heart beats is used to predict the arrival of the next heart cycle. A reliable prediction is only possible if the heart rhythm is regular.



Use this dose-saving sequence mode for patients with a lower and stable heart rate.

9.4.1 Setting up phase start

- 1 In the **Phase Start** area of the **Trigger** sub-task card, select % (relative time) or ms (absolute time) as the unit in the second list.



2 Specify the phase starting time in the first box.

Phase Start



- A negative value (unit: %) defines starting points before R-peaks.
- For a scan with an irregular heart rate, try % instead of ms.

9.4.2 CorCTA sequence scan prerequisites

Make sure that the following pre-requisites are fulfilled before you start a CorCTA scan in the sequence mode:

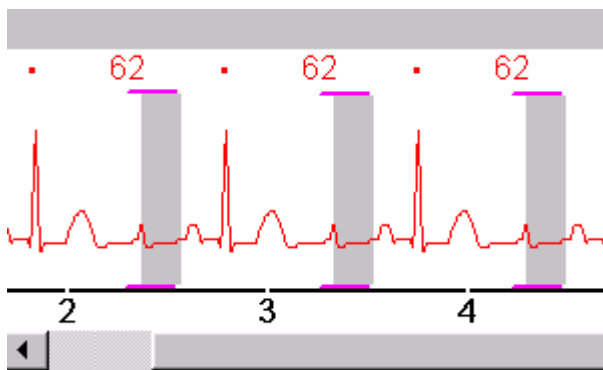
- The patient is positioned properly.
- ECG electrodes are placed properly and stable ECG signals are available.
- A proper cardiac scan protocol has been chosen.
- Bolus timing has been planned.
- A topogram is available, on which the heart range is defined for the CorCTA scan.
- Phase start has been set up.

9.4.3 CorCTA sequence scan information

First, please observe the pre-requisites of a CorCTA sequence scan (→ Page 163 *CorCTA sequence scan prerequisites*).

A coronary CTA scan is carried out in a similar way as a typical contrast-enhanced examination. For a CTA scan, bolus timing is important. The scan should be initiated when the vessel of interest has an optimal contrast enhancement.

After a cardiac scan is done in the sequence mode, the ECG that is used for the cardiac scan is updated on the **Trigger** sub-task card. Dashed pink lines appear to indicate the periodical radiation exposure over a few heart beats.



The scan duration in the sequence mode depends on the heart rate. The more irregular the heart rate, the longer the cardiac scan will last.

9.4.4 CorCTA sequence recon

In the sequence mode, scan data is only acquired in a pre-defined cardiac phase. This is done by specifying the **Phase start** before a scan. Therefore, after the scan, you cannot reconstruct the images in a different cardiac phase.

Functional analysis of the heart is *not* possible. However, you can work with general recon parameters, for example, setting up the slice width and recon increment for coronary evaluation.

10 Examination reports

Examination results can be reported in several ways:

- Filming & printing
- Dose report
- Patient protocol
- E-logbook



- To open a report, you can double-click the report in the Patient Browser.
- If you double-click a series, the series is loaded into the active task card application.
- If you send an SR-series to a network node with images copied to the report before, it may occur that the images are not transferred. Send also the series including the images copied to report, to display the report correctly.

10.1 Filming and printing

You can expose the images of an examination on film or print them on paper for documentation and reporting.

The regular page mode is the default print mode. The scalable page mode provides images with reduced quality and does not support original size.

CAUTION

Using scalable page mode may reduce the image quality. The resulting image quality may no longer be sufficient. The scalable page mode does not support real size!

Possible wrong diagnosis.

- ◆ Be aware of image quality reduction if scalable page mode is configured.

CAUTION

The printed flag is set if the images are successfully transferred to the printer control. Not all printers, for example, paper printers, may be able to solve printing problems themselves!

Possible loss of image printout.

- ◆ Verify that the printouts are available before you delete images.

CAUTION

Use of paper printouts for diagnosis!

Wrong diagnosis due to wrong image information.

- ◆ Only use images on film for diagnostic purpose.



Do not open the print dialog while radiation is applied.

10.1.1 Camera

Various cameras are supported to commit the CT images to film. You can obtain an up-to-date list of cameras supported from your Siemens Healthineers representative.

**CAUTION**

Exposing images with a non-verified camera!

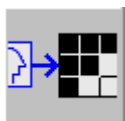
Wrong basis for diagnosis.

- ◆ Only use cameras released by Siemens Healthineers or by your system owner.

10.1.2 Loading images to Filming

There are two methods for sending images to the **Filming** task card:

- Set up **Auto filming** on the **Auto Tasking** sub-task card before you start images reconstruction. Images are transferred to the **Filming** task card as soon as they have been reconstructed.
- After images have been reconstructed, open Patient Browser and load the series for the images into the **Filming** task card.



You can also send images to **Filming** by clicking the **Copy to Film Sheet** icon on the Image Manipulation toolbar.



If you add image text or perform measurements close to the image segment border, both might not be visible anymore after sending to the film sheet. To solve this problem, you can reposition the evaluation and the image text, or change the image zoom.

10.1.3 Editing film sheets

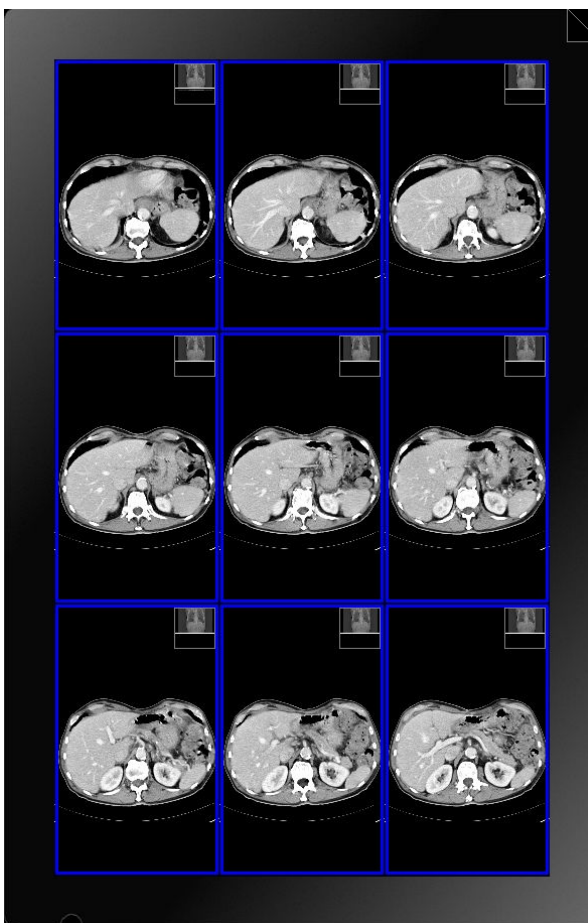
On the **Filming** task card, you can delete, move, or copy selected images. If you want to add new images from the **Patient Browser** to a film job, you can do that easily by drag-and-drop.

Then, you can do several tasks such as:

- Applying a film layout
- Changing the reference image
- Windowing images
- Showing or hiding image text/graphics

10.1.4 Selecting images in Filming

- ◆ Do one of the following:
 - Click the border of the film sheet to select all the images it contains.



- You can select all the film sheets with all images of one film job. Click the opened film job. Or, double-click the border of a film sheet.



The folder icon of a selected film job is highlighted (blue). All the images of this job are shown with a continuous blue line border.



- Use the **Ctrl** key to select non-consecutive images.
- Use the **Shift** key to select consecutive group of images.

10.1.5 Deleting images in Filming

- 1 Select one or more images or film sheets.
- 2 Click the **Delete** icon or push the **Del** key on the keyboard.



- You can also delete images with **Film > Clear Document(s)** in the main menu.
- Turn on the **Film > Repack** option to automatically reorganize the film sheet or the entire film job (gaps are filled again) after deletion.

10.1.6 Moving and copying images in Filming

- 1 Select the images/film sheets that you want to copy or move.
- 2 Do one of the following:
 - To move selected images, select **Edit > Cut** in the main menu.
 - To copy selected images, Select **Edit > Copy** in the main menu.
- 3 Select the image in front of which you want to insert the images to be copied or moved.
- 4 Insert the images with **Edit > Paste** in the main menu.

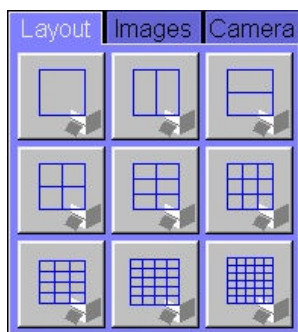


If you only want to move images within a film sheet, you can also do that easily by drag-and-drop. In this case, deselect the **Zoom/Pan** icon.

10.1.7 Applying film layout

With the layout of the film sheet, you define the number and size of the segments on a film sheet.

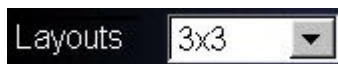
- 1 Select individual images or a film sheet or select an entire film job.
- 2 Click the **Layout** subtask card on the **Filming** task card into the foreground.



- 3 Click the icon for a film layout to display the selected images in this format.

– or –

From the **Layouts** selection list in the lower right-hand control area, select a layout.



It is not possible to change the film layout of film sheets with different layouts at the same time. You can only change the film layout for selected images or series if they have the same layout. Otherwise, you have to unify the layout of the selected images first before changing the layout.

10.1.8 Exposing film sheets

- 1 Select one or more film sheets in the **Filming** task card.
- 2 Click the **Expose Film Task** icon.



You can call up the **Film Job Status** dialog box by clicking the **Film Job Status** icon on the **Filming** task card. In the dialog box, you can get information about film jobs that are waiting to be executed in the queues for each camera and printer.



10.1.9 Printer

A Siemens Healthineers CT system is a medical device where under no circumstances other software than the released syngo CT software may be installed. This also applies to printer drivers for postscript printers.

Your software includes the generic Adobe Postscript Driver. With this generic printer driver, different postscript printers can be connected to a Siemens Healthineers CT system without the need of installing any specific printer driver for the individual printer.

CAUTION

The image quality on paper printouts might be reduced so that the printouts may not be sufficient for diagnosis!

Possible wrong diagnosis.

- ◆ Only use images on film for diagnostic purposes.
- ◆ You may use printouts for diagnosis if the printer has been evaluated and released for this purpose by the manufacturer.

Siemens Healthineers only allows the connection of a postscript printer via a TCP/IP network port or locally at the parallel interface. Connected postscript printers shall be used primarily for printing the reports that are created in the different applications available on the system. Postscript Printers might also be used to print medical images. These printouts are not for diagnostic use.

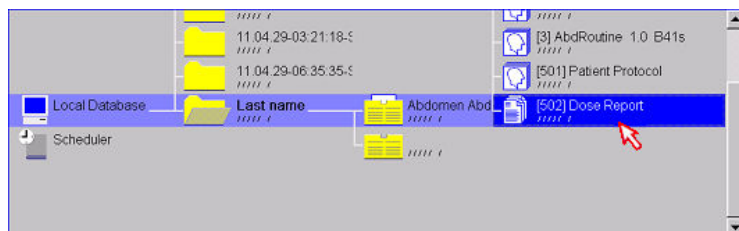


More information about the recommended printers can be obtained from your Siemens Healthineers representative.

10.2 Dose report

A dose report is introduced according to the DICOM standard for dose evaluation purposes.

A dose report can be automatically generated at the end of an examination and cannot be modified. It is available as a separate series in Patient Browser.



The dose report uses DICOM terminologies that vary from the phrases appearing on the user interface of the *syngo* CT software.

A dose report contains the following information:

- Patient data
For example, patient name and ID.
- Administrative data
For example, referring physician and ward number.
- Dose parameters
For example, $CTDI_{vol}$ and DLP.

10.2.1 Setting up the CT dose report

You can configure the system to automatically create the CT dose report at the end of an exam.

- 1 Choose **Options > Configuration** from the main menu.
- 2 Click the **Examination** icon to open the **Examination Configuration** dialog box.
- 3 On the **Dose** tab, select the **Activate dose report** check box.



You can also set up auto transferring by selecting the **Auto transfer dose report** check box.

10.2.2 Viewing dose report

- 1 In Patient Browser, select a dose report.
- 2 Double-click the dose report icon.



The dose report is opened using the SR Viewer.

- 3 Scroll down to view the detailed dose information.

CAUTION

If there is no specific style sheet available locally for display of a SR document, then the generic style sheet is applied instead!

Not all information might be present or displayed correctly as a consequence, especially when private attributes are included in the structured reports.

- ◆ In the display of the report and in exported documents it will be indicated that only a generic mechanism has been applied.

**CAUTION**

When opening a SR document for further processing, an old version of a structured report document is accidentally opened!

Existing processing results within the latest version of the document might get lost because the currently opened document can now be saved as latest version.

- ◆ When opening and editing a structured report document, always assure that you use the latest version. The version information is recognizable within the *syngo* Patient Browser.

**CAUTION**

The image quality of images which are viewed within the Structured Report viewer might not be sufficient for reading purposes!

Possible wrong diagnosis.

- ◆ Do not base your diagnosis on images which were only displayed in the Structured Report viewer application.

10.2.3 Exporting dose report

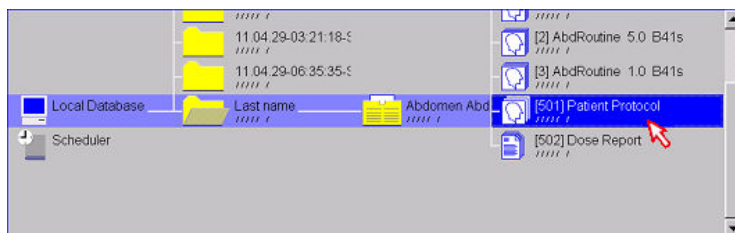
- ◆ In the SR Viewer, choose **File > Export > Dicom SR** (or **PDF, HTML, XML**).

The dose report is exported from the SR Viewer.

10.3 Patient protocol

A patient protocol summarizes the information about a finished examination and can be used to document examinations, for example, in case the MPPS (Modality Performed Procedure Step) feature is not available.

A patient protocol can be automatically generated at the end of an examination and cannot be modified. It is available as a separate series in Patient Browser and can be sent to **Viewing** or **Filming**.



A patient protocol consists of the following information:

- Patient data
For example, patient name and ID.
- Administrative data
For example, referring physician and ward number.
- Examination data
For example, patient position and total mAs.
- Contrast data
For example, CM quantity and injection parameters.
- Scan protocol data
For example, scan type.
- Dose parameters
For example, $CTDI_{vol}$.

10.3.1 Setting up patient protocol

You can let the system automatically create the patient protocol at the end of an exam.

- 1 Choose **Options > Configuration** from the main menu.
- 2 Click the **Examination** icon to open the **Examination Configuration** dialog box.
- 3 On the **Workflow** tab, select the **Patient Protocol** check box.



You can also set up auto filming and auto transferring by selecting the **Auto film patient protocol** check box and the **Auto transfer patient protocol** check box.

10.3.2 Viewing patient protocol

- 1 Switch to the **Viewing** task card, if necessary.
- 2 Open Patient Browser.
- 3 Double click a patient protocol.

The patient protocol appears in the **Viewing** task card.

10.4 E-Logbook

E-Logbook is a tool used to collect, view, and export examination information for statistics and documentation. It offers an alternative to a paper logbook.

The examination information recorded by E-Logbook has two types:

- System fields are collected automatically and shown read-only, for example, patient name, ID, and examination date.
- Manual fields are entered by you at the end of an examination, for example, history, storage medium, and custom fields.



Use of E-Logbook requires a license.

10.4.1 Activating E-Logbook

While E-Logbook is turned on, it prompts you to input the information about an examination when you close the examination.

- 1 Choose **Options > Configuration**.
- 2 Double-click the **E-Logbook Configuration** icon to open the **Configuration Overview** dialog box.



- 3 On the **General** tab, make sure to select *both* of the following check boxes:

- **Activate E-Logbook**
- **Show E-Logbook**



If some manual fields are defined under the **Manual Entries** tab, the **Show E-Logbook** check box is selected by mandatory.

10.4.2 Inputting E-Logbook records

- ✓ E-Logbook is activated.

- 1 Click the **Close current patient** icon.



In the sub-task card area, the E-Logbook input panel appears.

No.	4	Patient ID	15.03.13-15.11.50...	Total mAs	4557
Date	3/13/2015	Date of birth	11/18/1958		
Patient name	15.03.13-15.11.50...	Scan protocol name	AbdomenRoutine (...)		
				OK	Cancel

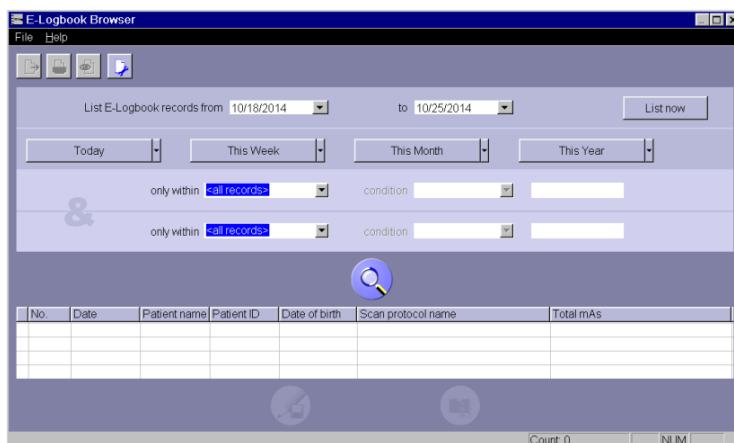
- 2 Specify the info about an examination.



- E-Logbook system fields are read-only; manual fields are editable.
- For text longer than the width of a box, “...” is displayed. To see the entire text, rest your mouse pointer on the box and a tool tip will appear.

10.4.3 E-Logbook Browser

On the *syngo* Workplace, you can use the E-Logbook Browser to view, search, edit, sort, export, and print E-Logbook records.



10.4.4 Opening E-Logbook Browser

- ◆ Choose **Patient > E-Logbook** from the main menu.

The **E-Logbook Browser** window appears.



- Every time it is opened, the E-Logbook Browser lists the records within the default date period that you define in **E-Logbook Configuration**.
- You can also open the **E-Logbook Browser** by pressing **F12** on the computer keyboard.

10.4.5 Editing E-logbook records

You can modify an E-Logbook record in E-Logbook Browser. E-Logbook manual fields appear in a white background and can be modified. E-Logbook system fields appear in a deeper color background and are read-only.

- 1 Right-click a manual field to enter the editing mode.
- 2 Type or select a new field value.
- 3 Click outside the field.

The background of the field turns green, indicating that the modification is made, but not saved.

- 4 Click the **Apply** icon.



All green fields are saved. Their background changes to white again.

10.4.6 Exporting E-logbook records

E-Logbook records can be exported to an XML (Extensible Markup Language) file. You can view the XML file on an external computer by using Internet Explorer or Microsoft Excel.

- 1 In the E-Logbook Browser, click the **Export** icon to open the **Save As** dialog box.



- 2 Specify a file name and click **Save**.

The current list of records in E-Logbook Browser is saved to *H:\SiteData\E-Logbook* in File Browser.



- You cannot save the records to a path other than *H:\SiteData\E-Logbook*.
- In addition to the XML file, an *.xslt file is generated. This file is required for viewing the content of an XML file properly.
- You can also choose **File > Export** in E-Logbook Browser to export records.

10.4.7 Printing E-logbook records

- ✓ A default printer and default print template have been set up.
- ◆ Click the **Print** icon ()



- You can use the **Print Preview** icon () to check the page layout before printing.
- You can use the **Page Setup** icon () to set up page properties before printing.

10.4.8 Searching E-Logbook records

You can search E-Logbook records in E-Logbook Browser by using the following methods:

- Search using a specified date range
Specify a date range (the starting date and the ending date) and click the **List now** button.
- Search using a pre-defined date period
Click *one* of the following buttons:
 - **Today/Yesterday**
 - **This Week / Last Week**
 - **This Month / Last Month**
 - **This Year / Last Year**

■ Search using specified E-Logbook field values

You can set up a search statement by specifying a field name, an operator, and a field value. Up to two search statements can be specified. For example:

- Field name: *Date of birth*
- Operator: *is less than*
- Field value: *28-Jan-1979*

You can use the following operators with E-Logbook fields defined as text strings:

- **wild card**
- **equals**
- **has substring**

You can use the following operators with E-Logbook fields defined as numbers:

- **is greater than**
- **greater or equal**
- **is less than**
- **less or equal**
- **equals**

11 Data storage and exchange

The SOMATOM CT system comes with functionalities for easy access and exchange of the patient data and exam data.

CAUTION

Use of external tools to alter the stored results!

Retrieved results may be potentially incorrect.

- ◆ Do not modify the stored results.

CAUTION

Wrong configuration of **Presentation State** support when configuring network nodes!

Loss of post-processing results.

- ◆ When configuring network nodes, enable **Presentation State** support only after verifying the remote node for PR support.

11.1 Storage media

Your *syngo* workstation is equipped with a DVD recorder and USB ports to connect to USB storage devices.

CAUTION

Switching off a device while writing to the device!

Possible loss of all data and damage of media.

- ◆ Never switch off the device while writing to the device.

CAUTION

When you simultaneously export data to a device (like CD-R/ DVD-R/Blu-Ray/USB/Network share drive) and try to read data from the same device with external (non-syngo-based) applications, one job or both jobs can fail (depending on the timing).

A loss of data is possible. When data is being written to a drive, any external access at the same time will destroy the current write job and could even damage the storage disk. Data already stored on a multi-session disk might also become unreadable.

- ◆ Do not try to access the device with non-syngo-based applications while it is writing data or reading data.
- ◆ To ensure successful writing or reading, check the **Local Job Status** to see if there are any write or import jobs in progress.

CAUTION

Low-quality storage media are used!

Data loss due to improper handling or low-quality CDs or DVDs.

- ◆ Use only quality media approved for medical applications when storing patient data.
- ◆ Observe proper handling guidelines for CDs and DVDs.

 CAUTION

DVD-R/Blu-Ray Medium used for export/archive may get corrupted or may not be readable with other DVD/Blu-Ray devices.

Loss of data or user perception of loss of data.

- ◆ Configure and use local media only as **Archive** media if the requirements regarding retention and readability are sufficient fulfilled, for example, the media is approved by the manufacturer for archiving.
- ◆ Verify readability of the data on the medium before the data is deleted in the database.

 CAUTION

Archiving to media already containing clinical data.

Loss of data due to corruption of removable media.

- ◆ Avoid recording on removable media which already contains clinical data.



- The following types of devices are not allowed with the system: Disk burner, for example, CD, DVD, Blu-ray; USB floppy drives; Cell phones; MP3 players; USB flash drives and hard disk drives with multiple partitions and enabled security features (for example, password protection); storage card reader; Printers with storing function. Do not use such devices with this system.
- Before restarting an application while an **Archnet** job is active, delete the completed jobs in the **Local Job Status** dialog box.

11.1.1 DVD recorder

With the DVD recorder, you can read and write data onto/from CDs and DVDs. You can export your CT acquisitions to recordable CDs (Compact Discs) and DVDs (Digital Versatile Discs).

From recorded CDs and DVDs you can, for example, import the examination data stored on them into your database. As soon as you have switched on the system or computer, the DVD recorder is also ready for operation.



- (1) DVD tray
- (2) LED
- (3) Opening for manual ejection of discs
- (4) Open/Close/Stop button

You can use the following types of discs on your system:

- CD-ROM
- CD-R/RW
- DVD-ROM
- DVD-R



- The appearance of the DVD recorder may vary and depends on the delivery date of the computer.
- Note the restrictions to DVD-R: Maximum number of sessions is limited to 30; The time of finalizing a DVD depends on the amount of data on the DVD (if the amount of essential data is less than 1 GB, the time to finalize may be several minutes).

Instructions for using DVD drive

Please observe the following instructions for using the DVD drive:

- Do not leave a disc in the drive when turning off your system. However, if a disc was left in the drive, remove the media shortly after you turned on the unit by pressing the release button.
- Make sure that the DVD drive is not subject to vibrations during operation. Vibrations can cause the drive to stop.
- Only open the drawer to insert or remove the disc.
- Do not place any objects on the open drawer.
- Never press down on the open drawer.

DVD session mode

The process of writing data on a CD or DVD is called "Recording". You can write new data onto recordable CDs (CD-R) and DVDs (DVD-R) in one action (single session mode) or in several recording steps (multi session mode).

CAUTION

Switching to multi-session mode may destroy data previously recorded on this medium!

Previously stored data can no longer be read.

- ◆ Only delete the data that you have exported on a CD-R from the local database after you have completed the session.

CAUTION

A recording error may make a medium unusable!

In multi-session mode, all data previously stored on that medium could be lost, too. If this happens, any archive flags set to this data in the database become invalid.

- ◆ Only delete the data that you have archived on a media from the local database after you have successfully finalized and verified the media.

The setting of multi-session or single-session mode can be changed by Siemens Healthineers service. Factory default is multi-session.



In Multisession mode, if you click the **Cancel** button in the **Medium full** dialog box, the job goes to failed state. To solve this problem, you can insert a new writeable CD and restart.

Inserting a disc

✓ The DVD recorder is switched on and the status LED lights up green.

1 Press the insert/eject key briefly.

The drawer moves out.

2 Insert the disc into the drawer with the label side up.

3 Press the insert/eject key again briefly.

The drawer is retracted.

As soon as the disc has been inserted and the drive is ready for reading and writing, the status LED lights up green.



If the status LED does not light up green after you have inserted the disc, the disc is damaged or dirty.

◆ Take the disc out of the drive.



After an auto job transfer to DVD/CD was started when no medium was inserted, the **No Medium** dialog box pops up. The **Eject** buttons are dimmed. To solve this problem, you can press the hardware button to open the DVD/CD drive, then click the **OK** button in the **No Medium** dialog box. Do not click the **Cancel** button because the job will go into error state.

Writing & reading disc data

It takes some time to write data onto a CD or DVD. During recording, the read/write status LED lights up green.

◆ Use the appropriate menu entries of the user interface to write to and read from the CD or DVD.



- Depending on the volume of data, the recording process can take several minutes.
- Do not write data onto CD or DVD during a patient examination.
- While data is being read, the read/write status LED lights up green.

Ejecting a disc



CAUTION

Taking out the CD-R, CD or DVD too early!

Loss of data and destruction of the CD-R, CD or DVD possible.

- ◆ Do not remove the CD-R, CD or DVD from the recorder until the recording process has been completed and the status LED has gone out.
- ◆ Use the **Transfer** menu to eject media.



CAUTION

A medium is not finalized.

Another system/device may show this medium as empty, or defective or the data of the last session is missing.

- ◆ Use **Eject with finalize** to complete the work on this medium and to ensure that the medium is readable on other systems/devices.
- ◆ If a medium on which you expected to find data seems to be empty or defective or which the last data is missing, try to use original or another *syngo* system to finalize the medium for further read-only access.

✓ Reading and writing has been completed and the drive is not being accessed.

✓ The read/write status LED has gone out.

1 Call up **Transfer > Eject from CD** in the main menu.

The drawer moves out.

- 2 Take the CD or DVD out of the drawer when the mechanism stops and put it back in its cover.
- 3 Press the insert/eject key.

The drawer is retracted.



If the status LED flashes red, a drive fault has occurred.

- ◆ Contact Siemens Healthineers service.

11.1.2 USB device

You can store data on USB (Universal Serial Bus) devices. The location of the USB ports may vary and depends on the delivery date of the computer. To provide the maximum storage space, USB devices must be NTFS formatted.



CAUTION

Plugging or unplugging USB devices during acquisition can make the system unstable which may affect other processes like running acquisition tasks!

Loss of data, acquisition process disturbed.

- ◆ Do not plug in or unplug USB devices during acquisition tasks or other critical processes.



Only use USB devices approved by Siemens Healthineers. More information about the supported USB devices can be obtained from your Siemens Healthineers representative.

Instructions for using USB device

It is recommended to use a USB disk without additional hardware function, for example backup, encryption, and so on. Please click **Cancel** if **Found New Hardware** window pops up, the USB disk function is still available.

 CAUTION

Use of non-integrated USB storage device!

Reboot request of the operating system.

- ◆ Ignore the reboot request and press **Cancel**.
- ◆ Finish your examination or application.
- ◆ Restart the system.

 CAUTION

Using USB devices without own power supply!

The USB controller can be permanently damaged.

- ◆ Use USB devices as recommended by the manufacturer of the device.

It is necessary to format your USB hard drive as NTFS file system to transfer raw data > 4 GB to your USB hard drive. The entire process consists of two main steps:

- Formatting the USB hard drive
- Setting **Full Control** access to group **Everyone**



- We recommend not to insert, or to remove USB devices during an interventional examination.
- For instructions using the USB device, please refer to the operator manual of the USB device manufacturer. More information can be obtained via internet.
- Always use **Export to off-line...** to export raw data to the USB device. Do not export raw data by copy and paste, or by drag-and-drop. Do not start scans while exporting raw data to the USB device.

Inserting a USB device

- ◆ Insert the USB device into the USB port at the front of the computer.

The USB icon appears in the status bar.



Removing a USB device



CAUTION

Removal of USB device without deactivation by the software!

Possible loss of data, damage of the operating system, and damage of media.

- ◆ Do not immediately unplug a USB device.
- ◆ Always use the software functionality for safe removal before unplugging.

✓ Reading or writing from or to the USB device has been completed.

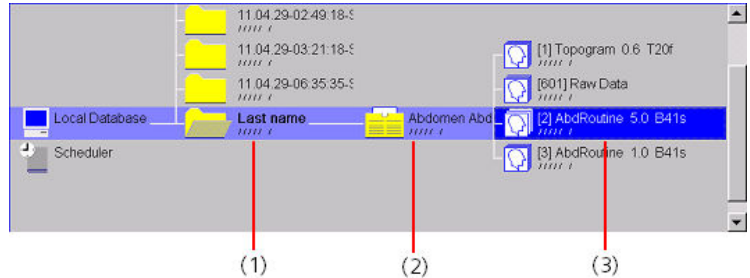
1 Double-Click the USB icon in the status bar, when USB icon disappears in the status bar, USB device can be safely unplugged.



2 Pull the device out of the USB port.

11.2 Patient Browser

In the **Patient Browser** dialog box, the hierarchical structure of the patient and examination data is displayed. Patient data from the databases and the connected drives is displayed.



Exam data in Patient Browser

- (1) Patient
- (2) Study
- (3) Series

! CAUTION

Patient folders containing data from other patient or series, for example, after patient merge, import of data from external media!

The diagnosis or treatment may be made on the basis of incorrect information.

- ◆ Check that data is assigned to the correct patient and series before using the data.

! CAUTION

Not all graphic and evaluation results created by one software can be interpreted.

Not all information from post processing available for diagnosis.

- ◆ A message box will inform you if not all information can be loaded and displayed.
- ◆ Please be aware when using such data for diagnosis.
- ◆ Use original application to view the post processing results.

CAUTION

Image size and quality of preview images provided in the patient browser are not sufficient for clinical diagnosis!

The diagnosis and treatment may be based on incorrect information.

- ◆ When performing clinical diagnosis only use the high resolution images in the imaging area.

11.2.1 Correcting or rearranging patient data

The images of a patient are grouped together by examination and series into hierarchical groups. If you find that the structure is not correct, you can regroup the examinations and series by moving them.

CAUTION

Correcting or rearranging objects containing references!

Possible loss of references.

- ◆ Rearrange the entire hierarchical group containing all objects with references in order to maintain the references.
- ◆ Only references found within the selection will be adapted.

CAUTION

Rearranging series/images to another series may lead to wrong image information if the selected images/series are not compatible.

Wrong diagnosis caused by wrong image information.

- ◆ Correct the attributes which do not correspond first and then rearrange the series/images.

You use the **Correct** dialog box for correcting or modifying patient data and information. Click the **Correct** icon on the tool bar.



The **Correct** dialog box opens.

You can view the modifications history of the patient data. Click the **History** icon on the tool bar.



With the function **Edit > Correct** you can correct, add, or rearrange patient and examination data. Use this function carefully and do not ignore warning messages. Some modifications may lead to inconsistent information within a patient study and wrong results in further processing or reading steps. Check all data of the patient after using **Edit > Correct**. In case of doubt please contact Siemens Healthineers customer service.

In some cases a complete and consistent correction of all data (images and raw data) is not possible.

For example: If you have scanned a colon patient prone and supine, but forgot to change the patient position accordingly, both raw data sets are within the same series. A partial re-orientation of only one raw data file is not possible and therefore a correction of the patient position is not possible for the raw data.

A possible solution can be:

- Reconstruct all necessary images.
- Control all images (axial images, oblique images, topograms etc.) series by series carefully and correct the patient position with **Edit > Correct**, if necessary.

We recommend to delete the raw data afterwards.

CAUTION

Inconsistent patient data or image information for *syngo* applications like Osteo CT, Pulmo CT, Volume, Circulation, Dental CT, LungCARE CT or Calcium Scoring!

Wrong basis for diagnosis.

- ◆ Do not modify patient data (**Edit > Correct** in the **Browser** menu) after results have been calculated.

11.2.2 Deleting patient data

You can explicitly delete patient data in the **Patient Browser** manually. Select the data that you want to delete in the navigation area or content area. Click the **Delete** icon on the tool bar.



You can turn off the confirmation dialog box, when selecting **Options > Configure Browser** from the main menu and deactivate the **Confirm Deletion** check box.

 CAUTION

Deletion confirmation deactivated!

Possible loss of data.

- ◆ The option in the **General** tab card to configure the **Delete Confirmation** dialog box should not be switched off.
- ◆ Be careful when setting permissions to delete documents without check if flags are not set.

 CAUTION

Deletion confirmation deactivated!

Possible loss of data.

- ◆ The option in the **General** tab card to configure the **Delete Confirmation** dialog box should not be switched off.
- ◆ Be careful when activating a filter for auto deletion of data sets that it fits to your workflow as well.

You can configure the automatic deletion in the **Auto Delete Configuration** dialog box to be performed regularly meeting selected criteria. To open the configuration dialog box select **Options > Configuration** from the main menu and select the **Auto Delete** icon.

 CAUTION

Deactivation of the **Delete Confirmation** dialog box for **Auto Delete**!

Loss of data, delay of diagnosis or therapy.

- ◆ The **Delete Confirmation** dialog box for **Auto Delete** should always be switched on, especially on modalities.
- ◆ If the **Delete Confirmation** dialog box for **Auto Delete** is switched off, **Auto Delete** will start without user confirmation.

CAUTION

The **Delete Confirmation** dialog box for auto deletion is kept open for a long time on scanner systems!

Loss of data if objects chosen for auto deletion are updated by acquisition when the Delete Confirmation dialog box is open.

- ◆ Do not keep the **Delete Confirmation** dialog box open for a long time in parallel during a patient examination, and close it before a patient registration.

CAUTION

Conflicting **Auto Delete Configuration** and **Transfer Configuration**!

Loss of data if both functionalities start in parallel.

- ◆ Do not use the **Sent** flag for rule **Delete all Series with flags** if auto transfer is configured to start when the **Sent** flag is set in combination.
- ◆ Use storage commitment whenever the sending and receiving systems supported it.
- ◆ If this combination is not avoidable, always crosscheck if the data chosen for auto deletion is actually stored on the receiving system before confirming auto deletion.

11.2.3 Exporting patient data

Patient data can be exported for exchanging it with other computers or to a server in your hospital's network, for the purpose of imaging reading, reporting, or archiving outside of the *syngo* workstations.

On the **Transfer** menu of Patient Browser, several commands for exporting data are available:

- **Export to** to transfer data, for example, to USB devices.
- **Send to** to transfer data to other workstations over network.

- **Export to Off-line** to export data to a local directory of File Browser.
- **Record Off-line Files** to burn data from the **CDBurn** folder in File Browser.

Please note that not all these functionalities are available and they depend on your system configuration.

**CAUTION**

Lossy JPEG compression is selected for data transfer.

Image quality of the lossy compressed images may no longer be adequate for diagnostic purpose.

- ◆ Do not use the lossy JPEG compression for sending data to a report or primary diagnosis workstation.
- ◆ Do not use the lossy JPEG compression for archiving.

**CAUTION**

Transferring manipulated non-square matrices or viewing segments!

Diagnostically relevant areas of images may be lost.

- ◆ When manipulated images are exported or sent to another workstation, the related original images should be sent, as well. It is strongly recommended to base the final diagnosis always on the original images and not on modified or manipulated images.



- If you transfer patient images from one *syngo* Acquisition Workplace to another with lower software version, DICOM Structured Report, e.g. Dose Report, will fail to be transferred, and the **Status** in **Network Job Status** opened by selecting **Transfer > Network Job Status** in Patient Browser become failed.
- For reasons of data protection, it is recommended not to create folders under the patient name in offline directories when exporting images.
- If lossy compressed images are sent to a network node, and the same instances are already stored uncompressed on that network node, the newly sent images will appear as duplicates in the database.
- Avoid using duplicated node names during network configuration.
- With fast network connections, the transfer of compressed images takes significantly longer than uncompressed. To assure full performance, you can deactivate the image compression.
- When selecting **Record Off-line Files** to export images from the **Patient Browser** to CD a dialog box pops up asking for the label although **Record Off-line Files** is running. The CD is invalid after starting the export jobs and the files from the **CDBurn** folder are removed, but not burned. To solve this problem: Make sure that no other processes are accessing to the CD drive, e.g., burning patient data automatically on CD; Do not start DICOM export to CD when **Record Off-line Files** is used. The process of burning offline data takes time. When the process is finished, a message appears in the status bar.
- The patient name is omitted in the naming of the exported file if the patient name contains Chinese characters and the images acquired with a former software version are exported via **Export to offline**. you should export a single patient only. You first select **Options > File Browser** from the main menu to open the **File Browser**, then create a file folder as target folder for exported files to identify clearly the exported file, and you correct the patient name in the offline directory immediately.

- It can happen, that it is not possible to export data to offline into a newly created directory. To solve this problem, you can reboot the system. After reboot, the directory can be created and the images will be exported into this directory.
- Export jobs to DVD/CD which are started before an intervention examination may fail. They cannot be restarted from the **Local Job Status** dialog box. To solve this problem, you can perform a new export job after finishing the intervention examination.

11.2.4 Importing patient data

Patient data from external sources (for example, DVD disc, USB device, & network node) can be imported into the local database of Patient Browser for you to view or process on *syngo* workstations.

- You can import patient data from DVD discs or USB devices by using **Import** or **Import from Off-line** on the **Transfer** menu of Patient Browser.
- With **Patient > Search** in Patient Browser, you can retrieve patient and examination data from network DICOM nodes into the local database of Patient Browser.

CAUTION

If *syngo* receives images but the study or series UID is already available in the local database, the data will be assigned to this locally existing patient. This happens regardless of the patient identification (name, date of birth, gender and patient ID) of the received images. Changes of patient information on other systems will not automatically change the patient information in *syngo*!

Data retrieved from PACS or other nodes seems to be misaligned or lost, but is stored elsewhere in the database.

- ◆ Imported data will always be appended to the already existing patient on the local system, based on matching DICOM UIDs.
- ◆ If you imported data, but cannot find them on the local system under a certain name, use other criteria from the study or series to search for the imported data, or filter the data.
- ◆ To completely avoid these inconsistencies, always delete the local patient data after you have successfully archived them.

CAUTION

If the Patient ID is not unique, studies may be listed which do not belong to the patient selected and displayed in the information area!

Import of wrong patient data possible.

- ◆ Always check the patient data before importing them into your system.

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- For the systems configured with Chinese language, some problems about patient name may happen when importing or exporting patient: If you import a patient from European language systems, you may not change the patient name to Chinese characters; If you export a patient to a system with different software version, the patient name might be shown incorrectly; If you export a patient to offline, the patient name may be shown incorrectly.
- If you try to import more than 10 000 images via DICOM, the import job is aborted. A misleading message is displayed in the status bar that the network job has been completed successfully. But if you look in the history of the status bar, you can see that the network job was aborted. To solve this problem: Do not import more than 10 000 images at the same time; Check the history of the status bar to ensure that all images have been imported; Import the missing images, if necessary.

11.2.5 Archiving patient data

You archive patient data for long-term storage. The data archiving can be done either by recording patient data to medical grade DVD discs or by sending the data to the archiving servers on your hospital's network.

There may be legal regulations pertaining to archiving of digital image data. The archive operator is responsible for observing these regulations.

When archiving to network nodes, we strongly recommend the use of the "Storage Commitment" service. It can be enabled in the Local Service software, if the remote system supports it.

CAUTION

Systems that do not use storage commitment only report back to the sending system when the data has been fully received (flags A and S). A user or an auto-delete mechanism at the sending system might subsequently delete the sent data. However, this response does not imply that the data has already been stored at the receiving system!

In the case that the receiving system cannot store the data, the data might irrevocably be lost.

- ◆ Double-check that the data is actually stored on the receiving system. Use storage commitment whenever it is supported by the sending and receiving systems.

CAUTION

Misleading or misinterpretation of the storage commitment flags AC/SC. Storage Commitment means storage to hard disk but this may not fulfill or guarantee regulatory requirements about long-term archiving. You can delete the committed objects!

Possible loss of data within the required period for retention.

- ◆ Observe the regulatory requirements regarding long-term archiving.

 CAUTION

Data are automatically deleted due to configurable settings!

Loss of clinically relevant data if rules are not correctly specified.

- ◆ Be very careful when creating "not to be archived" rules for data. This data can be automatically deleted and cannot be recovered.
- ◆ Make sure that all data (images and reports) necessary for medical purposes are completely and successfully sent to an archive.
- ◆ Do not use automatic deletion if the archive node does not support Storage Commitment.

- The saving of patient and examination data to local exchangeable data media via the **Archive to...** function is not possible anymore. It is still possible to save data to a data medium with the **Export to...** function. Please note that the word "Archive" means to transfer data to an archive station, not an archive media.
- **Storage Commitment** within a standard *syngo* configuration means storage on a hard disk. Be aware of the fact that any end user can delete these objects.

11.3 Data set conversion

The **Data Set Conversion** tool prepares DICOM images of other manufacturers that are already imported into the *syngo* database, making them accessible for CT applications. The tool can be started alongside other applications.

Additionally, the tool splits up a series from Siemens Healthineers CT multi-phase heart reconstruction into series of different heart phases or slice positions.

11.3.1 Setting up lossy image compression

In case pixel values of the original data exceed the range compatible with existing Siemens Healthineers systems, the original images are lossy compressed during data set conversion.

CAUTION

Images with lossy compression are used for diagnosis!

Wrong basis for diagnosis.

- ◆ Always verify your evaluation results with the original DICOM images (first reader duty).
- ◆ Never use lossy compressed images for primary diagnosis.
- ◆ Check the image text for information on lossy compression before starting an evaluation.

- 1 Select the **Options > Configurations** menu item.

The **Somaris/5 – Configuration Panel** appears.

- 2 Double-click the **Conversion** button.



The **DICOM Converter Configuration** dialog box appears.

- 3 Deselect the option **Mark images as lossy compressed**, if indispensable, if you do not want images with modified pixel range to be marked.



As the option **Mark images as lossy compressed** is activated by default, images with modified pixel range are labeled with, for example, CQ!, CF 10 (Compressing Factor 10). If marked as lossy compressed, images are not accepted for further processing by CT applications.

11.3.2 Converting data set

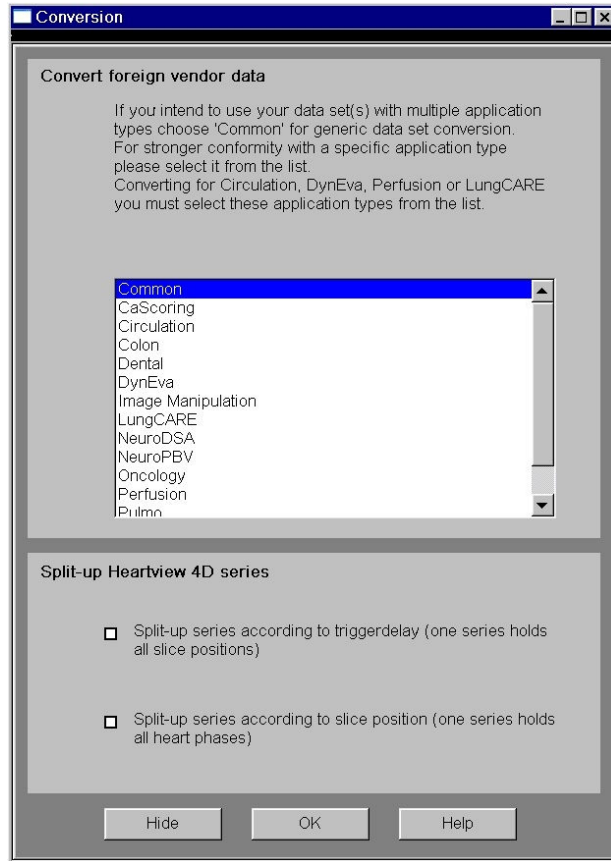
You start **Data Set Conversion** via the **Patient Browser**.

- 1 Call up the **Patient Browser** by selecting **Patient > Browser** from the main menu.
- 2 Select one or more series of a study in the navigation or content area of the **Patient Browser**.

If you want to convert more series simultaneously, merge the series first. You can select all the series you want to convert, and drag-and-drop the selected series onto the remaining series. One series has to remain unselected.

- 3 Select **Applications > Conversion** from the menu of the **Patient Browser**.

The **Conversion** dialog box opens.



- 4 Do one of the following:
 - Select **Common**, if you intend to use the converted series for more than one application type.
 - Select a specific application from the list for which you want to convert the data, for example, **DynEva**, or **Perfusion**.
- 5 For cardiac data, do one of the following:
 - Select **Split-up series according to trigger delay**, to split-up the series of multi-phase heart reconstruction according to the different heart phases included.
 - Select **Split-up series according to slice position** to split-up the series according to the different slice positions it contains.

6 Click **OK** to start the conversion.

You are informed of the status of conversion on the status bar, where any conversion risks are indicated by error messages.

11.4 HIS/RIS integration

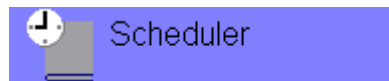
A HIS/RIS system is an integrated information system designed to manage the administrative, financial, and clinical aspects of a hospital, and it is used by radiology departments to store, manipulate, and distribute patient radiological data and imagery. HIS/RIS system generally consists of patient tracking and scheduling, result reporting, and image tracking capabilities.



Some manufacturers of Radiology Information Systems (RIS) do not support the **Requested Procedure Description** and **Scheduled Procedure Step Description** in the **Modality Worklist**. In these instances, no information about the kind of procedure is shown in the worklist. You can contact your RIS manufacturer for support.

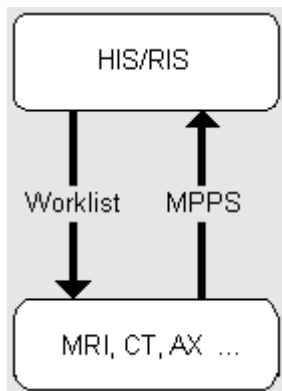
11.4.1 HIS/RIS patient registration

If your CT modality is linked to, licensed, and registered for an HIS/RIS system, a patient can be registered via the HIS/RIS system. After registration, the patient information is saved in the **Scheduler** of Patient Browser.



11.4.2 MPPS

The exchange of information based on Modality Performed Procedure Step (MPPS) closes the gap between the modalities and the HIS/RIS. The information documents which actions have been performed for a patient. The DICOM worklist, generated in the RIS, was handed over to the modalities. A MPPS object containing certain information can be sent back from the modalities to the HIS/RIS.



The purpose of MPPS is to provide features that will help to integrate the modalities with the supervising information system in the hospital. Thus a smooth clinical flow is ensured, and billing, dosage, and other information is generated within the hospital system.

When you register a patient, the system creates an MPPS object. An MPPS object is generated according to the DICOM Standard for archiving and evaluation purposes.

If a study is continued with additional scans, a new MPPS is created.

During the examinations and during the post-processing of the examination results all MPPS relevant information (protocol information, radiation dose information, film consumption information, etc.) is automatically filled into the related MPPS object.



- The MPPS function is available by an optional software license. If this license is not available, only local MPPS management is possible, and no feedback is sent to the HIS.
- Registration by DICOM worklist provides the **Scheduled Procedure Step** description. It will be mapped to an existing scan protocol. If mapping does not exist yet, the study item on the registration card is empty. To solve this problem, you can select a scan protocol now to keep the mapping in future.

The MPPS window

Select the patient, the examination, or one of the associated series or images from the **Patient Browser**; choose **Patient > Show MPPS** to open the **Modality Performed Procedure Step** window.

Modality Performed Procedure Step

PATIENT

Patient name : Tester
 Patient ID : 14 08 17-16 29 10-STD-1.3.12.2.1
 Date of birth : 8/17/1994
 Sex : male

STUDY

Study description :
 Modality : CT
 Accession No. :
 Station : CTDSCDRL
 Study date : 8/17/2014
 Study time : 4 29:17 PM
 Study ID : 1

MPPS

Description :
 Status : IN PROGRESS
 Type :
 End date :
 End time :
 Comments :
 Previous
 Next

Performed actions

Protocol Code	Protocol Item

Performed series

Series Description	Protocol	Performing Physician	Operator	Retrieved AE Title

ApplyToAll Completed Discontinued Send Save Cancel Help

- (1) **PATIENT** and **STUDY** areas
- (2) Upper part of the **MPPS** area
- (3) Lower part of the **MPPS** area

Examination data sorted by various aspects and displayed in three tab cards (not all of them may be available at your modality):

- The performed persistent series are listed on the **Action** tab card of the MPPS (for example, RTD series are not referenced).
- The dose information is automatically filled in the comment field on the **Dose** tab card.
- The contrast media data is filled in the **Billing** tab card.

Contrast media information for billing, for example, name and catalog ID, is set in the **Contrast** tab card of the **Examination Configuration** dialog box.

MPPS status In the **Patient Browser**, the MPPS series are marked by the following flags:

- IN PROGRESS
- DISCONTINUED
- COMPLETED

Viewing & editing MPPS You can also add information, for example, volume & contrast medium, to an MPPS object.

- 1 Open the **Modality Performed Procedure Step** window. By default, the latest MPPS object is displayed.
- 2 Do one of the following:
 - Click the **Previous** button to have the earlier MPPS object displayed again.
 - Click the **Next** button to have the later MPPS object displayed.
 - Click the **Save** button to save your changes of the document.

Saving an MPPS does not complete the examination. No message is sent to the HIS/RIS. If you consider further steps, you can at first save the document, and send it later.



- In case you cannot edit the MPPS information, you do not have appropriate access rights. Security in *syngo* ensures that access to functions and data is always provided in accordance with the privileges and permissions granted to a user.
- Click the **Cancel** button to close the dialog box if you only have opened the document for printing or viewing.

Sending MPPS Information about status and changes of the MPPS is sent to the HIS/RIS every time you click one of the following buttons in the **Modality Performed Procedure Step** dialog box:

- Completed
- Discontinued
- Send

◆ Do one of the following:

- Click the **Completed** button when all data shown in the performance document are correct, and complete the examination.
- To complete all the MPPS documents of a patient or a study, click the **Complete MPPS** icon in the **Patient Browser**.



- Click the **Discontinued** button if the examination is canceled.
- Click the **Send** button to mark the MPPS as being processed and to inform the HIS/RIS about changes in the MPPS. The **Send** button is displayed only, if an MPPS node is configured.

A corresponding message is forwarded to the HIS/RIS.



- The usage of the DICOM with MPPS service depends on the availability of the corresponding RIS system. Even if no RIS with MPPS service is connected, the **MPPS** icon is displayed in the **Patient Browser**.
- If an MPPS node is configured, make sure that at least one series is associated with the MPPS object, also do not delete the empty series to keep the association with the MPPS.
- If no series was associated to the MPPS during the examination and an MPPS node is configured, one empty series is automatically created at the end of the examination which is associated with the MPPS.

MPPS auto-sending

In the **Workflow** tab card of the **Examination Configuration** dialog box, you can activate the checkbox **Auto Complete MPPS**. All MPPS are then automatically set to **COMPLETED** after all recon jobs were reconstructed and the patient was closed.



- Make sure that you printed all film sheets before closing the patient. Otherwise MPPS does not consider the film sheets anymore.
- If autofilming is active, all the images sent to **Filming** will be inserted in the MPPS and not the ones that were really printed.

MPPS messages

In the status bar of any task card, you can easily confirm whether the message was successfully sent from MPPS to HIS/RIS.

Click in the center of the status bar of the task card to check the corresponding entry in the History of Events. Look for an entry in the list concerning the MPPS. If you do not find such an entry, it is already removed, because the information was sent successfully to the HIS/RIS. If you find an entry concerning the MPPS, click it.

If an error occurred, a message appears. Check also the status bar at the bottom of the task card for any error information.

11.5 Teampay

Teampay is a network that connects medical institutions and their imaging devices as a virtual team. This allows users to get information about all the radiology modalities located in different departments of an organization.

12 Quality assurance

To ensure a constantly high image quality, you must perform quality measurements regularly. Quality tests are usually performed in several steps:

- Daily quality measurements
- Monthly quality measurements (constancy test)
- Camera test
- CTDI test (once every six months)

You require the set of phantoms for the quality measurement and possibly additional accessories for the constancy test.



We recommend having the constancy test performed by Siemens Healthineers customer service as part of a service contract.

12.1 Daily Quality

It is recommended that you perform a quality check of the CT scanner after long pauses, or whenever ring artifacts occur in images.

The daily quality measurements only require checking of three parameters on the water phantom:

- The CT value of water is calculated in Hounsfield units (HU).
- The Pixel noise of images is calculated as a standard deviation.
- Tube voltages are measured directly on the x-ray tube.

These measurements are carried out for all available kV values.

CAUTION

Wrong correction tables!

X-ray not, or only partially, usable.

- ◆ Perform the daily quality tests every day before you start the actual examinations.



The CT value of water can be measured in Constancy Test \Homogeneity Test.

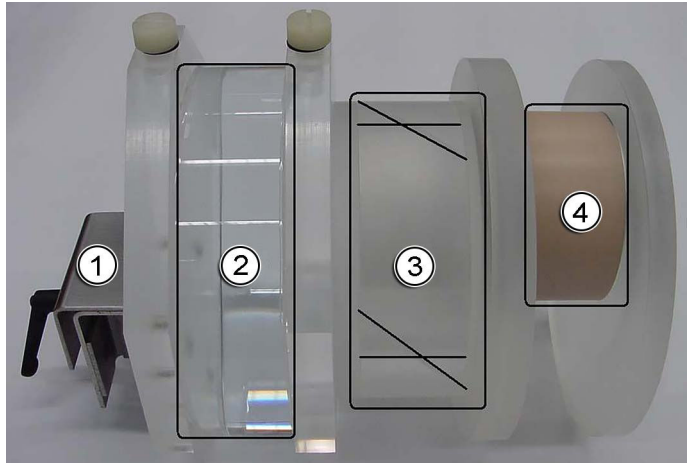
12.1.1 Equipment for Daily Quality

The following components are required for the daily quality measurement:

- Phantom holder
- Set of phantoms

12.1.2 CT Quality Phantom Setup

The CT quality procedure must be performed using the CT quality phantom that comes standard with the system. This phantom is also used for acceptance testing and constancy tests. This phantom has a uniform water section, a section for measuring CT slice thickness, and a wire section.



- (1) CT Quality Phantom Clamp
- (2) Water Section
- (3) Slice Thickness Section with etchings (highlighted in black) for positioning with the laser markers.
- (4) Wire Section

To attach the CT quality phantom, perform the following steps:

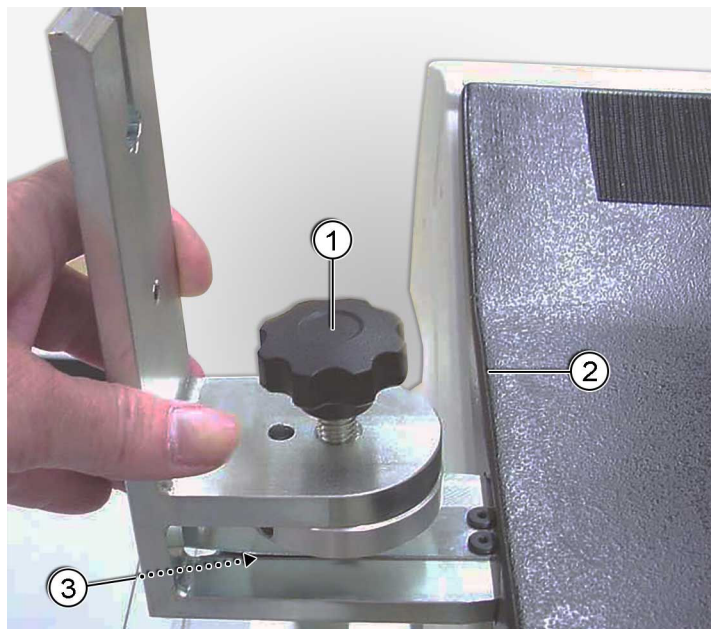
- 1 Remove the pallet pad from the pallet.
- 2 Insert the bracket into the slot at the end of the patient bed.

The bracket attaches itself to the pallet when inserted. A circular button beneath the bracket can be pressed to release the bracket.

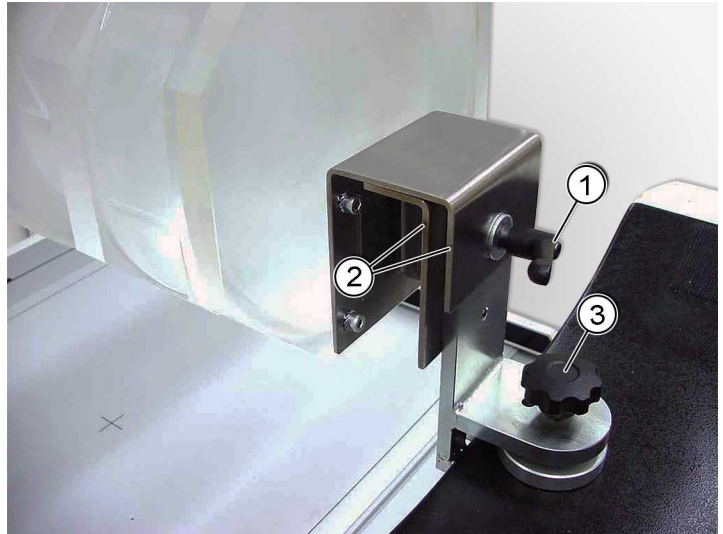


This is the same slot used by the head holder for brain studies.

- 3 Spin the bracket clamp down to tighten it against the pallet.



- (1) Bracket is hand tightened to pallet to secure phantom.
 - (2) Foot End of Patient Bed
 - (3) Bracket Release Button (located beneath the bracket)
- 4 Place the CT quality phantom's clamp over the bracket, and tighten the clamp using the knob.



- (1) The phantom's clamp has a knob to tighten the clamp to the bracket attached to the pallet.
- (2) The bracket attached to the pallet holds the phantom in place.
- (3) The bracket clamp has a knob to tighten its clamp to the pallet.



Always attach the phantom *bracket* to the patient bed first, and then clamp the CT quality phantom onto the bracket.

12.1.3 Positioning the CT Quality Phantom

- 1 Adjust the bed height so that the bed index stage clears the detectors (approximately 16.5 cm).

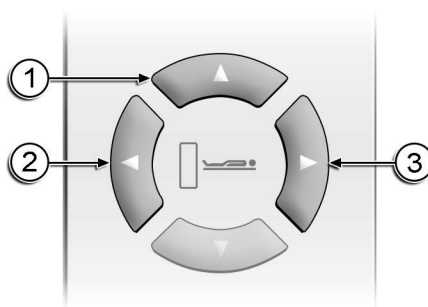


Some screens display values in centimeters, while others millimeters.

- 2 Using the hand control, press the **Bed Index** button twice.



3 Press the **CT Brain (Bed Up)** button.



(1) CT Brain SPECT Study Position

(2) CT Body Study Position

(3) SPECT Study Position

The bed index stage of the patient bed moves into position.

4 Double click the **Laser** button. The lasers activate.



5 Use the hand control to move the patient bed until the quality phantom's slice thickness section etchings are aligned with the laser.

6 Record the patient bed's height and distance so that it can be used for aligning the phantom's slice thickness section etchings next time.

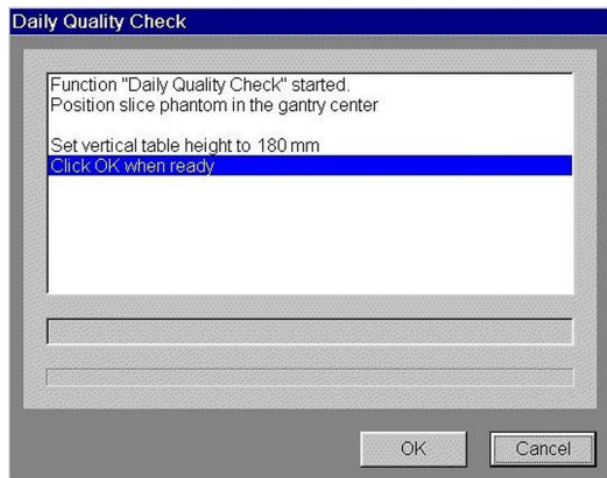
7 When the phantom is aligned, perform the CT Quality measurements.

12.1.4 Performing Daily Quality

You must perform calibration before you start quality measurement. Position the phantom set. Call up **Setup > Quality**.

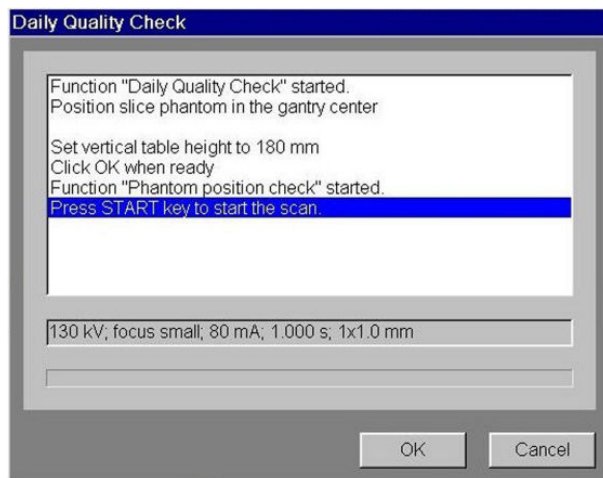
1 Call up **Setup > Quality**.

The **Daily Quality Check** dialog box is displayed.



2 Click on the **OK** button.

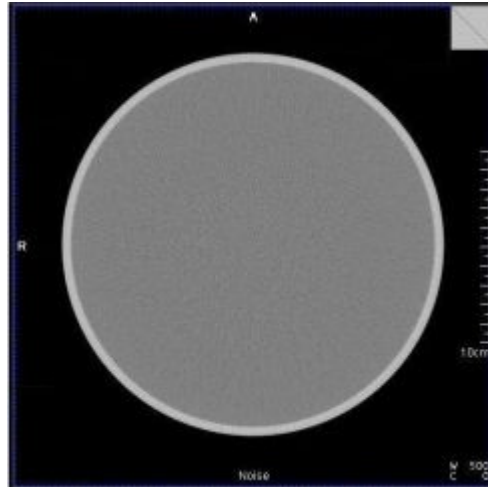
The "Quality Assurance Patient" is automatically selected. The patient table is positioned such that the water phantom is in the scan plane.



- 3 Press the **Start** key on the control box.



The position of the phantom set is checked and displayed. The measurements are started automatically. For each tube voltage (kV step), two measurements are performed. After each measurement, a CT-image of the water phantom is displayed in the active segment.

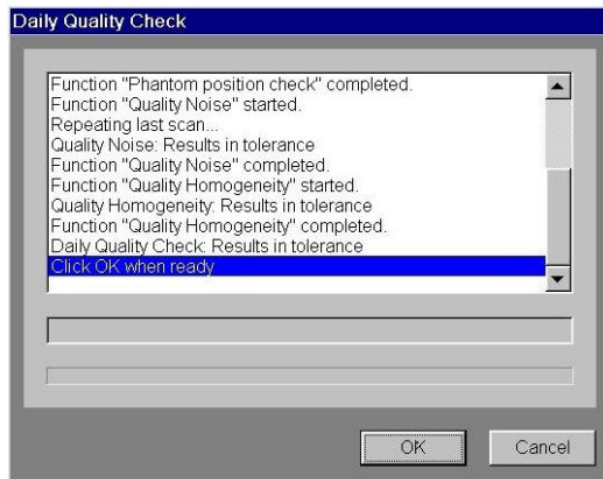


The difference between the first and the second measurement is calculated and displayed. The following evaluations are shown:

- ROI
- Mean value of the CT value
- Sigma value (measure for pixel noise)



All measurements were performed and evaluated. Test results are displayed in the **Daily Quality Check** dialog box.



4 Click on the **OK** button.



- All parameters and images shown are examples. Only the parameters displayed by your system are definite.
- For incorrect positioning of the phantom set, a message is displayed. Correct the position, if necessary.
- You can cancel quality measurement in any phase. Click on the **Cancel** button. Then, you have to repeat the quality measurement from the beginning before you start the actual examinations.



Test results of the Daily Quality measurement are out of tolerance.

- ◆ Call your Siemens Customer Service.

12.1.5 Daily Quality report

The images of the **Daily Quality** measurements are stored as images of the "Quality Assurance Patient" in Patient Browser. New images overwrite older images.



All values of the **Daily Quality** measurements are stored in the constancy test reports.

12.2 Constancy tests

Together with other characteristic values, the daily test values are checked for constancy monthly. These tests are performed on the complete set of phantoms. The monthly quality test is performed via the **Quality Constancy** dialog box of the **Local Service** window.

In addition, the CTDI (CT Dose Index) must be measured at least once every six months. The CTDI must also be measured after any maintenance work that might have altered the test results.

This section describes how to perform the following constancy tests:

- Phantom and phantom position check; **Phantom check**
- Position of the lightmarker; **Lightmarker**
- Position of the sagittal and coronal lightmarker; **Sag/Cor. Lightm**
- Automatic positioning of the tomographic plane using a preview image; **Prev. Image Pos.**
- Real slice thickness; **Slice**
- Homogeneity; **Homogeneity/Water**
- Pixel noise; **Noise**
- Modulation transmission function; **MTF**
- Position of the patient table; **Table Position**

For the monthly constancy test, you require the following phantoms and accessories:

- Phantom holder
- Set of phantoms
- 70 kg (154 lb) weight
- Ruler, 300 mm or longer



We recommend having the constancy test and the CTDI measurement performed by proper personnel. If you want to conduct the constancy test and the CTDI measurement yourself, you must perform all steps of the constancy test or the CTDI measurement under your own responsibility.

CAUTION

The system requires preventative quality control measures to be performed to ensure system performance!

There is a potential for misdiagnosis.

- ◆ Perform calibrations and image checks on a regularly scheduled basis. If a defect is detected or system damage found, do not use the system. Contact your Siemens Customer Service Representative.

CAUTION

Lightmarkers can be out of position!

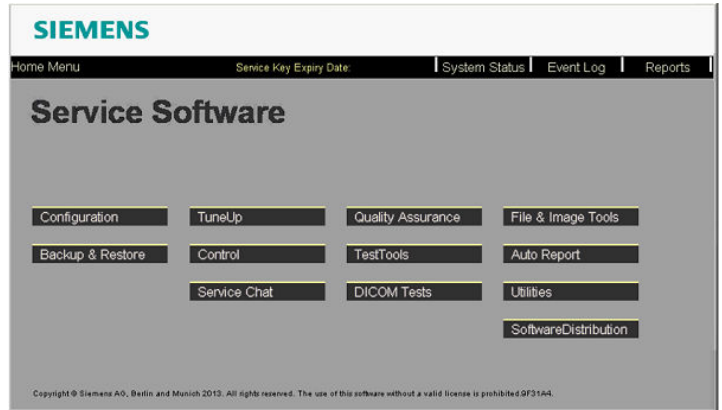
Patient injury can result.

- ◆ Perform the Lightmarker test monthly to prevent imaging the wrong field of view.

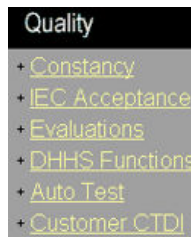
12.2.1 Calling up constancy tests

- 1 Call up **Options > Service > Local Service**, delete the password entries, and click **OK**.

The **Home Menu** dialog window is displayed.



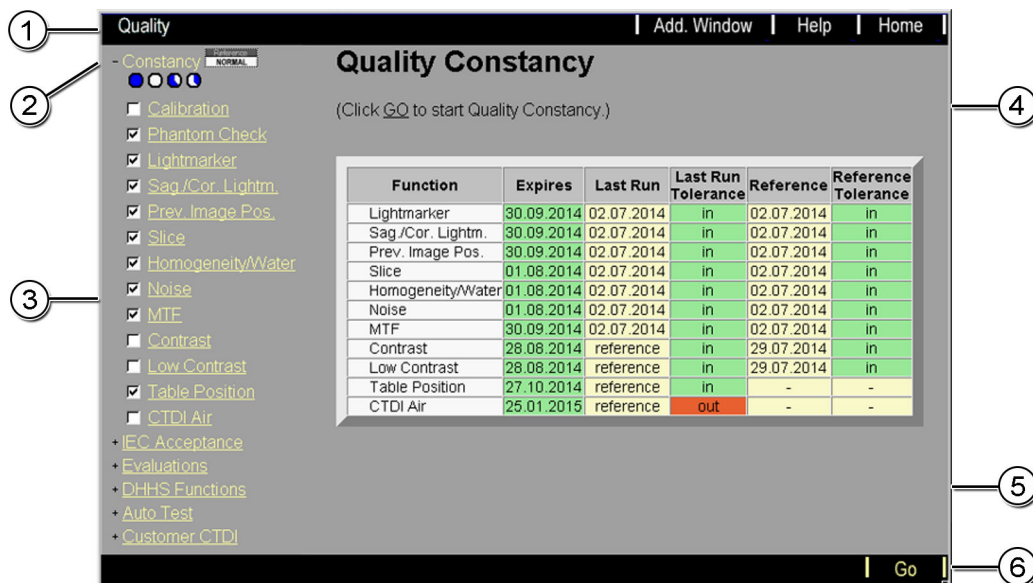
- Click the **Quality Assurance** button to display the **Quality** dialog window.



Quality/Constancy

- Select the **Constancy** procedure.

The **Quality Constancy** dialog box is displayed.



The Quality Constancy window

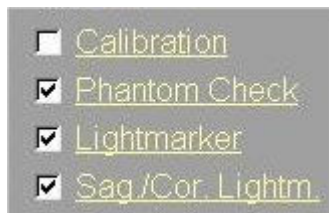
- (1) Title bar
- (2) Mode button
- (3) Quality functions
- (4) Content area
- (5) Status and error message area
- (6) Command buttons

4 Do one of the following:

- Click the tests that you want to include in addition in this constancy test.

The system calibration and phantom position check are typically required and run before any other individual constancy test.

- Deselect the tests that you do not require by clicking them again.



i

Every time the Quality Constancy dialog is opened, all constancy tests are selected by default. Siemens recommends that only a subset of these tests be selected for monthly Quality Constancy, but individual tests can be added or removed, based on need and the availability of system accessories.

The recommended subset of constancy tests for your Symbia system is:

- Phantom check
- Lightmarker
- Sag/Cor. Lightm.
- Prev. Image Pos.
- Slice
- Homogeneity/Water
- Noise
- Table Position

5 Confirm selection of the constancy test with **Go**.

You are prompted to enter the name of the tester and the serial numbers of the phantoms.

Quality Constancy

Enter name of tester:

Enter serial numbers:

Slice phantom:

Water phantom:

Wire phantom:

Low Contrast phantom:

CTDI Body phantom:

CTDI Head phantom:

Dosimeter:

CT-Ionisation chamber:

• Click **GO**, when ready.

6 Click **Go** when you have completed your preparations.



- After you called up the constancy test, follow the instructions given on the screen. The system starts the first quality measurement. After successful completion of the first test the next test is started automatically without pressing the **Start** key.
- After you finish the constancy test, click **Done** and then **Home**. You return to the home page of Local Service.
- You can interrupt each test within a constancy measurement. Click the **Cancel** button. The current measurement is aborted.

12.2.2 Normal or reference test

Evaluation of the constancy test results are based on a comparison with reference data that are stored in your system. These are determined by Siemens Healthineers service.

If you have any of the system parts replaced (for example, tubes), the reference data must be determined again. You can do this with a reference measurement.



- ◆ Click the field **Reference/NORMAL** to toggle between a normal constancy measurement and a reference measurement.



In reference mode, the individual selected tests are marked by an "R". The test results will be used as reference data in the future.

A reference measurement follows the same sequence as a normal constancy test.



- In **Reference** mode, click the **Save** button first before terminating the constancy test with **Done**. Without **Save** the old reference values will be retained.
- We recommend that Siemens Healthineers service always performs your reference measurements.

12.2.3 Reference Measurements for Upgraded Systems

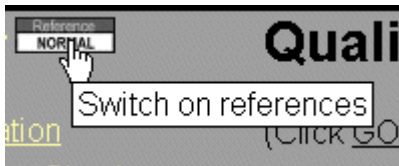
If your system has been upgraded to software version VB2xx, you must obtain reference measurements from the first constancy calibration performed with the new software. Non-reference constancy calibrations will not function until the required reference measurements are recorded. A single reference run is needed for each calibration, and the system will allow measurements that are within tolerance to be saved.



- If a reference measurement has not been recorded, an error message will be displayed: **The required reference measurement was never done.**

To record reference measurements, from the constancy calibration, perform the following steps:

- 1 Navigate to the Quality Constancy dialog box by selecting **Options** > **Service** > **Local Service** > **Constancy**.
- 2 Select the reference option.

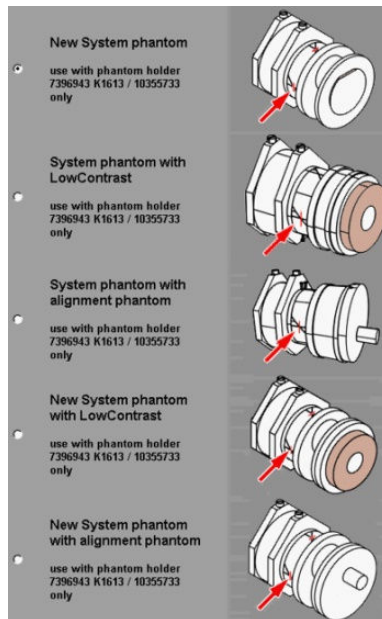


- 3 Select the desired constancy tests.
- 4 Click the **Go** button to start the constancy test.

12.2.4 Phantom check

Before the individual test starts, phantom check may be required by the system.

- ✓ The phantom check step is initiated.
 - ✓ The set of phantoms is positioned in such a way that the inner light marker hits the reference marking on the slice thickness phantom.
- 1 Click the phantom type corresponding to the phantoms that have been positioned on the patient table. The available sets of phantom for you to select depend on the CT system configuration.



- 2 Click **Go**.
- 3 Press the **Start** key on the control box.



Scanning is started.

The result of the scan will be displayed in the status and error message area.

Scanning ...
Phantom position is in range.
Function "Phantom position check" completed.

12.2.5 Inner light marker test

With the quality measurement, you determine the deviation of inner light marker from the current slice plane.

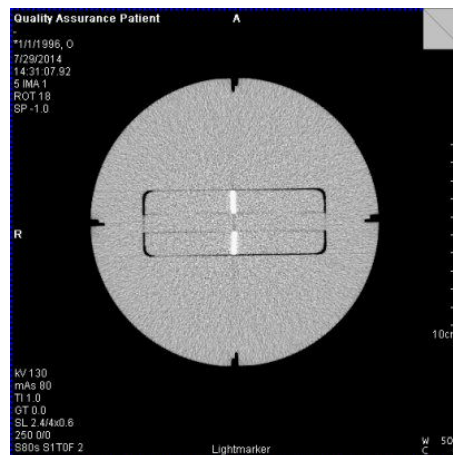
- ✓ The set of phantoms is positioned in such a way that the inner light marker points on the reference marking on the slice thickness phantom.
- ✓ The **Lightmarker** test mode is loaded.
- ✓ You are prompted to press **Start**.

1 Press the **Start** key on the control box.

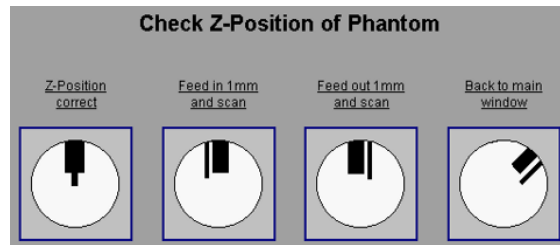


Scanning is started.

In the image area, the phantom is displayed as a circle in the image. For the light marker test, both the short and the long strip in the 12 o'clock position are important.



- 2 Compare the position of the short/long strip on the image with the options displayed in the content area of the **Quality Constancy** dialog box.



- 3 Correct the table position by clicking **Feed In** or **Feed Out**. With **Feed In** and **Feed Out**, you can move the table by 1 mm.



A new image is acquired.

- 4 Repeat step 2 to step 3 until you have set the correct z-position.
5 Click **Continue**.

The internal z-position is set to 0, the display of the table position shows 0. The result of the test is output in the content area of the **Quality Constancy** dialog box.



In the status and error message area, the result of this test (in tolerance or out of tolerance) is output. Use the vertical scroll bar to view all test results.

Please compare image with the pictures above. Select table movement or continue

Quality Lightmarker: Results in tolerance

Function "Quality Lightmarker" completed.



All parameters and images shown are examples. Only the parameters displayed by your system are definite.



Test is out of tolerance.

◆ Contact Siemens Healthineers customer service.

12.2.6 Sagittal and coronal light marker test

With the quality measurement, you determine the position of the sagittal/coronal lightmarker.

- ✓ The slice thickness phantom is positioned in such a way that the sagittal and horizontal lightmarkers mark the horizontal and vertical center.
- ✓ The **Sag/Cor. Lightm.** test mode is loaded.
- ✓ You are prompted to press **Start**.
- ◆ Press **Start** on the control box.



Scanning is started. In the image, the middle of the phantom is determined. The result is the deviation from the horizontal and vertical middle of the image to the middle of the phantom. The result of the test is output in the content area of the **Quality Constancy** dialog box.

Quality Sagittal/Coronal Lightmarker		
Results 		
	Sagittal	Coronal
Reference [mm]:	0.0	0.0
Tolerance [mm]:	+/- 2.0	+/- 2.0
Result [mm]:	-1.0	-6.0 <

In the status and error message area the result of this test (in tolerance or out of tolerance) is output. Use the vertical scroll bar to view all test results.

Scanning ...
 Quality Sagittal-Coronal Lightmarker: Results out of tolerance
 Function "Quality Sagittal-Coronal Lightmarker" completed.



All parameters and images shown are examples. Only the parameters displayed by your system are definite.



Test is out of tolerance.

- ◆ Contact Siemens Healthineers customer service.

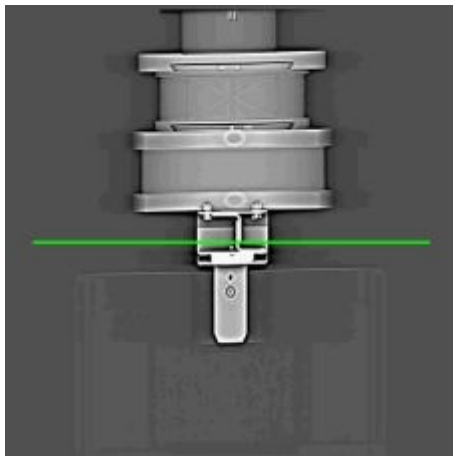
12.2.7 Preview image positioning test

With the quality measurement, you determine the automatic positioning of the tomographic plane using a preview image.

- ✓ The set of phantoms just before the scanplane. The internal lightmarker must not hit the phantom.
 - ✓ The **Prev. Image Pos.** test mode is loaded.
 - ✓ You are prompted to press **Start**.
- 1 Press **Start** on the control box.



In the image area, the Quality Assurance Patient image is displayed. You can see a horizontal line in the image and the corresponding table position to the line position is shown.

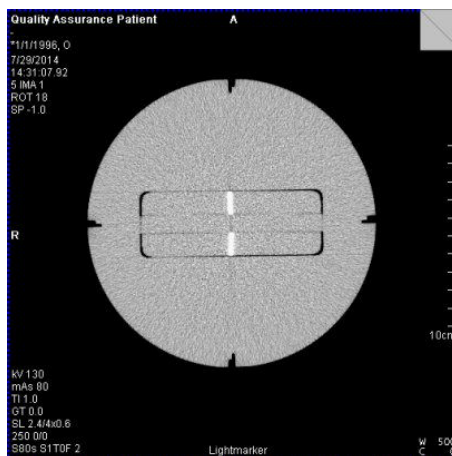


- 2 Move the horizontal line and mark the position of the middle of the slice thickness phantom.
- 3 Click **Go**.

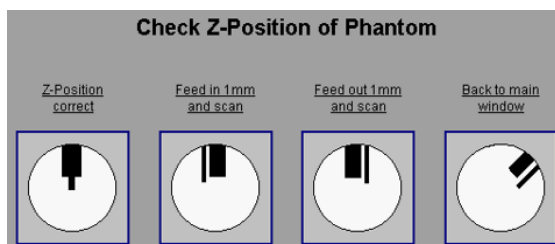


The patient table moves automatically to the marked position. Scanning is started.

In the image area, the phantom is displayed as a circle in the image. For the preview image test, both the short and the long strip in the 12 o'clock position are important.



- 4 Compare the position of the short/long strip on the image with the options displayed in the content area of the **Quality Constancy** dialog box.



- 5 Correct the table position by clicking **Feed In** or **Feed Out**. With **Feed In** and **Feed Out**, you can move the table by 1 mm.

Feed In

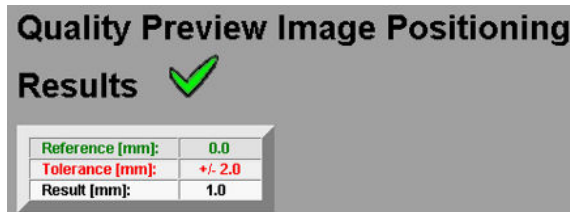
Feed Out

A new image is acquired.

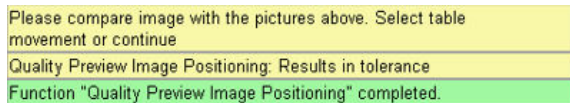
- 6 Repeat image recording and correction of the table position until you have set the correct z-position.
- 7 Click **Continue**.

Continue

The result of the test is output in the content area of the **Quality Constancy** dialog box.



In the status and error message area, the result of this test (in tolerance or out of tolerance) is output. Use the vertical scroll bar to view all test results.



All parameters and images shown are examples. Only the parameters displayed by your system are definite.



Test is out of tolerance.

- ◆ Contact Siemens Healthineers customer service.

12.2.8 Slice thickness test

The slice thickness is determined with the slice thickness phantom. For each slice thickness, a tomogram of the phantom is recorded and the real slice thickness is calculated from this.

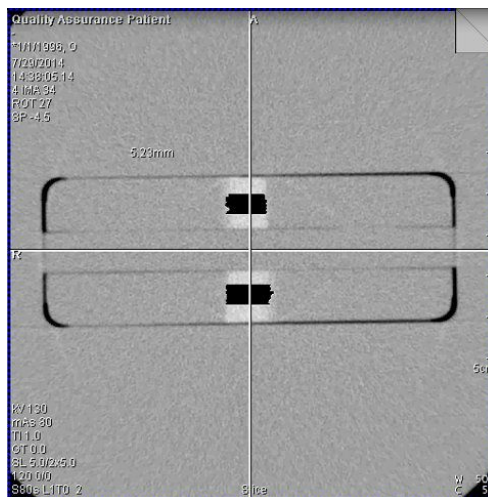
After the previous test has been completed, the **Slice** test is started automatically.

- ◆ Press the **Start** key on the control box, if prompted.



Scanning is started.

An image of the slice thickness phantom is displayed in the image area and the slice thickness is calculated. The procedure is repeated automatically.



After all slice thicknesses have been evaluated, the result of the test is output in the content area of the **Quality Constancy** dialog box.

Quality Slice Results ✓

	Nominal [mm]	0.60	0.60	0.60	5.00	1.20	8.00	1.20
	Tolerance [mm]	+/- 0.30	+/- 0.30	+/- 0.30	+/- 0.50	+/- 0.50	+/- 0.80	+/- 0.50
row 1	Result [mm]	0.64	0.72	0.74	5.20	1.29	8.05	1.43
row 2	Result [mm]	0.66			5.23		8.11	
row 3	Result [mm]	0.65						
row 4	Result [mm]	0.60						
row 5	Result [mm]							
row 6	Result [mm]		0.69			1.41		
row 7	Result [mm]		0.68			1.43		
row 8	Result [mm]			0.70				1.39
row 9	Result [mm]			0.70				1.38
row 10	Result [mm]							

In the status and error message area the result of this test (in tolerance or out of tolerance) is output. Use the vertical scroll bar to view all test results.

Scanning ...
 Quality Slice: Results in tolerance
 Function "Quality Slice" completed.



All parameters and images shown are examples. Only the parameters displayed by your system are definite.



Test is out of tolerance.

- ◆ Contact Siemens Healthineers customer service.

12.2.9 Homogeneity test

With this test, you measure the homogeneity of the CT values in five regions of the water phantom. After the previous test has been completed, the **Homogeneity** test is started automatically.

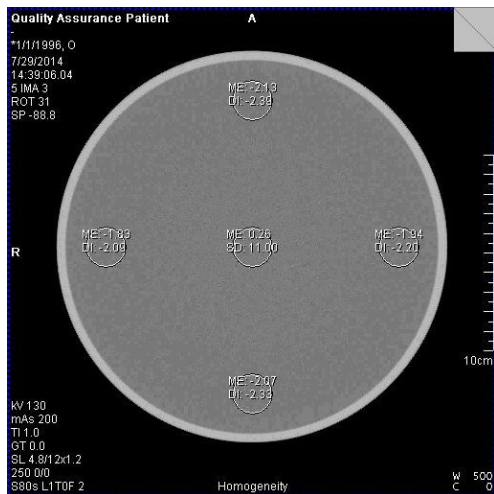
- ◆ Press the **Start** key on the control box, if prompted.



Scanning is started.

An image of the water phantom is recorded. 5 ROIs are marked in the image, one central and four peripheral.

In addition to the ROIs, the mean values and standard deviations of the CT values are displayed. The difference between the mean values of the central ROI and the peripheral ROIs is displayed.



The measurement is repeated automatically for other tube voltages. The results of the test are output in the content area of the **Quality Constancy** dialog box.

Quality Homogeneity Results

ROI		Mean	Diff.	Diff.	Diff.	Diff.
		center/water	3	6	9	12
all rows	Nominal [HU]:	0.00	0.00	0.00	0.00	0.00
	Tolerance [HU]:	+/- 4.00	+/- 4.00	+/- 4.00	+/- 4.00	+/- 4.00
130.0 [kV]						
row 1	Result [HU]:	-0.48	-1.07	-0.90	-1.13	-0.91
row 2	Result [HU]:	0.24	-2.19	-2.06	-2.35	-2.49
row 3	Result [HU]:	0.26	-2.20	-2.33	-2.09	-2.39
110.0 [kV]						
row 1	Result [HU]:	-0.36	-1.29	-1.32	-1.37	-0.85
row 2	Result [HU]:	-0.21	-1.82	-1.73	-2.12	-1.64
row 3	Result [HU]:	0.25	-2.79	-2.65	-2.33	-2.70

In the status and error message area, the result of this test (in tolerance or out of tolerance) is output. Use the vertical scroll bar to view all test results.

Scanning ...

Quality Homogeneity: Results in tolerance

Function "Quality Homogeneity" completed.



All parameters and images shown are examples. Only the parameters displayed by your system are definite.



Test is out of tolerance.

- ◆ Contact Siemens Healthineers customer service.

12.2.10 Pixel noise test

The pixel noise is determined from two tomograms of the water phantom in the same way as for the daily quality measurement.

After the previous test has been completed, the **Noise** test is started automatically.

- ◆ Press the **Start** key on the control box, if prompted.



Scanning is started.

Two images of the water phantom are recorded with identical parameters. The difference between the first and the second measurement is shown.

The following evaluations are displayed:

- ROI
- Mean value of the CT value
- Sigma value (pixel noise)



The measurement is repeated automatically for every tube voltage. The results of the test are output in the content area of the **Quality Constancy** dialog box.

Quality Noise Results

		Sigma [HU]	Voltage [kV]
KV Step 1	Reference:	---	80.0
	Tolerance:	---	+/- 8.0
	Result:	---	79.6
KV Step 2	Reference:	---	110.0
	Tolerance:	---	+/- 11.0
	Result:	---	109.8
KV Step 3	Reference:	4.60	130.0
	Tolerance:	+/- 0.46	+/- 13.0
	Result:	4.64	129.6
KV Step 4	Reference:	3.50	130.0
	Tolerance:	+/- 0.35	+/- 13.0
	Result:	3.46	129.8

In the status and error message area, the result of this test (in tolerance or out of tolerance) is output. Use the vertical scroll bar to view all test results.

Scanning ...
 Quality Noise: Results in tolerance
 Function "Quality Noise" completed.



All parameters and images shown are examples. Only the parameters displayed by your system are definite.



Test is out of tolerance.

- ◆ Contact Siemens Healthineers customer service.

12.2.11 MTF test

The *spatial resolution* is characterized through the modulation transmission function (MTF). It describes the contrast amplitude of bar patterns with different local frequencies shown in the image relative to an ideal transmission.

The 2 % value of the MTF is the local frequency at which the contrast of periodical bar patterns of this local frequency is reduced to 2 %. In practice, the MTF values are determined using a single thin wire.

After the previous test has been completed, the **MTF** test is started automatically.

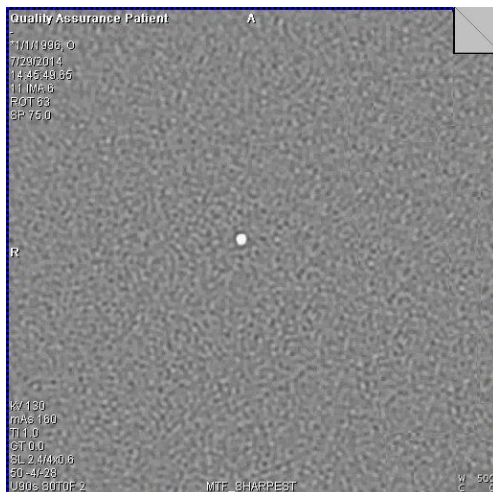
- ◆ Press the **Start** key on the control box, if prompted.



Scanning is started.

The MTF is represented graphically as a function of the local frequency (LP/cm). In addition, three characteristic values, 50 % MTF, 10 % MTF and 2 % MTF are displayed.

The scan is displayed in the image area. The result image is calculated and stored in the local database.



The evaluation is done for all detector rows using different kernels. The results of the test are output in the content area of the **Quality Constancy** dialog box.

Quality MTF Results 			
	50%	10%	2%
wire in plastics; 2x5.0mm; B31s			
Reference [lp/cm]:	3.51	6.83	8.30
Tolerance [lp/cm]:	+/- 0.35	+/- 0.68	+/- 0.83
row 1	3.35	6.59	8.21
row 2	3.39	6.63	8.23
wire in plastics; 2x5.0mm; H41s			
Reference [lp/cm]:	3.30	6.79	8.56
Tolerance [lp/cm]:	+/- 0.33	+/- 0.68	+/- 0.86
row 1	3.27	6.75	8.53
row 2	3.26	6.71	8.54
wire in plastics; 1x2.4mm; U90s			
Reference [lp/cm]:	9.58	13.29	15.15
Tolerance [lp/cm]:	+/- 0.96	+/- 1.33	+/- 1.51
row 1	9.57	13.49	15.54

In the status and error message area, the result of this test (in tolerance or out of tolerance) is output. Use the vertical scroll bar to view all test results.

Scanning ...
 Quality MTF: Results in tolerance
 Function "Quality MTF" completed.



All parameters and images shown are examples. Only the parameters displayed by your system are definite.



Test is out of tolerance.

- ◆ Contact Siemens Healthineers customer service.

12.2.12 Contrast test

With this test, you measure the contrast scale by a measurement in air and water.

After the previous test has been completed, the **Contrast** test is started automatically.

- ◆ Press the **Start** key on the control box, if prompted.



Scanning is started.

A central ROI is evaluated, the corresponding mean value (ME) and standard deviation (SD) are displayed, see the following image.



The results of the test are output in the content area of the **Quality Constancy** dialog box.

	CT Air [HU]	Contrast scale
Reference:	-1000.00	1.92e-004
Tolerance:	+/- 10.00	+/- 3.00e-006
row 1	-1000.38	1.92e-004

In the status and error message area, the result of this test (in tolerance or out of tolerance) is output. Use the vertical scroll bar to view all test results.

Scanning ...
Quality Contrast: Results in tolerance
Function "Quality Contrast" completed.



All parameters and images shown are examples. Only the parameters displayed by your system are definite.



Test is out of tolerance.

- ◆ Contact Siemens Healthineers customer service.

12.2.13 Table position test

With this quality measurement, you test whether the actual table position matches the table position displayed.

The **Table Position** test is performed without triggering radiation. It is started with **Go** in the **Quality Constancy** dialog box.

After the previous test has been completed, the **Table Position** test is started automatically.

- 1 Place approximately 70 kg (154 lb.) weight on the patient bed.

This is optional for customer testing. However, Siemens Customer Service performs the test with weight.

- 2 Click the **Constancy** procedure.

- 3 Click **Table Position**.

- 4 Draw a line down the center of a piece of tape and attach it across the pallet and the CT bed index beneath it. The tape's line will mark the 0 mm position on the patient bed.



- 5 Cut the tape between the patient bed and pallet.



Example showing the tape's line marking the 0 mm position.

- 6 Start measurement with **Go**.



The horizontal table position is set to 0. The table top moves 300 mm in the direction of the gantry.

- 7 Measure the distance between the lines that were drawn on the tape.
- 8 On the syngo MI Workplace, enter the distance between the lines, measured in **mm**, in the entry field of the content area.

Quality Table Position

Enter actual value at 300 mm position:

Click CONTINUE, when ready.



For combined SPECT and CT systems, the value on the *syngo* MI Workplace (CT user interface) will be 19 mm different than the actual measured position.

- 9 Click **Continue**. The table top now moves stepwise into the gantry (300 mm) and back.



- 10 Enter the actual values at the 300 mm and the 0 mm position in the entry field again.

Remove tape from patient bed and pallet.

The deviations of the actual from the displayed positions are calculated. The results of the test are output in the content area of the **Quality Constancy** dialog box.

Quality Table Position Results								
Position:	A	B	C	D	A	B	C	D
Move:	continuous				stepwise			
Reference [mm]:	-300.0	0.0	300.0	0.0	-300.0	0.0	300.0	0.0
Tolerance [mm]:	±1.0	±1.0	±1.0	±1.0	±1.0	±1.0	±1.0	±1.0
Result [mm]:	300.0 >	0.0	300.0	0.0	-300.0	0.0	300.0	0.0

In the status and error message area, the result of this test (in tolerance or out of tolerance) is output. Use the vertical scroll bar to view all test results.

Tablepositioning: Results in tolerance
Moving table outside the gantry.
Function "Tablepositioning" completed.



All parameters and images shown are examples. Only the parameters displayed by your system are definite.



Test is out of tolerance.

- ◆ Contact Siemens Customer Service.

12.2.14 CTDI Air test

With this test, you measure the dose in the system axis.

- ✓ The ionization chamber is centered in the axial direction in the system axis. The slice plane has to run through the center of the chamber. An appropriate dosimeter is connected to the ionization chamber.
 - ✓ The **CTDI Air** test mode is loaded.
 - ✓ You are prompted to press **Start**.
- 1 Press the **Start** key on the control box.



Scanning is started.

- 2 Read off the value on the dosimeter and enter it in the protocol.

Quality CTDI Air

Enter the result as displayed by dosimeter [mGycm]:

Click **CONTINUE**, when ready.

- 3 Continue with the next kV values.

The results of the test are output in the content area of the **Quality Constancy** dialog box.

Quality CTDI Air Results ✓

Slice [mm]	0x3.00	0x2.00	1x10.00	0x1.00	0x0.50	1x1.00
	130.0 [kV]					
Nominal [mGy/100mAs]:	30.80	32.90	29.60	30.40	30.50	30.60
DLP [mGycm]:	85.51	38.88	28.44	19.36	7.62	2.10
Result [mGy/100mAs]:	31.60	32.17	25.55	40.33	32.58	26.25

In the status and error message area, the result of this test (in tolerance or out of tolerance) is output. Use the vertical scroll bar to view all test results.

CTDI Air = 35.00 mGy/100mAs
 Quality CTDI Air: Results out of tolerance
 Function "Quality CTDI Air" completed.



All parameters and images shown are examples. Only the parameters displayed by your system are definite.



Test is out of tolerance.

◆ Contact Siemens Healthineers customer service.

12.2.15 Low contrast test

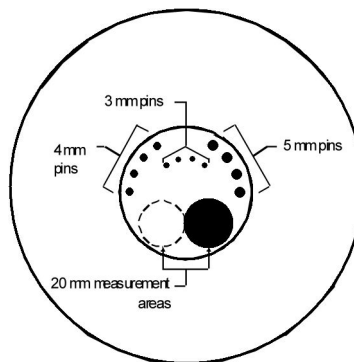
The Low Contrast test is performed in the **Quality Constancy** dialog box of the **Local Service** window.



The availability of the low contrast test is country-specific. It has to be activated by Siemens Healthineers service.

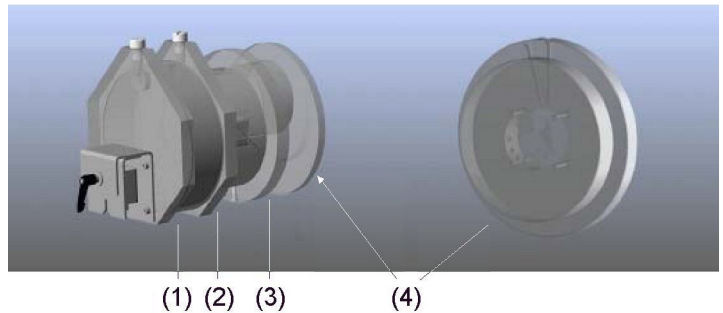
Low contrast phantom

The low contrast phantom section consists of a 165 mm diameter, 25 mm thick, water-equivalent, brown plastic cylinder. A clear plastic insert at the center contains three sets of low contrast pins (with diameters of 3, 4, and 5 mm, each set with 4 pins), together with two 20 mm diameter measurement areas.



The two 20 mm diameter measurement areas are used to establish low contrast differences between the pins and the surrounding area. The rated contrast between the pins and the surrounding area is 6 HU $\pm$ 1 HU (0.6 % $\pm$ 0.1 % contrast).

- 1 Attach the low-contrast phantom section to the Siemens Healthineers phantom combination, using the mounting bracket and four mounting screws.



- (1) Water phantom
 - (2) Slice thickness phantom
 - (3) Wire phantom
 - (4) Low contrast phantom
- 2 Mount the phantom assembly on the patient table at normal body height.
 - 3 Position the phantom set.

Acquiring low contrast image

- 1 Call up **Options > Service > Local Service**, delete the password entries, and click **OK**.
- 2 Click the **Quality Assurance** button to display the **Quality** dialog window.

- 3 Select the **Phantom Check** and the **Low Contrast** test.



- ☒ Phantom Check
- ☐ Lightmarker
- ☐ Sag /Cor. Lightm.
- ☐ Prev. Image Pos.
- ☐ Slice
- ☐ Homogeneity/Water
- ☐ Noise
- ☐ MTF
- ☐ Contrast
- ☒ Low Contrast
- ☐ Table Position

- 4 Confirm selection of the constancy test with **Go**.



You are prompted to enter the name of the tester and the serial numbers of the phantoms.

- 5 Enter your name and the data required.



Quality Constancy

Enter name of tester:

Enter serial numbers:

Slice phantom:

Water phantom:

Wire phantom:

Low Contrast phantom:

CTDI Body phantom:

CTDI Head phantom:

Dosimeter:

CT-Ionisation chamber:

• Click GO, when ready.

- 6 Move the table until the inner light marker points on the reference marking on the slice thickness phantom.
- 7 Start the low contrast test with **Go**.

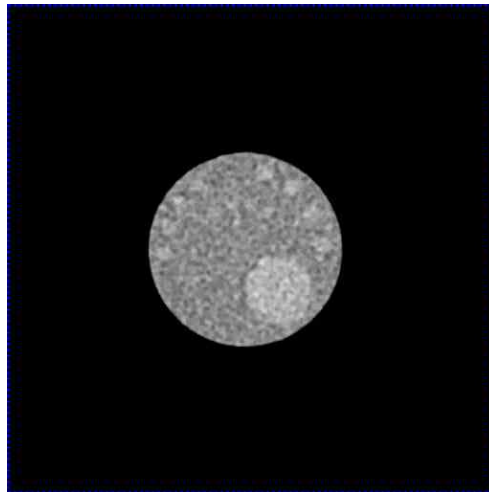


You are prompted to press **Start**.

- 8 Press the **Start** key on the control box.



The phantom position and the gantry inclination are checked. The image of the low contrast phantom is displayed in the image area.



Evaluating low contrast image

Normally one scan is sufficient to discern the pins in the row with the smallest diameter (3 mm). However, the procedure may be repeated if desired. Low contrast measurements are directly affected by the level of noise ("sigma") in the CT system, which may vary from scan to scan within a specified range. Low contrast estimations involve visual verification methods that are somewhat subjective. It is normal to expect some variations in assessing low contrast of the smallest (3 mm) pins.

- 1 Adjust the window width and center to improve visualization (adequate starting values: 40 width/90 center).
- 2 Observe the region where the small pins are located just below the two 20 mm measurement areas.
- 3 Enter the resolution in the corresponding entry field of the content area.

Quality Low Contrast

Enter the diameter [mm] of the smallest circle that can be seen on the scanned phantom:

The low contrast test is in tolerance if the resolution is at least 5.0 mm.

- 4 Click **Continue**.

Continue

The result of the test is output in the content area of the **Quality Constancy** dialog box.

Quality Low Contrast Results ✓

Nominal [mm]:	5.00
Result [mm]:	4.00

In the status and error message area the result of this test (in tolerance or out of tolerance) is output. Use the vertical scroll bar to view all test results.

Please enter value and click CONTINUE button
 Warning: table moves automatically after CONTINUE.
 Quality Low Contrast: Results in tolerance
 Function "Quality Low Contrast" completed.



- It may help to observe the image in a semi-darkened room and from distance of up to 5 feet.
- All parameters and images shown are examples. Only the parameters displayed by your system are definite.

**Test is out of tolerance.**

- ◆ Contact Siemens Healthineers customer service.

12.2.16 Constancy test results

A defined number of results are stored. Older values are overwritten by new values.

The images recorded during measurement are stored as images of the "Quality Assurance Patient".

Viewing QA test report

You can view the results of the daily and monthly quality test with the **Report** function of the **Local Service**.

- 1 Call up **Options > Service > Local Service**, delete the password entries, and click **OK**.
- 2 Click **Reports** in the **Home Menu** dialog window.



The **Report Files** dialog box is displayed.

- 3 Select **Quality Assurance**.

A list of the quality measurements performed (daily quality tests and constancy tests) is displayed sorted by date.

Report - Windows Internet Explorer provided by Siemens Healthcare Sector

Report Files Service Key Expiry Date: Help Exit

Tune-Up/QA/Tests

- Quality Assurance
 - Constancy
 - Daily

Tube

Current History

Third Party SW

Open Source SW

Double click onto report to view its details.

Function	Date	Time	Result	Coverage
CTDI Air	29.07.2014	14:54:01	cancel	partial
Table Position	29.07.2014	14:49:29	out tol.	complete
LowContrast	29.07.2014	14:47:41	in tol.	complete
Contrast	29.07.2014	14:46:55	in tol.	complete
MTF	29.07.2014	14:41:08	in tol.	complete
Noise	29.07.2014	14:39:27	in tol.	complete
Homogeneity	29.07.2014	14:38:53	in tol.	complete
Slice	29.07.2014	14:37:17	in tol.	complete
Prev. Image Pos.	29.07.2014	14:33:57	in tol.	complete
Sag. Cor. Lightm.	29.07.2014	14:31:50	out tol.	complete
Lightmarker	29.07.2014	14:30:34	in tol.	complete
Constancy	29.07.2014	14:26:14	cancel	reference
CTDI Air	29.07.2014	14:11:54	out tol.	complete
Constancy	29.07.2014	14:11:50	out tol.	reference
CTDI Air	29.07.2014	14:01:16	out tol.	complete
Table Position	29.07.2014	13:56:11	in tol.	complete
Constancy	29.07.2014	13:56:07	out tol.	reference
LowContrast	29.07.2014	13:49:54	in tol.	complete
Contrast	29.07.2014	13:47:56	in tol.	complete
Constancy	29.07.2014	13:47:51	in tol.	reference
Constancy	29.07.2014	13:46:50	error	*
MTF	02.07.2014	15:08:29	in tol.	complete
Noise	02.07.2014	15:06:56	in tol.	complete
Homogeneity	02.07.2014	15:06:21	in tol.	complete
Slice	02.07.2014	15:04:40	in tol.	complete
Prev. Image Pos.	02.07.2014	15:02:49	in tol.	complete

Copy Selected Reports to Disk

Extract Details

- 4 Double-click a quality test in the list to have more detailed information displayed.

The results of the selected quality test are displayed in the content area. Any test results outside the tolerance range are marked "<" or ">".



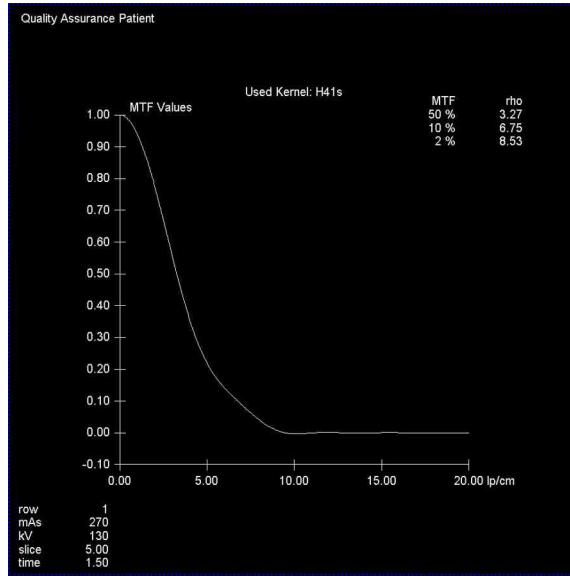
- A click of the **Extract** button generates a compressed file of report data and stores this file in a specific folder. This function is for service purposes only.
- Click **Home**. You return to the **Home** window.

Viewing QA test images

You can view the quality images (i.e., the result images of the MTF test) in the **Viewing** task card.

- 1 Open the **Patient Browser**.
- 2 Select the desired images of the "Quality Assurance Patient".
- 3 Call up **Patient > Load into Viewing**.

The images are displayed in the **Viewing** task card.



Result images of the MTF test

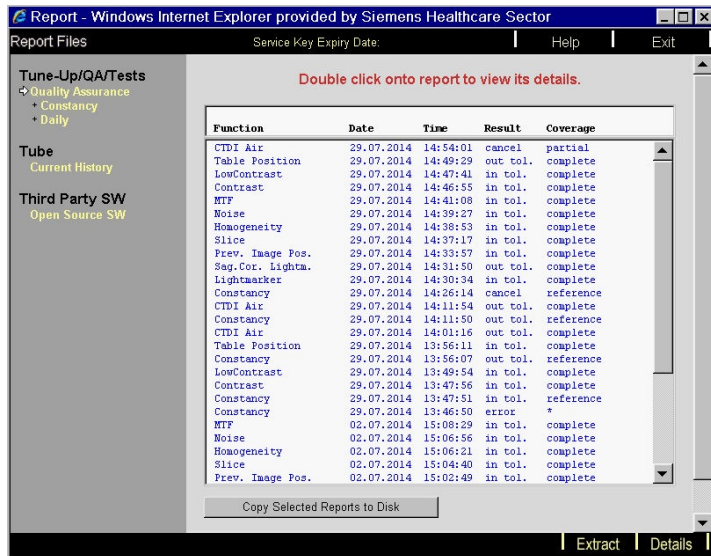


All parameters and images shown are examples. Only the parameters displayed by your system are definite.

Documenting QA test report

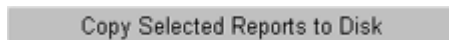
- 1 Call up **Options > Service > Local Service**, delete the password entries, and click **OK**.
- 2 Click **Reports** in the **Home Menu** dialog window.

The **Report Files** dialog box is displayed. Select **Quality Assurance**.



3 Do one of the following:

- You can copy individual report files to diskette. In the Report Files dialog box, select the quality tests in the list. Click **Copy Selected Reports to Disc.**



The selected reports are transferred to the diskette drive.

- You can print out an individual report. Double-click a report. Then click **Print.**



The displayed report is sent to the connected printer.



- You can enter the test results in the forms supplied in System Owner Manual.
- Test films and test forms must be stored for at least two years.
- Click **Home**. You return to the **Home** window.

12.3 Camera test

In the camera test, the image quality and the camera settings are checked. You can also film test images of the quality measurement in order to meet requirements for documentation.



CAUTION

Missing camera test!

Wrong basis for diagnosis.

- ◆ Perform the camera test regularly at the recommended intervals.

Please note the following points:

- Use the correct film type (emulsion number).
- Keep the developer bath at a constant temperature.
- Use the correct chemicals.
- Follow the development instructions supplied by the film manufacturer.



If the equipment or the emulsion number of the film is changed, the camera must be readjusted by a customer service technician who is authorized to do so. After this, another camera test is required.

12.3.1 Camera test references

The camera was set for optimum image quality during installation and should not be changed.

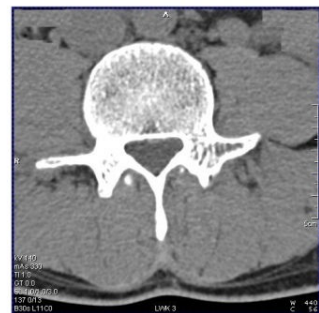
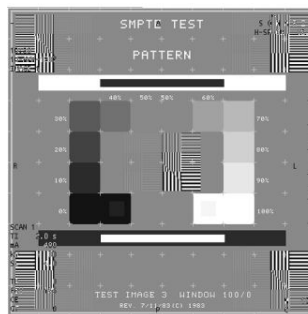
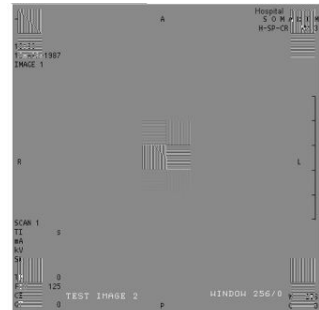
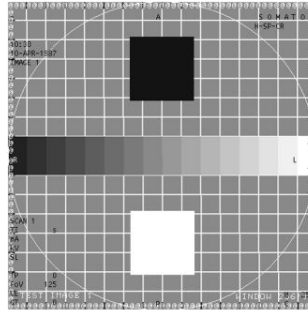
After installation, a test film is recorded and evaluated. The results are summarized in a table. These values are used as reference values for the daily quality check of the camera.

12.3.2 Performing the camera test

With the camera test (Film Demo), you can check the settings of the camera by filming the images of the “Reference Images” patient. You also have the option of filming the images of a previous constancy test to check the camera settings.

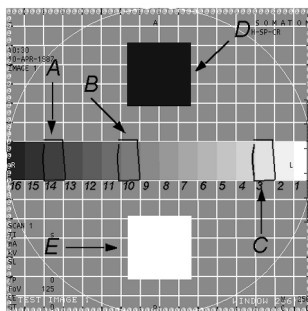
- 1 In the **Patient Browser**, transfer the test images of the “Reference Images” patient to the **Viewing** task card.

- 2 In the **Viewing** task card, film the test images of the “Reference Images” patient.



Examples of test images

- 3 Measure the fields marked in the first test image with the densitometer X-RITE 331.



Example of measurement fields marked by technologist

- 4 Enter the measured values in the constancy test report.
- 5 Compare the test results with the reference values.



Test is out of tolerance.

- ◆ Contact Siemens Healthineers customer service.

12.3.3 Camera test report

You will find a form for the test report in System Owner Manual.

12.4 Monitor test

Monitors are subjected to a constancy test to ensure that the image quality and measured values lie within a defined range.

In the **Patient Browser** window, you can select the test images stored in the database under **Local database /Reference /Images**. Load the test image of your choice in the **Viewing** task card.



To ensure that the criteria of the constancy test are met, make sure that the requirements for the room lighting are fulfilled or restored according to the acceptance test performed during initial start-up.

12.4.1 Monitor test requirement

Constancy test requirements for image display devices (monitors) may vary according to different national regulations. The constancy test described herein complies with the German X-ray regulations (Deutsche Roentgenverordnung) and is based on requirements given in DIN V 6868-57.



CAUTION

Missing constancy test of the monitor!

Wrong basis for diagnosis.

- ◆ Perform the monitor test regularly at the recommended intervals.

The acceptance test performed during initial start-up commissioning ensures optimum image quality of the device. The results of these measurements are documented in an acceptance test report and serve as reference values for the follow-up constancy test.

If any alterations are made to the device that affect the image quality, for example, repairs, replacement of parts, or readjustments, the acceptance test according to the Germany X-ray regulations must be repeated.

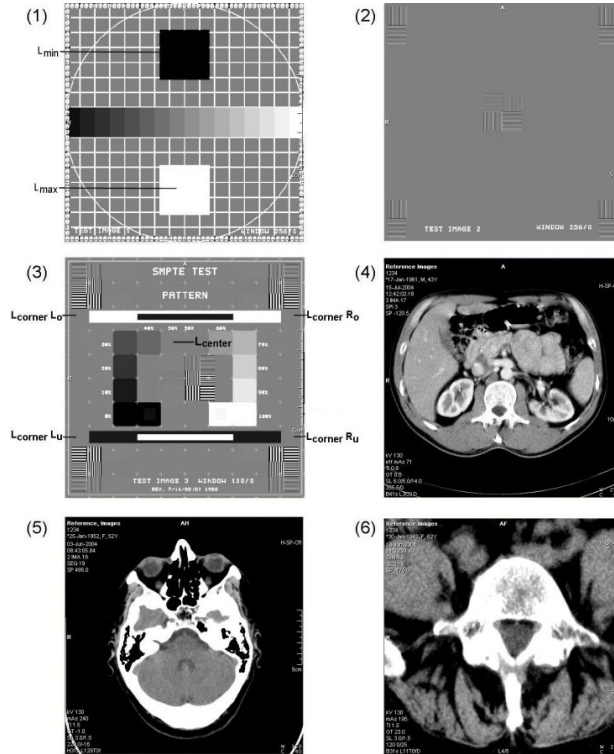
12.4.2 Monitor test equipment

The constancy test is performed using the following test equipment:

- Luminance meter for LCD monitors

Specification: Class B (DIN 5032-7) with valid calibration, range $0.05 \text{ cd/m}^2 - 1000 \text{ cd/m}^2$ (recommended device: Spotmeter SMfit Act for LCD monitors)

- Test image source (imaging device)
- Test images



- (1) Technical images
- (2) Clinical reference images

12.4.3 Measuring monitor luminance

In this test the veiling luminance (environmental lighting) and the maximum contrast are tested.

- 1 Switch the monitor off and wait at least 30 seconds.
- 2 Adjust then the room lighting to the operating conditions using the dimmer switch.
- 3 Measure veiling luminance L_s at the center of the screen with the luminance meter.
- 4 Switch on the monitor.

- 5 Select test image 1 to measure the minimum and maximum luminance, with the luminance meter.

In test image 1, the squares for the maximum grayscale value ($L_{\max}$, bright field) and the minimum grayscale value ($L_{\min}$, black field) are used.

The ratio $L_{\max}/L_{\min}$ is calculated as maximum contrast MK ($\geq 40:1$).



- When taking measurements with the luminance meter, observe the distance recommended by the manufacturer (40 – 60 cm).
- You can simply increase the contrast ratio by reducing the environmental lighting intensity. However, make sure that enough light is available for reading and writing.



Test is out of tolerance.

- ◆ Contact Siemens Healthineers customer service.

12.4.4 Monitor spatial resolution test

The visual resolution is checked in this test.

- 1 Select test image 2 or test image 3.
- 2 Check the lines of the grid with a modulation of 100 % high contrast (HC) at the center and in the four corners.

The lines of the grid must be recognizable.

12.4.5 Monitor homogeneity test

This test ascertains any deviations in the luminance (L) within the image.

- 1 Select test image 3.
- 2 With the luminance meter measure a point close to the center (L_{center}) and four measuring points in the corners (L_{corner}).
- 3 Calculate the deviation in % = $100 \times (L_{\text{center}} - L_{\text{corner}})/L_{\text{center}}$.

Maximum deviation of the corner points: Flatscreen ± 20 % (max.)



Visual interference such as ghosting, artifacts, overshoots, light or dark spots, flickering, and so on, encountered.

◆ Please consult your service technician.



Test is out of tolerance.

◆ Contact Siemens Healthineers customer service.

12.4.6 Monitor test report

Enter the test results in the forms supplied in System Owner Manual. All measurement and test results must be confirmed with the date and initials.



We recommend that the constancy test and collection of the reference values is performed and documented by authorized personnel.

13 Glossary

Algorithm	General computation procedure. <i>In this case:</i> image reconstruction from the measured <i>raw data</i> .
Anode	Positive electrode in the x-ray tube (source of radiation).
AP	A nterior- P osterior from front to back.
API	A utomatic P atient I nstruction; breathing instructions to the patient.
Artifact	Structure in a medical image which does not represent any anatomical structure and which impairs the representation of the anatomical structure (e.g., streaks).
Attenuation	The ratio of x-ray beam intensity prior to and after passing through an attenuating object.
Attenuation profile	Local distribution of attenuation values from one direction of projection of the fan beam.
Beam hardening artifact	<i>Artifact</i> caused by the change in the x-ray spectrum as radiation passes through the object being scanned.
Bolus tracking	A CARE application (optional). With the bolus tracking function, you can directly monitor the contrast medium bolus in the ROIs of a reference scan defined by the user. As soon as the preset contrast medium threshold is reached, the acquisition is automatically started.
Calibration	Procedure for adjusting the sensitivity of the individual detector channels; used for correction of measurement data.
CARE	CARE (C ombined A pplications to R educe E xposure) aims to achieve optimum image quality while reducing the radiation and contrast medium dose for the patient.
CD-R	Compact Disk Recordable.

Cluster	A chain of slices acquired rapidly in Sequence CT mode within a breath-hold.
Convolution	Mathematical procedure for image reconstruction in computed tomography.
CPU	Central Processing Unit, part of the computer that performs the calculations. The speed of the CPU not only determines factors such as the <i>RAM</i> configuration but also the processing speed of the computer.
CT	Computed Tomography.
CTA	CT Angiography Display of the vascular system using CT.
CTDI	CT Dose Index, calculated as absorbed dose to PMMA (Lucite) according to 21CFR, § 1020.33. The Integral of the dose profile is calculated along a line perpendicular to the tomographic plane (from -7T to +7T, T is the nominal tomographic slice thickness).
CTDI 100	CT Dose Index, calculated as absorbed dose to air according to IEC 60601-2-44. The Integral of the dose profile is calculated along a line perpendicular to the tomographic plane (from -50 mm to +50 mm).
CTDI vol	The CT Dose Index volume (IEC 60601-2-44) is the approximation of the average dose over a single slice in the standard head or body CT dosimetry phantom, expressed in terms of absorbed dose to air (mGy).
CT value	Number used to quantify the beam attenuation capability of an object. The scale uses water as a reference with a CT value of 0 (air : CT value of -1000).
Density determination	Quantitative evaluation of the <i>CT values</i> in a CT image.
Detector	Component of the scanning system which measures the beam intensity after radiation has passed through the object being scanned.
Detector array	System of <i>detectors</i> for acquiring radiation intensities.
DICOM	D igital I maging and C ommunications in M edicine. Standard for exchanging medical images.

Digital camera	Camera to which image data is digitally transferred.
DLP	Dose Length Product. When examining with CARE Vision CT the DLP value is displayed in the Patient Protocol. The DLP is calculated as follows: $DLP [mGy \times cm] = CTDI_{vol} [mGy] \times SliceWidth \times NoOfSlicePositionsPerScan$
Documentation	Transfer of image results to transparent film.
Dose	The total effect of radiation on a defined absorption volume of the body.
Dual Topo	Combination of <i>two Topograms</i> with different tube positions (AP and lateral).
Dynamic Multiscan	Scanning technique of continuous data acquisition of <i>multiple scans</i> at the same slice position.
Exporting	Image data can be transferred to an external medium (<i>CD-R, MOD, magnetic tape, PACS</i>) for storage.
Fan beam device	CT system in which an arc-shaped detector array and an x-ray tube generating a fan-shaped beam rotate about the patient.
File	Unit of data stored on the storage medium in the computer. A file might be an image, a database or a program.
Flying focal spot	Electromagnetic deflected focal spot for sampling each object detail from two slightly different perspectives. Used to reduce sampling artifacts.
Focal spot	Focal spot of an x-ray tube.
FOV	(Reconstruction) Field Of View , Used as an aid to calculate the tomogram.
Gantry	Scanning unit of a CT system including the x-ray tube and the <i>detector array</i> .
Hard disk	Part of the computer on which data and programs are stored for quick access.
Hardware	Technical physical equipment of a computer system, in contrast to software.

High contrast resolution	Spatial resolution; ability to display minute structures.
Highlighting	Bright display of a range of CT values during windowing.
HIS	Hospital Information System. Network for transmitting patient data and images within a hospital.
Histogram	Graphic and numerical display of frequency of CT values in a selected region.
Homogeneity	Characteristic of image quality which describes how accurately CT values are measured when a homogeneous object is scanned and displayed by the CT system.
Hounsfield unit	Unit used to measure the <i>CT value</i> .
HU	Abbreviation of Hounsfield Unit .
ICT	Interventional Computed Tomography. Acquisition method providing more space between the patient table and gantry, e.g., for use of a C-arm.
Image control system (ICS)	Computer on which the CT scanner is controlled (Navigator).
Image data compression	Data is compressed when storing images for more efficient use of the storage capacity of the data carrier. The <i>image matrix</i> is maintained.
Image matrix	Size of a medical image in <i>pixels</i> , e.g.: 512 x 512, 1024 x 1024.
Image mode	In image mode filming, the transfer of the images to the camera takes place image by image (see also <i>page mode</i>).
Image reconstruction	Computation of the CT image from the measured <i>raw data</i> .
Image reconstruction system (IRS)	Computer for reconstruction of the CT images from the measured <i>raw data</i> .
Isocenter	Center of the <i>scan plane</i> .
kV	Kilovolt (1 kV = 1 000 Volt); unit used to measure the tube voltage.
Laser camera	Camera which utilizes a laser beam for exposing images.

LED	L ight E mitting D iode, semi-conductor element that lights up like a bulb.
Low contrast resolution	Resolution of minute image details when there is little differentiation in density. For example, in soft tissue.
mA	Milliampere (1 mA = 1/1000 A); unit used to measure the tube current.
mAs	Milliampere per second product; unit used to measure the product of the tube current (mA) and scan time (s). Equivalent to the applied radiation dose.
MCA	M otion A rtifact C orrection A lgorithm: algorithm correcting motion artifacts.
Measurement field	The field from which the <i>detector array</i> acquires data in the <i>gantry</i> .
MinIP	M inimum I ntensity P rojection, like <i>MIP</i> but visualization of the darkest <i>voxels</i> (areas of least absorption).
MIP	M aximum I ntensity P rojection is a method for 3D display: Along the viewing direction through a volume, the voxel with the strongest absorption is displayed in the result image in each case.
MOD	Magneto Optical Disk resp. Drive; system for exchange data.
Motion artifact	<i>Artifact</i> caused by patient movement during the scan.
Mouse	Operating element at the computer to direct the input cursor on the monitor and to adjust window values.
MPR	M ulti P lanar R econstruction : using an image data set you have the ability to perform retrospective reconstruction in different planes of orientation (i.e., from axial images create coronal, sagittal, and paraxial images).
MTF	M odulation T ransfer F unction ; the frequency-dependent ratio of object contrast to image contrast. The MTF permits qualitative determination of the spatial resolution of an imaging system.
Multiformat camera	Camera with selectable segmented film format; allows for exposures of multiple images on a single film.

Noise	Unwanted interference in the signal caused by random factors and containing no information regarding the object being scanned.
Operating system	Group of computer programs required for general operation of the computer and its periphery.
Osteo CT	Semiautomatic bone mineral density measurement and evaluation method for the quantitative assessment of bone mineral content in the lumbar spine.
PA	Posterior Anterior.
PACS	Picture Archiving & Communication System. Siemens Healthineers network for archiving and transmitting medical images.
Page Mode	In page mode filming, the entire film sheet is composed in the computer and then transferred to the camera (see also <i>image mode</i>).
Partial scan	<i>Scan</i> for which only a section (240°) of the revolution of the X-ray tube is used to measure a body slice.
Partial volume artifact	<i>Artifact</i> caused by bone structures which are only partially included in the slice being scanned.
Phantom	Measuring object whose weakening properties is similar to human; Reference body used for functional and quality assurance tests.
Pitch factor	Term applicable only for spiral CT. It is defined as the ratio of table feed per rotation to the collimated width of the entire detector. Example: $64 * 0.6 \text{ mm collimation, table feed } 96 \text{ mm/sec, rotation time } 0.6 \text{ s. This gives a pitch factor of } (96 * 0.6) / (64 * 0.6) = 57.6 / 38.4 = 1.5.$ Term applicable only for spiral CT. It is defined as the ratio of table feed per rotation to the collimated width of the entire detector. Example: $16 * 0.75 \text{ mm collimation, table feed } 18 \text{ mm/sec, rotation time } 1 \text{ s. This gives a pitch factor of } 18 / (16 * 0.75) = 18 / 12 = 1.5.$
Pixel	Image element of the two-dimensional image.
PSB	The Power Switch Box is a small power distribution device to connect the console to the main power.

RAM	Fast, non-permanent memory (R andom A ccess M emory) with which the computer operates. The size of the available RAM determines other factors, including the processing speed. Data are stored long term on <i>hard disk</i> , <i>MOD</i> or <i>CD-R</i> .
Raw data	Preprocessed measurement data. A raw data file corresponding to each scan is used to reconstruct the CT image.
Real Time Display	Displaying of the measured image synchronous to the scanning and just after the acquisition of a slice (rotation) is finished.
Real Time Recon	Reconstruction synchronous to scanning with an additional delay for displaying the first image.
Reconstruction FOV	see FOV.
Rendering	General procedure for constructing images using a computer.
RIS	R adiology I nformation S ystem. Network for transmitting patient data and displays to radiology departments, see also HIS.
ROI	A region of the CT image which can be selected with respect to position, size and shape, and in which quantitative evaluations are performed (R egion O f I nterest).
RTD	R ead T ime D isplay, see there.
Scan	CT acquisition of a slice.
Scan FOV	Determines the size of the area that you expose during acquisition. The Scan FOV is the maximum possible reconstruction FOV.
Scanning footswitch	Component that can trigger scanning as an alternative to using the start key. It is connected to the gantry.
Scan plane	Plane in which the X-rays penetrate the patient's body.
Scan time	Period in which radiation is released during a CT scan.
Scatter radiation	Radiation whose direction of dispersion is modified through the interaction with a material.

Secondary slice	Calculation of a selectable slice from a series of adjacent or overlapping slices.
Sequence CT	Scanning technique with single scans and optional table feed between the scans.
Slice thickness	Thickness of slice to be reconstructed. The slice thickness determines the collimation during scanning.
Software	Programs of a computer.
Spiral CT	Scanning technique with continuous rotation, continuous data acquisition and continuous table travel.
SSD	Shaded Surface Display: Rendering technique for the 3D surface construction ; only <i>voxels</i> the <i>CT values</i> of which exceed a defined threshold value are considered.
Tomogram	<i>Scan</i> of a slice perpendicular to the longitudinal axis of the patient.
Topogram	Frontal or lateral survey <i>scan</i> , similar to a conventional x-ray exposure.
UFC	Ultra-Fast Ceramic. <i>Detectors</i> from Siemens Healthineers with short decay time for fast <i>scans</i> .
UPS	Uninterrupted Power Supply ensures operation even during short power failures.
VAR	Volume Artifact Reduction. Procedure for reducing <i>partial volume artifacts</i> .
Volume Pitch	Term applicable only to <i>Spiral CT</i> . It is defined as the ratio of table feed per rotation to slice width of one detector row.
Voxel	Mapped volume element determined by the pixel size of the image matrix and the slice thickness.
VRT	Volume Rendering Technique. Rendering technique for 3D visualization.

Windowing Display of a selectable portion of the CT values using optimized contrast range of the monitor.

Working memory see *RAM*.

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The original language of this document is English.

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Date of Issue: 2019-10

**Siemens Healthineers
Headquarters**
Siemens Healthcare GmbH
Henkestr. 127
91052 Erlangen
Germany
Phone: +49 9131 84-0
siemens-healthineers.com

EU Authorized Representative
Siemens Healthcare GmbH
Henkestr. 127
91052 Erlangen
Germany

Legal Manufacturer
Siemens Medical Solutions USA,
Inc.
Molecular Imaging
2501 N. Barrington Road
Hoffman Estates, IL 60192
USA

Material No. 11517070.02

Print No. MI-CTEXSC.621.01.01.02 | © Siemens Healthcare GmbH, 2019

www.siemens.com/healthcare